The Statistical Value of Adaptive Feature Queries

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Abstract

Questionnaires, clinical protocols and budgeted classifiers cap how many features each record may reveal. We ask how many more records a collector needs when it fixes each record's questions in advance instead of choosing each question after seeing earlier answers. The target is a known contrast of low Fourier degree, such as the accuracy of a frozen screening rule; labels are observed for every record, and no complete reference is trusted to transport to the target. For any finite menu of query protocols, the first-order minimax constant is a convex program whose converse covers biased and sequential procedures. In the worst case over populations, pre-committed forms need at most 2^{b-1} times as many records as branching interviews, and this is sharp. With r batches of questions the loss is at most 2^{b-r}, and for every fixed r this is tight up to polynomial factors in b. At a given population the loss lies between two computable quantities, and a column-generation solver whose cost does not grow with the 2^m profiles certifies optimal fixed, r-round and branching designs. In exact computations on random-forest updates and on screening trees with up to 21 survey items, pre-commitment costs a factor between one and 1.98 in records, an optimal second round of questions recovers most of it, and imputation from complete source records is preferable when its transport assumption holds but loses coverage under pre-specified population shifts.

1. Introduction

Population quantities are often estimated from records that each reveal only a few features. Questionnaires cap respondent burden, clinical protocols cap the tests per patient, and budgeted classifiers acquire a few inputs per case (Turney, 1995; Trapeznikov & Saligrama, 2013; Nan & Saligrama, 2017). A collector can pre-commit to each record's feature set, as in split questionnaires, matrix sampling and fixed laboratory panels (Raghunathan & Grizzle, 1995; Chipperfield & Steel, 2009), or it can choose each question after seeing earlier answers, as in skip logic and sequential testing. This paper asks how many more records pre-commitment costs when the target is a population quantity, such as the accuracy of a frozen screening rule or the accuracy gain of a model update.

We study the question in a design-based regime. The label of every sampled record is observed and charged separately, the target is one known contrast of degree at most b, answers do not depend on question order, and no complete reference is trusted to transport to the target population. When such a reference exists, imputation is the better tool, and when order effects exist, they bias every design; Section 6 quantifies both.

Classical adaptivity results count queries per input. A depth-b decision tree is simulated non-adaptively by reading all of its $2^b - 1$ nodes, and the huge-object model and interactive local privacy separate protocol classes by query or sample complexity (Goldreich & Ron, 2022; Adar & Fischer, 2024; Joseph et al., 2019; 2020). Under a hard per-record cap the price of non-adaptivity is paid in records instead, and the questions become quantitative: how large is the price, how much of it do a few rounds of questions recover, and how large is it at the population at hand?

Contributions.

- *Exact characterization (Theorem 1).* For any finite menu of interviews, the first-order minimax constant is a finite convex program. Its converse holds for biased estimators, for adaptation across records and for interview choices that see the current label; a finite-sample version is tight within a universal factor for every contrast.
- *Worst case and rounds (Theorems 2 and 3).* Pre-committed forms lose at most $\min\{2^{b-1}, \binom{m-1}{b-1}\}$, with equality at an address contrast. Batches of questions in r rounds lose at most 2^{b-r}, and a binary pointer tree shows a loss of $2^{b-1}/(3b)^r$. At small caps

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the solver computes the pointer tree's constants: two rounds lose 1.80 and 3.32–3.33 at $b \in \{3, 4\}$, close to the upper bound 2^{b-2}.

- *Population value and computation (Theorem 4, Propositions 5 and 6).* The loss at a population lies between the inverse energy explained by the best form and an optimized path-set design, both computable from reference data. A column-generation solver with exact dynamic-programming pricing certifies the optimal design of every protocol class at cost $O(\binom{m}{b}4^b)$ per iteration, and a two-phase design learns it from the audit itself without bias.
- *Experiments (Section 6).* Certified constants for forest updates at caps two to four and for screening trees on 12 and 21 survey items: pre-commitment costs a factor between one and 1.98, the exact optimal two-round design recovers a median 100% of the log-gap on BRFSS at caps four and five and six, the path-set upper end is within a median 2.0% of the optimum, and under pre-specified population shifts imputation from complete source records loses coverage while design-based scores keep it.

2. Estimand and Acquisition Protocols

Let X take values in $\mathcal{X} = \{-1, 1\}^m$ and let $Y \in \{-1, 1\}$ be a label. Freeze two rules $p_0, p_1 : \mathcal{X} \to [0, 1]$ that predict $+1$ with probabilities $p_j(x)$, put $d = p_1 - p_0$, $t(x, y) = yd(x)$, and

$$\Delta(P) = \mathbb{E}_P[t(X, Y)] = \mathbb{E}_P[Yd(X)]. \tag{1}$$

This is the accuracy difference of the two rules; $p_0 \equiv 1/2$ makes it affine in the accuracy of p_1 alone, and Brier improvements are handled in the same way (Appendix G.3). Labels are recorded for every sampled record and charged separately; the cap b bounds the feature queries per record and is *binding* when $m > b$.

Protocols. An interview ι maps a profile x to a transcript $\ell_\iota(x)$. A *fixed form* queries a coordinate set chosen before the record is seen. An *r-round interview* queries batches $B_1, B_2(x_{B_1}), \ldots, B_r$, each chosen from the answers to earlier batches, with $\sum_s |B_s| \le b$. A *branching interview* is a decision tree of depth at most b. The menus satisfy $\mathcal{A}_{\text{fixed}} = \mathcal{A}_1 \subseteq \mathcal{A}_2 \subseteq \cdots \subseteq \mathcal{A}_b = \mathcal{A}_{\text{branch}}$. Across records, a protocol may use all previous records to choose the next interview, and our lower bounds even allow the current label to be seen first. $R_{\mathcal{A}}(d, n)$ is the minimax mean squared error over protocols and estimators that use n records and interviews from $\mathcal{A}$.

Identification. Let $\mathcal{H}_{\mathcal{A}}$ be the span of functions constant on the transcript cells of $\mathcal{A}$. The contrast is identifiable for all populations exactly when $d \in \mathcal{H}_{\mathcal{A}}$; for fixed forms of size b and for depth-b trees this is the same condition, Fourier degree $\deg(d) \le b$ in the characters $\chi_S(x) = \prod_{j \in S} x_j$. It holds for every probability average of depth-b trees, such as a depth-limited random forest. Thresholds and nonlinear links can raise the degree; Appendix G treats that case.

Table 1. The constants of the paper. Ratios of V compare protocol classes over worst-case populations; ratios of W are the ratios of records needed at a population P.

$V(d, \mathcal{A})$	worst-case first-order constant, $\lim_n nR_{\mathcal{A}}(d, n)$ (Thm. 1)
$W_{\mathcal{A}}(P)$	smallest variance of a pointwise-unbiased design at P; local minimax risk times n (Eq. 4)
$M_{\mathcal{A}}(P)$	$W_{\mathcal{A}}(P) + \Delta(P)^2$, the second moment of that design
$K_{m,b}$	$\min\{2^{b-1}, \binom{m-1}{b-1}\}$, the sharp worst-case gain (Thm. 2)
Γ_P	largest fraction of $\mathbb{E}_P[t^2]$ that one form explains; $1/\Gamma_P$ lower-bounds the gain (Thm. 4)
$N_P(\tau)$	Rényi-1/2 effective number of a tree's question paths; upper-bounds the gain (Thm. 4)

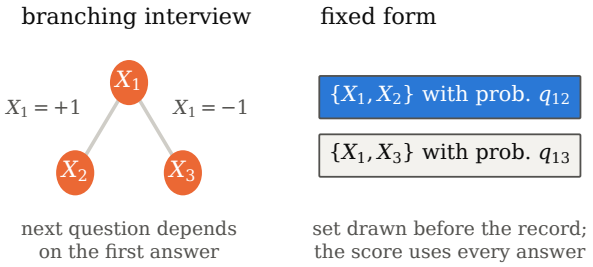


Figure 1. The worked example. A two-query branching interview asks X_2 after $X_1 = +1$ and X_3 after $X_1 = -1$. A fixed form commits to $\{1, 2\}$ or $\{1, 3\}$ before the record is seen and completes the tree's path only for half of the profiles.

A worked example. Let $m = 3$, $b = 2$ and $d(x) = [(1 + x_1)x_2 + (1 - x_1)x_3]/2$: the first answer selects which of two others is the target (Figure 1). Branching reads x_1 and then the selected coordinate, so d is computed and $V_{\text{branch}} = \max_x d(x)^2 = 1$. Under the uniform law every form of size two explains exactly half of $\mathbb{E}[d^2]$, so $\Gamma_{\text{U}} = 1/2$, and the tree's two path sets are equally likely, so $N_{\text{U}} = 2$. Theorem 4 then forces $M_{\text{fixed}}/M_{\text{branch}} = 2 = K_{3,2}$: the best pre-committed design draws $\{1, 2\}$ or $\{1, 3\}$ with probability $1/2$ and doubles the score when the drawn form contains the selected coordinate, and it needs twice as many records.

3. An Exact Characterization

Choose interview probabilities q_ι and transcript tables g_ι. The score $Yg_\iota(\ell_\iota(X))/q_\iota$ is unbiased for $Yd(X)$ at every

profile exactly when $\sum_\iota g_\iota(\ell_\iota(x)) = d(x)$ for all x. Define

$$V(d,\mathcal{A}) = \inf_{q,g:\ \sum_\iota g_\iota = d} \max_{x\in\mathcal{X}} \sum_\iota \frac{g_\iota(\ell_\iota(x))^2}{q_\iota}, \qquad (2)$$

where a zero-probability interview has a zero table.

Theorem 1 (Exact characterization)**.** *Let $\mathcal{X}$ and $\mathcal{A}$ be finite and $d \in \mathcal{H}_\mathcal{A}$. Then $nR_\mathcal{A}(d,n) \to V(d,\mathcal{A})$, and an iid design with the unbiased score attains V/n for every n. The converse applies to biased estimators, to designs that adapt across records, and to interview choices that depend on the current label. Moreover*

$$V(d,\mathcal{A}) = \sup_{\mu,h} \frac{\langle d,h\rangle_\mu^2}{\max_{\iota\in\mathcal{A}} \|\mathbb{E}_\mu[h \mid \ell_\iota(X)]\|_\mu^2},$$

over strictly positive profile laws μ and witnesses h with nonzero denominator. For every contrast d, identifiable or not, and every n, the minimax risk $R_\mathcal{A}(d,n)$ lies within a universal factor 104 *of*

$$\Phi_n(d,\mathcal{A}) = \inf_{q,g} \max_{x\in\mathcal{X}} \Big[\big(d(x) - s_g(x)\big)^2 + \frac{1}{n}\sum_\iota \frac{g_\iota(\ell_\iota(x))^2}{q_\iota}\Big], \qquad (3)$$

where $s_g(x) = \sum_\iota g_\iota(\ell_\iota(x))$ is the mean score at profile x.

Formal statements and proofs are in Appendix B. The dual compares the signal $\langle d,h\rangle_\mu$ of a perturbation of the label mean with the largest projection energy that one interview can capture; a change of measure, the KL chain rule and a local van Trees argument turn this into minimax risk. These are established tools for linear functionals (Donoho, 1994; Juditsky & Nemirovski, 2009; Polyanskiy & Wu, 2026) and for estimation with a prescribed channel family (Acharya et al., 2019; Barnes et al., 2020); what is new is that the witness is controlled simultaneously for every interview of a menu, including label-dependent choices, which yields exact constants for *comparing* query classes. For $d \notin \mathcal{H}_\mathcal{A}$, (3) decreases to the squared identification floor $\delta^2 = \inf_{v\in\mathcal{H}_\mathcal{A}} \|d - v\|_\infty^2$, so the risk is $\delta^2 + \Theta(1/n)$ (Appendix C.3.3); the exact first-order constant of that biased regime is open.

Population-specific constants. For a fixed law P of (X,Y), let

$$W_\mathcal{A}(P) = \inf_{q,H:\,\mathbb{E}[H|X,Y]=t(X,Y)} \mathrm{Var}_{P,q}(H), \qquad (4)$$

where the score must be unbiased at every profile and label but may be tuned to P. At a full-support P, $W_\mathcal{A}(P)/n$ is the local asymptotic minimax risk over arbitrary estimators that use the capped records alone (Proposition C.7). Ratios $W_{\mathrm{fixed}}(P)/W_{\mathrm{branch}}(P)$ therefore compare protocol classes at that population for every such estimator; estimators that import complete records from another source lie outside this information set.

4. How Much Can Adaptivity Be Worth?

Theorem 2 (Sharp gap)**.** *For $1 \le b \le m$ and a nonzero contrast of degree at most b on $\{-1,1\}^m$,*

$$1 \le \frac{V_{\mathrm{fixed}}(d,b)}{V_{\mathrm{branch}}(d,b)} \le K_{m,b} = \min\Big\{2^{b-1}, \binom{m-1}{b-1}\Big\}. \qquad (5)$$

The supremum over dimensions is 2^{b-1}, attained at $m = b - 1 + 2^{b-1}$. For every $b \ge 2$ there is an example with $m \le b^2$ and ratio at least $2^{b-b/(\kappa+1)-\kappa}$, where $\kappa = \lfloor \log_2 b \rfloor$.

Proof idea. Pad every root-to-leaf path of a tree to b distinct coordinates. Its first $b-1$ answers determine the path's coordinate set, so a tree has at most $K_{m,b}$ distinct sets. A fixed form draws one set uniformly, follows the tree when it can, and multiplies a completed score by the number of sets (Figure 2b); the mean is unchanged and the second moment grows by at most $K_{m,b}$. For sharpness, let $\kappa = b-1$ address bits select one of $D = 2^\kappa$ data bits, $d(A,Z) = Z_A$ (Figure 2a); this is the worked example of Section 2 with $\kappa = 1$. Branching computes d, so $V_{\mathrm{branch}} = 1$. Under the uniform law, a form T with s address and r data bits explains energy

$$\|\mathbb{E}[d \mid X_T]\|_2^2 = r2^{s-2\kappa} \le 2^{-\kappa}, \qquad r + s \le \kappa + 1, \quad (6)$$

and the dual in Theorem 1 gives $V_{\mathrm{fixed}} \ge D$. Products of address blocks give the polynomial-dimensional example (Appendix C.4.1). Both constants in (5) are worst cases over populations taken separately; Section 5 compares the classes at one population. The gap requires that *every* record obey the cap: under a mean budget of b queries, querying all m features with probability b/m bounds the ratio by m/b, and the exact interpolation between the two budgets is in Appendix C.5.1.

Theorem 3 (Rounds of adaptivity)**.** *Let V_r denote the constant for r-round interviews with at most b queries.* (a) *For every nonzero contrast of degree at most b and $1 \le r \le b$, $V_{\mathrm{branch}} \le V_r \le \min\{2^{b-r}, \binom{m-r}{b-r}\} V_{\mathrm{branch}}$.* (b) *For $r, \kappa \ge 1$ and $b = r\kappa + 1$, the pointer-chasing contrast $N_{r,\kappa}$ on $m = \Theta(2^{r\kappa})$ coordinates satisfies $V_{r+1}(N_{r,\kappa}) = V_{\mathrm{branch}}(N_{r,\kappa}) = 1$ and $2^\kappa/b \le V_r(N_{r,\kappa}) \le 2^\kappa$.* (c) *For $b \ge 2$, the binary pointer tree B_{b-1} on $m = 2^b - 1$ coordinates satisfies $V_{\mathrm{branch}}(B_{b-1}) = 1$ and, for $2 \le r \le b$,*

$$V_r(B_{b-1}) \ \ge\ \frac{2^{b-1}}{r\,b\,(1 + 2b/(r-1))^{r-1}} \ \ge\ \frac{2^{b-1}}{(3b)^r}.$$

Hence $2^{b-2}/(b(2b+1)) \le \sup_{m,d} V_2/V_{\mathrm{branch}} \le 2^{b-2}$, and the exponent of the r-round gap lies between $b - 1 - r\log_2(3b)$ and $b - r$.

In $N_{r,\kappa}$, a κ-bit register points to one of 2^κ registers at the next level, r levels deep, and the last pointer selects a data bit; following one level per round computes the contrast

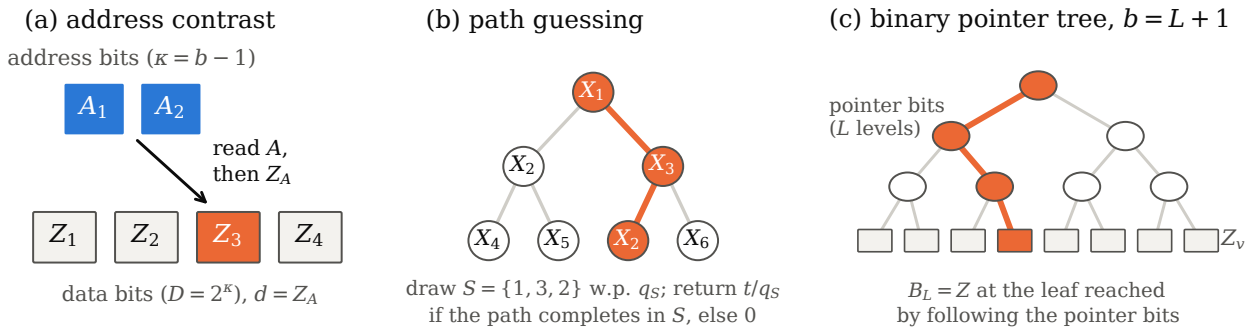


Figure 2. The three constructions behind Theorems 2 and 3. (a) Address contrast: $\kappa = b - 1$ address bits select one of $D = 2^\kappa$ data bits; branching reads the address and then the selected bit, while any fixed set of b coordinates explains at most a fraction $1/D$ of the target's energy under the uniform law. (b) Path guessing: a fixed form draws one of a tree's at most 2^{b-1} padded path sets and rescales the tree's score by $1/q_S$ when the path completes inside it; at a population, $q_S \propto w_S^{1/2}$ gives the effective number of paths N_P. (c) Binary pointer tree: $L = b - 1$ levels of pointer bits select a child and the target is the data bit at the reached leaf; a round that reads a pointer bits raises the conditional probability of the most likely leaf by at most a factor $1 + 2a$.

in $r + 1$ rounds, while in r rounds the selected data bit is read only by a lucky guess (Appendix D.1). The binary pointer tree is the tree evaluation instance of Cook et al. (2012), for which Li et al. (2026) show that exact evaluation with B queries per round needs $L/\lceil \log_2(B + 1) \rceil$ rounds; under a total per-record cap exact evaluation in few rounds is impossible for every strategy, and the question is how much variance the shortfall costs. Let M_t be 2^L times the conditional probability of the most likely leaf after t rounds. A round that reads a_t pointer bits satisfies $\mathbb{E}[M_t \mid \text{past}] \le (1+2a_t)M_{t-1}$, because a newly read bit can at most double the probability of a leaf below it; a supermartingale argument bounds the probability of ever reading the selected data bit, and the dual of Theorem 1 converts it into the bound on V_r. Against part (a), the bound is tight up to the factor $2^{r-1}(3b)^r$, a polynomial in b for a fixed number of rounds. It is informative for $b \ge 10$; at small b the solver of Section 5 computes the constants of B_{b-1} directly: at $b \in \{3, 4\}$ fixed forms lose exactly 4 and 8, two rounds lose 1.80 and 3.32–3.33 against the upper bounds 2 and 4 of part (a), and $b - 1$ rounds lose 1.80 and 1.80 (Appendix D.4). Questionnaire pages, clinic visits and laboratory batches are rounds in this sense.

5. The Value of Adaptivity at a Population

The worst-case constants are attained by extremal populations. For a fixed P we bracket the gain by interpretable quantities. Let τ be a depth-b tree that computes d, let $S_\tau(x)$ be the padded coordinate set of the path of x, and put $w_S = \mathbb{E}_P[t^2 \mathbf{1}\{S_\tau(X) = S\}]/\mathbb{E}_P[t^2]$. Define

$$N_P(\tau) = \Big(\sum_S w_S^{1/2}\Big)^2, \qquad \Gamma_P = \max_{|T|=b} \frac{\mathbb{E}_P\big[\mathbb{E}_P[t \mid X_T, Y]^2\big]}{\mathbb{E}_P[t^2]}.$$

N_P is the exponential Rényi entropy of order $1/2$ of the path weights, the effective number of question sets the tree uses at P; Γ_P is the largest fraction of the target's energy that any single form explains.

Theorem 4 (Population-specific bracket)**.** *If a depth-b tree τ computes d, then $M_{\text{branch}}(P) = \mathbb{E}_P[t^2]$ and*

$$\frac{1}{\Gamma_P} \le \frac{M_{\text{fixed}}(P)}{M_{\text{branch}}(P)} \le N_P(\tau) \le K_{m,b}.$$

On the variance scale, with $\sigma_P^2 = \mathbb{E}_P[t^2] - \Delta^2$ and $\Delta = \Delta(P)$,

$$\frac{\mathbb{E}_P[t^2]/\Gamma_P - \Delta^2}{\sigma_P^2} \le \frac{W_{\text{fixed}}(P)}{W_{\text{branch}}(P)} \le \frac{N_P(\tau)\mathbb{E}_P[t^2] - \Delta^2}{\sigma_P^2}.$$

Both second-moment inequalities are equalities, with value D, for the address contrast under the uniform law. For every menu $\mathcal{A}$ and witness h, $M_{\mathcal{A}}(P) \ge \mathbb{E}_P[th]^2 / \max_{\iota \in \mathcal{A}} \mathbb{E}_P[\mathbb{E}_P[h \mid \ell_\iota(X), Y]^2]$.

The upper bound is path guessing with $q_S \propto w_S^{1/2}$, the lower bound is Cauchy–Schwarz applied to the decomposition $\sum_\iota g_\iota = t$, and the variance scale follows because every design in (4) is unbiased (Appendix E.1). Adaptivity pays when the tree's questions diverge over many well-populated paths and no fixed set of b questions explains most of the target. Path guessing discards every answer on a form except a completed path; optimizing the scores on the same small menu tightens the upper end at no enumeration cost.

Proposition 5 (Optimized path-set design)**.** *Let $M_{\text{PS}}(P)$ be the optimum of (4) over the fixed-form menu of the padded path sets of τ. Then $M_{\text{fixed}}(P) \le M_{\text{PS}}(P) \le N_P(\tau) M_{\text{branch}}(P)$, and the program has at most $2^{b+1} K_{m,b}$ score variables.*

Computing every protocol class exactly. Pad every transcript to exactly b coordinates, so that an interview is a partition of the cube into cells $\{x : x_T = u\}$ with $|T| = b$. Pointwise unbiasedness is then equivalent to matching the Fourier coefficients of degree at most b, and the masses of all $\binom{m}{b}2^{b+1}$ cells are computed once from records. Column generation solves a restricted master problem over a few interviews. For its dual witness η, the information of a cell (T, u) is a Walsh transform of η restricted to T, and the most informative interview of any class $\mathcal{A}_r$ is found by dynamic programming over partial assignments: a batch $B \supsetneq A$ is chosen to maximize the summed value of its extensions, singleton batches give trees, and one full batch gives forms. Any normalized witness certifies a lower bound through the last display of Theorem 4, and the master value is an upper bound. One iteration costs $O(\binom{m}{b}4^b)$ and never touches the 2^m profiles (Appendix E.3).

Optimal pre-committed designs. On the cube, pointwise unbiasedness of form tables, $\sum_T g_{T,y}(x_T) = y d(x)$, is equivalent to $\sum_{T \supseteq S} \widehat{g}_{T,y}(S) = y\widehat{d}(S)$ for $|S| \le b$, so the fixed-form design problem

$$\min_{q,g} \sum_{T,u,y} P_T(u,y)\,\frac{g_{T,y}(u)^2}{q_T} \text{ s.t. these identities, } \sum_T q_T = 1 \tag{7}$$

is a second-order cone program. Its tables use every answer on a form and include all shared-coordinate control variates (Proposition E.4); in coarsening terms it computes the optimal augmentation of inverse-probability weighting jointly with the form probabilities (Robins et al., 1994; Tsiatis, 2006). The identities hold at every profile, so a design fitted on reference marginals stays exactly unbiased when the reference is wrong.

Proposition 6 (Two-phase self-calibration)**.** *Let a pilot of $n_1 \to \infty$, $n_1 = o(n)$, records be interviewed with the uniform design on $\mathcal{A}$, let the cell masses of every interview be estimated from the pilot and smoothed by a mass $\alpha_n \to 0$ with $\alpha_n n_1^{1/2} \to \infty$, and let the remaining records use the design that solves (4) at the estimated masses with a vanishing floor on the interview probabilities. The resulting estimator is exactly unbiased for every population, and n times its variance converges in probability to $W_{\mathcal{A}}(P)$.*

Every cell mass is estimable from the records assigned its interview, and the pointwise constraints keep every fitted design unbiased, so learning the design costs variance only (Appendix E.8).

6. Experiments

We compute the constants of Sections 3–5 exactly for trained models with the solver of Section 5. Every population P is a held-out or target empirical law mixed with a uniform mass of 1% over profiles and labels, which gives full support; every design is pointwise unbiased; and every reported constant is certified by primal and dual bounds that agree to a relative error of at most 10^{-4}. We report W_{fixed}/W_r, the factor by which the best design of a class needs more records than the best branching design. Oracle designs are optimized at P; fitted designs are optimized on separate reference records and evaluated at P. Protocols, sample sizes, per-task values and sensitivity analyses are in Appendix F.

Table 2. Exact records ratio $W_{\text{fixed}}/W_{\text{branch}}$ for independently retrained forest updates of depth b with cap b on WDBC, Spambase and Ionosphere: median [10th, 90th percentile] over binding tasks, medians by number of candidate features m, median ratio realized by designs fitted on training records, and median share of the log-gain captured when branching may use only the forests' own trees; – marks cells not computed.

b	tasks	median [p10, p90]	m=6	m=9	m=12	fitted	own
2	45	1.31 [1.21, 1.44]	1.31	1.25	1.34	1.31	49%
3	180	1.51 [1.33, 1.74]	1.38	1.52	1.64	1.67	17%
4	15	1.24 [1.15, 1.31]	1.24	–	–	1.27	26%

Forest updates across caps. On WDBC, Spambase and Ionosphere (Wolberg et al., 1993; Hopkins et al., 1999; Sigillito et al., 1989), an old and a new random forest of twenty depth-b trees are trained on disjoint records with $m \in \{6, 9, 12\}$ median-binarized features, and the remaining 269, 3,101 and 171 records define P. Table 2 reports the exact ratio for caps 2, 3 and 4. At cap three the best pre-committed design needs a median 1.51 times as many records as the best branching design (maximum 1.85), and the ratio grows from 1.38 at $m = 6$ to 1.64 at $m = 12$. The medians at caps 2 and 4 are 1.31 and 1.24. Designs fitted on training records realize ratios within a median 3% of the exact ones on the same tasks at every cap. Branching restricted to the forests' own trees captures a median of only 17% of the log-gain at cap three: the optimal branching designs mix many trees that neither forest contains.

Screening questionnaires. A depth-b screening tree asks at most b questions per person, so a validation study under the same burden must respect the cap. We train class-balanced CART trees on twelve yes/no items of the CDC Diabetes Health Indicators, derived from the Behavioral Risk Factor Surveillance System (BRFSS; 253,680 respondents, target diabetes or prediabetes) (Centers for Disease Control and Prevention, 2017), and on twelve binarized thyroid items (7,200 patients, abnormal function) (Quinlan, 1986), and validate them on a disjoint random half or on respondents of the other sex. Branching asks the tree's own questions and attains the full-information variance. Table 3 shows that the gain depends on how many questions the screening decision needs. On BRFSS the best pre-committed design needs a median of 1.35–1.58 times as

Table 3. Screening trees: records needed relative to branching, which asks the tree's own questions (medians over targets and seeds). Best, Fitted: optimal fixed design at the target law and on source records. Two: exact optimal two-round design (at $b = 6$ for 1 of six trees; not computed for 21 items, –); H2: following the root question, then the best fixed form. Lower, PS, N_P: the lower end of Theorem 4, the optimized path-set end of Proposition 5 and the completion-only end.

Data, b	Best	Fitted	Two	H2	Lower	PS	N_P
BRFSS, 4	1.49	1.57	1.00	1.14	1.18	1.53	3.6
BRFSS, 5	1.35	1.42	1.00	1.21	1.08	1.36	5.9
BRFSS, 6	1.58	1.68	1.00	1.39	1.13	1.61	11.4
Thyroid, 3–6	1.00	1.04	1.00	–	1.00	1.00	5.0
BRFSS-21, 3	1.00	1.00	1.00	–	1.00	1.00	1.5
BRFSS-21, 4	1.46	1.54	–	–	1.17	1.50	3.8

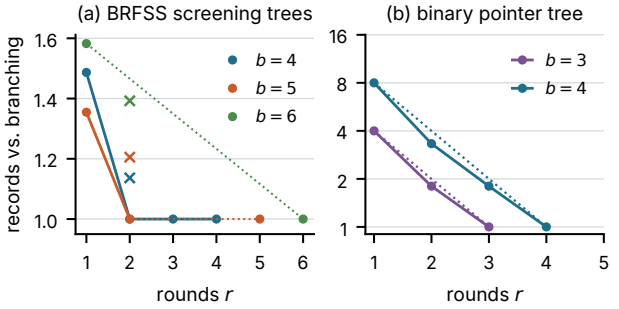


Figure 3. Records needed relative to branching as a function of the number of rounds ($r = 1$: fixed forms; $r = b$: branching). (a) BRFSS screening trees, exact optimal r-round designs (median over targets and seeds); crosses mark the heuristic second round of following the root question; dotted segments join round counts that were not computed. (b) Binary pointer tree under the uniform law, a lower bound on V_r (Appendix D.4); dotted lines are the upper bounds 2^{b-r} of Theorem 3(a).

many respondents at caps 4 to 6. One more round removes most of this cost: at caps 4, 5 and 6 (at $b = 6$ for 1 of the six trees) the exact optimal two-round design needs a median 1.00 times as many, recovering a median 100% of the log-gap and at least 90% in each of the 13 runs with a gain, against 33%–63% for following the root question and then a fixed form. Over the 21 runs with a gain, the optimized path-set design of Proposition 5 is within a median 2.0% (at most 3.9%) of the best fixed design, so the bracket between the lower end and this upper end is narrow and computable from reference data, whereas the completion-only end N_P overstates the loss by a median factor 4.3. On thyroid every tree, up to depth six, depends on at most b items, so $\Gamma_P = 1$ and pre-commitment costs nothing intrinsic; the lower end detects this from reference data, and the rule *reference lower end at most 1.05 ⇒ no material gain* has no misses over all screening runs and forest tasks (Appendix F.1).

All 21 survey variables. With all 21 BRFSS variables the profile cube has 2^{21} elements and the form program has 1,330 forms at $b = 3$ and 5,985 forms at $b = 4$; the solver certifies the optimal fixed design without enumerating the cube, in a median of 6 minutes at $b = 4$ on one core. At $b = 3$ every tree depends on at most b items, so $\Gamma_P = 1$ and there is no gain. At $b = 4$ the best pre-committed design needs a median 1.46 times as many respondents (maximum 1.55), the fitted design 1.54, and the optimized path-set end lies at 1.50 (Table 3, Appendix F.4).

Information versus design. Imputation from complete source records uses information that the capped audit does not collect, and it is only as good as the transport of the conditional law of unasked items. When that law transports, as under the random-half and sex splits, regression imputation and multiple imputation attain RMSE 0.89–0.99 times that of branching with coverage 0.940–0.993, and they are the better tool. Under the education shift at both caps and the poor-health shift at $b = 5$, regression imputation is biased by up to 1.73 branching standard deviations and covers 0.515–0.902 of the time, multiple imputation covers 0.667–0.935, and branching and the fitted fixed form stay unbiased with coverage 0.940–0.960. Under the cost-barrier shift at both caps and the poor-health shift at $b = 4$ imputation stays close to nominal (bias at most 0.23 standard deviations). The design-based scores need neither the reference nor the transport assumption, and that is the setting in which within-record adaptivity matters (Table 10).

Order effects. If the last of b answers flips with probability 0.1 and earlier ones proportionally less often, both designs are biased: on BRFSS the median absolute bias at $n = 1000$ is 1.54 clean branching standard errors for branching and 2.17 for the fitted fixed form, comparable to the sampling gains above, and branching keeps the smaller RMSE in 24 of 24 runs (Appendix F.6).

A real planned-missing archive. PISA 2012 assigned three questionnaire forms at random, and no U.S. student answered all nine predictors of our block-averaged forest ensembles; every block pair appears on a form, so a Brier contrast is identifiable from the actual menu, and optimized form scores, a block-pair Horvitz–Thompson score and multiple imputation agree (Appendix H). The archive demonstrates identification from real fixed forms and has no branching counterfactual.

7. Related Work

Adaptivity in query models. Within-input adaptivity is classical in decision-tree and property-testing complexity, with round hierarchies for testers (Canonne & Gur, 2018), pointer chasing in communication (Nisan & Wigderson, 1993) and round–query tradeoffs for evaluating a pointer tree exactly (Cook et al., 2012; Li et al., 2026); these count

queries or rounds on a worst-case input, whereas we measure the variance that a shortfall of rounds costs under a per-record cap. In the huge-object model, adaptive and non-adaptive testers of long strings are separated by query complexity (Goldreich & Ron, 2022; Chakraborty et al., 2023; Adar & Fischer, 2024), and Horvitz–Thompson weighting over random query paths appears in lower bounds for adaptive relaxed locally decodable codes (Goldberg et al., 2026). Adaptivity gaps in stochastic probing compare policies that evaluate a function on one input at minimum expected cost (Gupta et al., 2016; Hellerstein et al., 2022).

Interaction in information-constrained estimation. Estimation under communication and local-privacy constraints fixes a per-sample channel family (Acharya et al., 2019; Barnes et al., 2020). In local differential privacy, protocols that may return to a user are separated from sequentially interactive ones, and pointer chasing separates numbers of rounds (Joseph et al., 2019; 2020); for learning and testing distributions under communication or privacy constraints, interaction across users does not help (Acharya et al., 2022), and similar questions are studied for adaptive one-bit mean estimation (Lau & Scarlett, 2026). These results concern noisy channels and sample complexity up to constants. Our setting is the noiseless analogue of the within-user versus across-user distinction with exact constants: within-record adaptivity is worth the constants of Theorems 2–4, while adaptation across records leaves the first-order constant unchanged (Theorem 1).

Partial attribute views. Restricted-focus-of-attention learning, learning with partially observed attributes and budgeted learning let a learner see a few attributes per example (Ben-David & Dichterman, 1998; Cesa-Bianchi et al., 2011; Hazan & Koren, 2012; Lizotte et al., 2003; Kapoor & Greiner, 2005). They fit predictors; we estimate population functionals and compare acquisition classes at their optima.

Missing data by design and adaptive testing. Split questionnaires and matrix sampling assign fixed forms (Raghunathan & Grizzle, 1995; Graham et al., 2006; Thomas et al., 2006; Chipperfield & Steel, 2009; Rhemtulla & Little, 2012), optimal split designs choose forms by information criteria for a working model (Adigüzel & Wedel, 2008), and known-coarsening theory gives efficient augmented estimators for a fixed coarsening mechanism (Robins et al., 1994; Tsiatis, 2006); our form program implements that augmentation over all forms of a menu. Adaptive two-phase, multiwave and responsive designs adapt sampling across waves (Groves & Heeringa, 2006; Tao et al., 2020; Chen & Lumley, 2020; 2022), whereas our branching happens within a record under a hard cap. Multistage adaptive testing routes students between blocks, so its stages are rounds in our sense, and its gains for population estimates are assessed under item-response models (Yamamoto et al., 2018; Fink et al., 2024; König et al., 2026); adaptive inventories and dynamic question design choose each respondent's next item by posterior variance or expected information under a model (Montgomery & Rossiter, 2020; Zhang et al., 2020; Fukuda & Inoue, 2026). These lines optimize one protocol under a model; we compare protocol classes at their minimax optima without a response model.

Evaluation and acquisition in ML. Active testing, active statistical inference and prediction-powered inference allocate labels (Kossen et al., 2021; Zrnic & Candès, 2024; Angelopoulos et al., 2023), which reverses our cost structure. Active feature acquisition and cost-sensitive classifiers choose features for individual predictions (Turney, 1995; Trapeznikov & Saligrama, 2013; Nan & Saligrama, 2017; Ma et al., 2019; Covert et al., 2023), and the value of such a policy is estimated from retrospective data by missing-data methods (von Kleist et al., 2025). We study the statistical cost of the acquisition class itself, for population functionals.

8. Discussion and Limitations

Within-record adaptivity has a sharp worst-case value of 2^{b-1}, accrues round by round with a hierarchy that is tight up to polynomial factors for a fixed number of rounds and nearly attained at small caps by the binary pointer tree, and at a population is bracketed by quantities computable from reference data. Exact computations show that ordinary models realize gains between one and 1.98 at caps of two to six, that an optimal second round recovers most of that gain on screening trees, that the bracket computed from reference data is narrow enough to plan with, and that branching designs restricted to the models' own trees miss most of the gain.

The framework covers one known low-degree functional per design, not the many estimands of a multipurpose survey. It assumes that answers do not depend on question order or mode, that requested answers are returned, and that every sampled label is observed. Order effects are the first-order practical obstacle: in the position model of Section 6, a 10% chance that the last answer flips biases branching and form scores by a median 1.54 and 2.17 branching standard errors at $n = 1000$. The identification results concern finite domains and low-degree contrasts; thresholded or link-transformed ensembles often exceed small caps, and their identification floors must be certified first (Appendix G.1). Population-specific designs use the target law, which a fitted design approximates from reference records or a pilot at the costs reported above. When a complete reference whose conditional law of unasked items transports is available, imputation matches or beats every

capped design, and when the cap can be relaxed and labels are expensive, complete records can be cheaper than any capped design (Appendix F.7). The solver's cost grows as $\binom{m}{b}4^b$ per iteration; we ran it for $m \leq 21$ and $b \leq 6$. The exact frontier of $V_{\text{fixed}}/V_{\text{branch}}$ at fixed (m, b), the polynomial factors of the round hierarchy, and the first-order constant of the biased regime remain open.

Impact Statement

Shorter adaptive questionnaires and test sequences can reduce respondent and patient burden. Branching exposes people to different questions; order effects, selective nonresponse and subgroup-specific burdens can bias estimates or hide unequal outcomes, and population-average gains do not imply subgroup gains, so deployments should monitor both.

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A. Guide, Notation and a Worked Example

Guide. Appendix B proves Theorem 1; Appendix C proves Theorem 2 and its extensions to finite samples, larger alphabets, polynomial dimension and mean budgets; Appendix D proves Theorem 3 and reports the exact round constants of the binary pointer tree at small caps; Appendix E proves Theorem 4 and Propositions 5 and 6 and describes the solver and the form program; Appendix F gives every experimental detail; Appendix G treats contrasts of higher degree and other proper scores; Appendix H analyzes a real planned-missing archive; Appendix I describes code and checks. Every experiment reported anywhere in this paper is described in Appendix F.

Notation.

Symbol	Meaning
$d = p_1 - p_0$, $t(x, y) = yd(x)$	Contrast of the two frozen rules and the per-record target; $\Delta(P) = \mathbb{E}_P[t]$
ι, $\ell_\iota(x)$, $\mathcal{A}$	Interview, its transcript, and a finite menu of interviews
$\mathcal{A}_r(b)$	r-round interviews with at most b queries; $\mathcal{A}_1$ fixed forms, $\mathcal{A}_b$ branching trees
q_ι, g_ι	Interview probabilities and transcript score tables of a design
$V(d, \mathcal{A})$, V_r	Worst-case first-order constant (2); its value on $\mathcal{A}_r(b)$
$\Phi_n(d, \mathcal{A})$	Finite-sample program (3), within a factor 104 of the minimax risk
$W_{\mathcal{A}}(P)$, $M_{\mathcal{A}}(P)$	Smallest variance and second moment of a pointwise-unbiased design at P; $M = W + \Delta^2$
Γ_P, $N_P(\tau)$	Energy fraction explained by the best single form; Rényi-$1/2$ effective number of path sets
$M_{\mathrm{PS}}(P)$	Optimum over the padded path sets of a tree computing d (Proposition 5)
$K_{m,b}$	$\min\{2^{b-1}, \binom{m-1}{b-1}\}$
κ, $D = 2^\kappa$	Address bits and data bits of the address contrast; L levels of the binary pointer tree
U	Uniform law on the profile cube
B; n_1	Larger hard cap in the mixed-budget result; pilot size in Proposition 6

The worked example in full. Let $m = 3$, $b = 2$ and $d(x) = [(1 + x_1)x_2 + (1 - x_1)x_3]/2$, and let P have uniform profiles and a label independent of them. The tree asks x_1 and then x_2 or x_3; it computes d, so $V_{\mathrm{branch}} = 1$ and $M_{\mathrm{branch}}(P) = \mathbb{E}[d^2] = 1$. The Fourier expansion is $d = \frac{1}{2}x_2 + \frac{1}{2}x_3 + \frac{1}{2}x_1x_2 - \frac{1}{2}x_1x_3$. Each coefficient carries a quarter of the energy, and every form of size two contains exactly two of the four sets: $\{1, 2\}$ captures x_2 and x_1x_2, $\{1, 3\}$ captures x_3 and x_1x_3, and $\{2, 3\}$ captures x_2 and x_3. Hence $\Gamma_P = 1/2$ and $M_{\mathrm{fixed}}(P) \geq 2$. The two padded path sets $\{1, 2\}$ and $\{1, 3\}$ each carry weight $1/2$, so $N_P = (2 \cdot 2^{-1/2})^2 = 2$, and drawing either form with probability $1/2$ and returning $2Yx_2\mathbf{1}\{x_1 = 1\}$ or $2Yx_3\mathbf{1}\{x_1 = -1\}$ attains second moment 2 at every profile. Thus $V_{\mathrm{fixed}} = M_{\mathrm{fixed}}(P) = 2 = K_{3,2}$, both ends of Theorem 4 coincide, and the optimized path-set design of Proposition 5 equals the optimum. Section 4 generalizes the example to κ address bits.

B. Identification and the Exact Characterization

B.1. General Statistical Guarantees

For a design distribution q on $\mathcal{A}$ and a table g_ι on each transcript alphabet, abbreviate $g_\iota(x) = g_\iota(\ell_\iota(x))$ and $s_g(x) = \sum_\iota g_\iota(x)$. Define

$$B_n(q, g; x) = (d(x) - s_g(x))^2 + \frac{1}{n}\sum_{\iota \in \mathcal{A}} \frac{g_\iota(x)^2}{q_\iota},$$
$$\Phi_n(d, \mathcal{A}) = \inf_{q,g} \max_{x \in \mathcal{X}} B_n(q, g; x). \tag{B.1}$$

A design of mass zero may only have a zero table, in which case its contribution is zero. The first term permits a controlled approximation of the update. The second accounts for observing only one of the tables per respondent.

Theorem B.1 (Finite-sample audit risk)**.** *For every finite $\mathcal{X}, \mathcal{A}$, every contrast d and every $n \geq 1$,*

$$\Phi_n(d, \mathcal{A})/104 \;\leq\; R_{\mathcal{A}}(d, n) \;\leq\; \Phi_n(d, \mathcal{A}).$$

The upper bound uses one iid design and the mean of $Yg_\iota(\ell_\iota(X))/q_\iota$. The lower bound allows the choice of interview to depend on all previous records.

The constant 104 is conservative. This is a convex quadratic-perspective program with a finite second-order cone formulation. Its size depends on enumerated profiles and interviews; Section E.4 instead uses a compact tree representation for a restricted menu.

Theorem B.2 (Exact leading variance)**.** *If $d \in \mathcal{H}_{\mathcal{A}}$, then $\lim_{n\to\infty} nR_{\mathcal{A}}(d,n) = V(d,\mathcal{A})$ for fixed $d, \mathcal{X}, \mathcal{A}$. An iid design and an unbiased score give risk at most $V(d,\mathcal{A})/n$ for every n.*

An optimal iid design already achieves the first-order worst-case coefficient. The lower bound even permits observing the current label before selecting an interview, so early labels do not improve this coefficient either. Within-person branching changes the available observation maps themselves. Population-specific performance can still benefit from learning a design from previous people.

Why the characterization holds. Conditioning on the complete record bounds the score mean's squared bias and variance by (B.1). For the converse, put $\Pi_\iota h = \mathbb{E}_\mu[h \mid \ell_\iota(X)]$ and $J_\mu(h) = \max_\iota \|\Pi_\iota h\|^2_{2,\mu}$. Convex duality gives

$$\Phi_n(d,\mathcal{A}) = \sup_{\mu,h} \frac{\langle d, h\rangle_\mu^2}{\|h\|^2_{2,\mu} + nJ_\mu(h)}. \tag{B.2}$$

A witness perturbs the conditional label mean. Its numerator measures the change in the target, while $J_\mu(h)$ limits information from each interview. A change of measure and the KL chain rule handle sequential choices; a local van Trees argument gives the exact coefficient. Appendix B.3 gives the duality and information proofs. These are established tools for linear-functional estimation; the sharp comparison of query menus is the main result.

B.1.1. An Exact One-Feature Audit

For an additive contrast $d(x) = c_0 + \sum_{j=1}^m c_j x_j$ on the unrestricted sign cube, put $M = \sum_{j=0}^m |c_j| = \|d\|_\infty$.

Proposition B.3. *For every $n \geq 1$ the minimax risk is $M^2/(1+\sqrt{n})^2$ both with full profiles and with at most one feature query.*

Choose j with probability $|c_j|/M$, interpret $x_0 = 1$, and return the sign $Y \operatorname{sgn}(c_j) x_j$. Its mean is Δ/M. Estimating that mean reduces to a binomial problem. A population concentrated at an extremal profile gives the same binomial problem to an evaluator who sees all features (Appendix C.8). Equality concerns worst-case populations. The best length at equal monetary cost can differ for particular populations.

B.2. Identifiability and Irreducible Error

Let L_ι lift a transcript table to $\mathbb{R}^{\mathcal{X}}$, so $(L_\iota u)(x) = u(\ell_\iota(x))$. The space $\mathcal{H}_{\mathcal{A}} = \sum_\iota \operatorname{im}(L_\iota)$ is closed. Put $\delta = \inf_{v\in\mathcal{H}_{\mathcal{A}}} \|d - v\|_\infty$. In fact $\lim_{n\to\infty} R_{\mathcal{A}}(d,n) = \delta^2$. For the upper bound choose an approximant v attaining δ, decompose $v = \sum_\iota L_\iota g_\iota$, and give every used interview positive probability. The mean score is uniformly within δ of Δ, and its variance tends to zero uniformly over P since all alphabets are finite.

If $\delta > 0$, finite-dimensional norm duality gives a signed vector w of ℓ_1 norm one annihilating $\mathcal{H}_{\mathcal{A}}$ with $\sum_x w(x)d(x) = \delta$. This follows by separating d from the open ℓ_∞ ball of radius δ around that subspace, or by duality of the linear program for uniform approximation. In particular $\sum_{x:\ell_\iota(x)=u} w(x) = 0$ for every ι, u. Give X the law $|w|$, and in the two populations set $Y = \pm \operatorname{sgn} w(X)$. For each y, each transcript cell has identical joint probability with $Y = y$ in the two populations. This remains true if y determines which interview is used. Sequentially conditioning on past records gives identical transcript laws for n interviews while the gains are $\pm\delta$. The average of the two squared risks is at least δ^2. The same reasoning gives necessity in the identification criterion.

For a binary coordinate mask T its cells span all characters χ_S with $S \subseteq T$ and no other characters, by the uniqueness of the multilinear expansion. A decision-tree leaf indicator is a product of up to b factors $(1 \pm x_i)/2$. Conversely the fixed query of S recovers χ_S for every $|S| \leq b$. This proves the common identification threshold.

B.3. The General Minimax Characterization

Let $\|\cdot\|_\mu$ denote the norm in $L^2(\mu)$, identifying functions equal on the support of μ. All objects below are finite-dimensional. We can take q strictly positive when computing infima. If a feasible boundary design has $q_\iota = 0$, its table is zero; mixing q

with a positive design and taking the mixture coefficient to zero recovers the same objective. Empty transcript cells can be discarded.

B.3.1. The Dual Form

For a fixed μ define, on the span of the transcript subspaces of $L^2(\mu)$,

$$N_\mu(v) = \inf_{\sum_\iota f_\iota = v} \sum_\iota \|f_\iota\|_\mu, \qquad f_\iota \in \mathrm{im}(L_\iota);$$

extend it by $+\infty$ outside that span. The dual seminorm on the full space is $N_\mu^*(h) = \max_\iota \|\Pi_\iota h\|_\mu$. Indeed $|\langle h, \sum_\iota f_\iota\rangle_\mu| \le \max_\iota \|\Pi_\iota h\|_\mu \sum_\iota \|f_\iota\|_\mu$, and a single normalized projection onto a maximizing subspace attains the bound. The closed unit ball of N_μ is the image of a compact product-space unit ball, so its squared norm is lower semicontinuous.

Maximizing over profiles in (B.1) is equivalent to maximizing its expectation over all μ. The objective is continuous affine in μ on the compact probability simplex and continuous convex in (q, g) for positive q. Sion's minimax theorem exchanges maximum and infimum (Sion, 1958). For fixed tables, Cauchy–Schwarz and $q_\iota \propto \|L_\iota g_\iota\|_\mu$ give

$$\inf_q \sum_\iota \|L_\iota g_\iota\|_\mu^2 / q_\iota = \Big(\sum_\iota \|L_\iota g_\iota\|_\mu\Big)^2$$

with zeros handled by limits. Thus at fixed μ the inner value is $\inf_v\{\|d - v\|_\mu^2 + N_\mu(v)^2/n\}$. By finite-dimensional Fenchel duality this equals

$$\sup_h\{2\langle d, h\rangle_\mu - \|h\|_\mu^2 - nN_\mu^*(h)^2\}.$$

To verify the factors, set $F(v) = \|d - v\|_\mu^2$ and $G(v) = N_\mu(v)^2/n$. Their conjugates satisfy

$$F^*(u) = \langle u, d\rangle_\mu + \|u\|_\mu^2/4, \qquad G^*(u) = nN_\mu^*(u)^2/4.$$

Fenchel duality gives $\inf_v(F + G)(v) = \sup_u[-F^*(-u) - G^*(u)]$; substituting $u = 2h$ gives the displayed expression. There is no duality gap because the first squared norm is finite and continuous everywhere and the second is proper convex. Optimizing over the scale of h proves (B.2).

Relation to fixed-channel constants. For a prescribed iid observation kernel, the χ^2-modulus bound of Polyanskiy & Wu (2026) brackets MSE between the squared modulus divided by $(1 + \sqrt{e})^2$ and the squared modulus, under its stated assumptions. Juditsky & Nemirovski (2009) instead formulate their main comparison using a confidence-level error criterion. Neither modulus is the joint-design objective in (B.1). Our factor 104 is a conservative conversion for that objective under sequential interview choice, not an improvement on fixed-channel constants. The additional argument is a witness whose information is bounded simultaneously for every eligible interview, including the two choices selected by the current label.

B.3.2. Finite-Sample Converse

The upper bound in Theorem B.1 follows from the conditional-score calculation in Section 3. For the lower bound fix μ, h and normalize $\|h\|_\mu^2 + nJ_\mu(h) = 1$. Write $C_\mu = 1 + \mathbb{E}_\mu|h| \le 2$ and set

$$\nu(x) = \mu(x)(1 + |h(x)|)/C_\mu, \qquad u(x) = \mu(x)h(x)/\nu(x)$$

on the support of μ, with $u = 0$ elsewhere. We have $\|u\|_\infty \le 2$, $\nu \ge \mu/2$, and $\mathbb{E}_\nu[du] = \langle d, h\rangle_\mu$. For every partition into transcript cells C,

$$\sum_C \nu(C)\,\mathbb{E}_\nu[u \mid C]^2 = \sum_{C:\nu(C)>0} \frac{\big(\sum_{x\in C}\mu(x)h(x)\big)^2}{\nu(C)} \le 2\|\Pi_\iota h\|_\mu^2.$$

For $\vartheta = 1/5$ define $P_\pm(x, y) = \nu(x)(1 \pm \vartheta y u(x))/2$. Their gains are $\pm\vartheta\langle d, h\rangle_\mu$. Condition on past interviews and on independent random coins. Under the null law with $X \sim \nu$ and Y fair and independent, the probability $p_0(z)$ of a leaf $z = (y, C)$ changes to $p_\pm(z) = p_0(z)(1 \pm \vartheta s_z)$ with $s_z = y\mathbb{E}_\nu[u \mid C]$. If the label selects interview ι_y, the null second moment is

$$\mathbb{E}_0 s_z^2 = \frac{1}{2}\sum_{y\in\{-1,1\}}\sum_{C\in\iota_y} \nu(C)\mathbb{E}_\nu[u \mid C]^2 \le \frac{1}{2}\sum_y 2\|\Pi_{\iota_y} h\|_\mu^2 \le 2J_\mu(h).$$

Also $|s_z| \le 2$. The conditional divergence satisfies

$$D_{\mathrm{KL}}(p_+\|p_-) \le \chi^2(p_+\|p_-) \le \frac{4\vartheta^2}{1-2\vartheta}\mathbb{E}_0 s_z^2 \le \frac{8\vartheta^2}{1-2\vartheta}J_\mu(h).$$

The bound is uniform in the realized past. Independent protocol randomness can be revealed at the start, then discarded by data processing. The KL chain rule bounds the divergence of the two full experiments by $8/15$. Pinsker's inequality bounds their total variation by $2/\sqrt{15}$. Classifying an estimate by the nearer of the two parameter values yields worst-case squared risk at least $\vartheta^2\langle d,h\rangle_\mu^2(1-2/\sqrt{15})/2 \ge \langle d,h\rangle_\mu^2/104$. Taking the supremum in (B.2) proves Theorem B.1. The numerical inequality is equivalent to $15\cdot 27^2 > 104^2$. The losses come from the change of measure, the KL-to-total-variation conversion, and testing two alternatives. The same calculation applies when the current label cannot select the form; this constant is not caused solely by label-dependent selection.

B.3.3. Exact First-Order Constant

Assume $d \in \mathcal{H}_{\mathcal{A}}$. Apply the same minimax exchange to (2), retaining the affine pointwise constraint $s_g = d$. For strictly positive μ the pointwise constraint is equivalent to equality in $L^2(\mu)$, so the fixed-μ optimum is $N_\mu(d)^2$. Restricting the supremum to such μ preserves its value. Indeed mixing any μ with ϵ times a positive law gives an inner objective at least $(1-\epsilon)$ times the original one for every feasible design. By duality,

$$V(d,\mathcal{A}) = \sup_{\mu>0,\, h:\, J_\mu(h)>0} \frac{\langle d,h\rangle_\mu^2}{J_\mu(h)}. \tag{B.3}$$

The zero contrast has value zero. Equation (B.2) and (B.3) also give $n\Phi_n \to V$: the upper bound is V by a zero-residual design, and every fixed witness in (B.3) gives the matching lower limit.

For a lower bound on $nR_{\mathcal{A}}$, fix positive μ, h and set $P_\theta(x,y) = \mu(x)(1+\theta y h(x))/2$, $|\theta| \le \eta < 1/\|h\|_\infty$. Then $\Delta(P_\theta) = \theta c$, with $c = \langle d,h\rangle_\mu$. For each history and deterministic choice of interviews the conditional leaf law has the form $p_\theta(z) = p_0(z)(1+\theta s_z)$. At the null $\mathbb{E}_0 s_z^2 \le J_\mu(h)$ and $|s_z| \le \|h\|_\infty$. Its Fisher information at θ is therefore at most $J_\mu(h)/(1-\eta\|h\|_\infty)$. Conditional likelihood scores are martingale differences, so the full information is bounded by n times this quantity, including for a randomized protocol with its independent seed revealed.

Use a smooth prior π_η on $[-\eta,\eta]$ vanishing at both endpoints with finite information $I(\pi_\eta)$, for example $\eta^{-1}\cos^2(\pi\theta/(2\eta))$. Integration by parts in the joint data–parameter law gives $\mathbb{E}[(\widehat{\Delta}-\theta c)\partial_\theta \log(p_\theta\pi_\eta)] = c$. Cauchy–Schwarz gives the van Trees inequality (Gill & Levit, 1995), here a lower bound of

$$\frac{c^2}{nJ_\mu(h)/(1-\eta\|h\|_\infty) + I(\pi_\eta)}$$

on the prior risk, and hence on worst-case risk. First take $n\to\infty$ after multiplying by n, then $\eta \downarrow 0$, then optimize (B.3). The means of the unbiased scores in (2) give the opposite inequality for every n, proving Theorem B.2.

B.3.4. When the Constant Is Attained in the Unbiased Class

For a fixed positive design law q such that the eligible cells span d, the analogous program $V_q(d)$ fixes q in (2). Its value divided by n is exactly the worst-case minimax MSE among globally unbiased estimators with iid design q. To see the lower bound, choose $X \sim \mu$ of full support and an independent fair label. For a symmetric unbiased statistic U_n of observations $O_i = (\iota_i, \ell_{\iota_i}, Y_i)$ put $H(O_1) = \mathbb{E}[U_n \mid O_1] - \mathbb{E}U_n$. Orthogonality of the one-observation projections gives $\mathrm{Var}\, U_n \ge n\mathbb{E}H^2$. Perturbing the joint law to $\mu(x)(1+tyh(x))/2$ and differentiating unbiasedness gives $\langle d,h\rangle_\mu = n\mathbb{E}[H(O_1)Y_1 h(X_1)]$ for all h. Thus $d = \sum_\iota L_\iota g_\iota$ with $g_\iota(\ell) = nq_\iota\mathbb{E}[YH \mid \iota,\ell]$. Jensen bounds $\mathrm{Var}\, U_n$ below by $n^{-1}\inf_{s_g=d}\sum_\iota \|L_\iota g_\iota\|_\mu^2/q_\iota$. Taking the supremum over full-support μ and exchanging extrema as above gives the claim. Symmetrization costs no risk with iid records.

C. The Sharp Gap and Its Extensions

C.1. The Sharp Universal Gap

We prove Theorem 2 and the program comparison

$$\Phi_n(d,\mathcal{A}_{\mathrm{fixed}}(b)) \le \Phi_{n/K_{m,b}}(d,\mathcal{A}_{\mathrm{branch}}(b)). \tag{C.1}$$

Here the program is defined for any positive real sample-size argument. All scores in the construction below depend only on the queried feature values; the estimator subsequently multiplies them by Y. Theorems B.1 and B.2 already compare these constructions with the larger class of history-dependent protocols, including branching choices based on the current label.

C.1.1. Simulating One Tree without Branching

Let τ be a deterministic tree of depth at most b on $m \geq b$ coordinates, and let $H_\tau(x)$ be a score constant on each of its leaves. A repeated query on a path can be removed because its answer is already known. At every leaf with fewer than b distinct queries, append unused coordinates in increasing index order until the path length is b. Copy the original score to all descendants. For a tree with no queries, start with any coordinate. The padded tree has exactly b distinct queries on every path and the same score at every x.

Let $S_\tau(x)$ be its realized set of coordinates and let $\mathcal{S}_\tau$ be the collection of distinct such sets. Put $L_\tau = |\mathcal{S}_\tau|$. All sets contain the root, giving $L_\tau \leq \binom{m-1}{b-1}$. The first $b-1$ answers fix all queried coordinates, even though the final answer can change the leaf score. Thus $L_\tau \leq 2^{b-1}$, proving $L_\tau \leq K_{m,b}$.

Independently of X, Y, choose S uniformly from $\mathcal{S}_\tau$ and observe X_S. Run the padded tree until a queried coordinate is unavailable or a leaf is reached. Let $M_{\tau,S}(X_S)$ be the indicator of reaching a leaf. This event is measurable from the fixed-form transcript. For each full profile x,

$$M_{\tau,S}(x_S) = 1 \quad \Longleftrightarrow \quad S = S_\tau(x).$$

Indeed equality allows every successive query to be answered; conversely completion means that all b distinct coordinates of the actual path lie in the chosen set of size b. Define the simulated score to be $\widetilde{H}_{\tau,S} = L_\tau H_\tau$ on completion and zero otherwise. Then, pointwise in x,

$$\mathbb{E}_S[\widetilde{H}_{\tau,S}(x_S)] = H_\tau(x), \qquad \mathbb{E}_S[\widetilde{H}_{\tau,S}(x_S)^2] = L_\tau H_\tau(x)^2. \tag{C.2}$$

C.1.2. Mixtures and the Finite-Sample Program

For any feasible branching design (q, g) take $H_\tau = g_\tau/q_\tau$ on its positive-mass trees. Draw the tree with probability q_τ, then draw its path set as above. The resulting fixed-form procedure has conditional mean $s_g(x)$ and conditional second moment at most

$$K_{m,b} \sum_\tau g_\tau(x)^2/q_\tau.$$

The same coordinate set can arise from several tree tags. To represent the construction using only the untagged fixed-mask menu, give S mass $q'_S = \sum_{\tau: S \in \mathcal{S}_\tau} q_\tau/L_\tau$ and replace its score by the weighted average of the tagged scores, with weights $(q_\tau/L_\tau)/q'_S$. The tag is drawn before the person, so these weights do not depend on the population or the answers. Writing $p_{\tau,S} = q_\tau/L_\tau$ and $\overline{H}_S = \sum_\tau p_{\tau,S}\widetilde{H}_{\tau,S}/q'_S$, the reduction satisfies, pointwise,

$$\sum_S q'_S \overline{H}_S = \sum_{\tau,S} p_{\tau,S}\widetilde{H}_{\tau,S}, \qquad \sum_S q'_S \overline{H}_S^2 \leq \sum_{\tau,S} p_{\tau,S}\widetilde{H}^2_{\tau,S}.$$

This is conditional Jensen averaging over the tag given the fixed set. Zero-mass forms are discarded.

For an unbiased branching design this gives $V_{\text{fixed}} \leq K_{m,b} V_{\text{branch}}$ after taking infima. Fixed forms are available as branching trees, so the reverse comparison $V_{\text{fixed}} \geq V_{\text{branch}}$ holds. Moreover, Jensen gives $V_{\text{branch}} \geq \|d\|_\infty^2 > 0$. For a possibly biased design its residual $d - s_g$ is unchanged and its second moment is multiplied by at most $K_{m,b}$. Applying the same comparison to (B.1) gives (C.1). No attainment assumption is needed: apply the construction to designs approaching the infimum.

C.1.3. The Address Function Attains the Bound

Let $\kappa = b - 1$, $D = 2^\kappa$ and $m = \kappa + D$. Identify the κ address bits with an index $A \in \{1, \ldots, D\}$ in a fixed binary order, and let the remaining bits be $Z_1, \ldots, Z_D$. Set $d = Z_A$. This Boolean contrast is realized by the two rules $p_1 = (1+d)/2$, $p_0 = (1-d)/2$. Its polynomial expression is

$$d(A, Z) = \sum_{j=1}^{D} Z_j \mathbf{1}\{A = j\}.$$

Each address indicator has degree κ, so $\deg(d) \le \kappa + 1 = b$ and both menus identify the contrast.

Under the uniform profile law U, fix a mask with s address bits and a set R of r observed data indices. For a realized partial address α, let $C(\alpha)$ be its $2^{\kappa-s}$ consistent indices. Conditional independence and the centered unobserved data bits give

$$\mathbb{E}_{\mathsf{U}}[d \mid X_T] = 2^{-(\kappa-s)} \sum_{j \in R \cap C(\alpha)} Z_j.$$

The cross terms vanish on averaging over the observed independent signs. Each $j \in R$ is consistent with the random partial address with probability 2^{-s}. Consequently

$$\|\mathbb{E}_{\mathsf{U}}[d \mid X_T]\|_2^2 = 2^{-2(\kappa-s)} r 2^{-s} = r 2^{s-2\kappa}.$$

The value is zero if $r = 0$. Otherwise $s \le \kappa + 1 - r$ and $r 2^{1-r} \le 1$ for every integer $r \ge 1$, so it is at most $2^{-\kappa} = 1/D$. Querying all κ address bits and one data bit attains equality. Thus $J_{\mathsf{U}}(d) = 1/D$ for fixed forms, whereas $\langle d, d \rangle_{\mathsf{U}} = 1$. The dual formula (B.3) gives $V_{\mathrm{fixed}} \ge D$. This is a minimax lower bound over all protocols in the model, not only over unbiased score estimators, by Theorem B.2.

For the upper bound choose J uniformly from $\{1, \dots, D\}$ before the interview, and observe all address bits and Z_J. The feature score $H = D Z_J \mathbf{1}\{J = A\}$ satisfies $\mathbb{E}_J[H \mid X] = d(X)$ and $\mathbb{E}_J[H^2 \mid X] = D$ at every profile. Hence $V_{\mathrm{fixed}} = D$. Branching observes the address and then its data bit, giving feature score d with second moment one. Jensen gives the matching lower bound, so $V_{\mathrm{branch}} = 1$. In fact branching has the full-information risk $1/(1 + \sqrt{n})^2$ for every n by the binary-experiment calculation in Appendix C.8. The construction also covers $b = 1$, with no address bits and one data bit. It proves the asserted equality for every b.

Finally, for either menu write $C = D$ or $C = 1$ as appropriate. The uniform witness $h = d$ in (B.2) has denominator $1 + n/C$ and numerator one. Shrinking the preceding unbiased score by $\gamma = n/(n + C)$ gives residual $(1 - \gamma)d$ and constant second moment $\gamma^2 C$. The lower and upper bounds on the program therefore coincide:

$$\Phi_n = \frac{C}{n + C}.$$

The finite-sample minimax risk is bounded using Theorem B.1; identifying its exact ratio for fixed and branching forms is not part of Theorem 2.

C.2. Additional Sharpness Results

Proposition C.1 (Finite-sample address gap)**.** *For the address contrast with $D = 2^{b-1}$, every $n \ge 1$ satisfies*

$$D \left(\frac{\sqrt{n} + 1}{\sqrt{n} + \pi\sqrt{D}} \right)^2 \le \frac{R_{\mathrm{fixed}}(b, n)}{R_{\mathrm{branch}}(b, n)} \le D.$$

In particular, $n \ge 60D$ suffices for a ratio at least $D/2$.

Appendix C.3.1 proves the result by a finite-width van Trees bound and a constant-risk shrinkage estimator.

Corollary C.2 (Finite answer alphabets)**.** *For features with at most $s \ge 2$ answers and nonzero identifiable d, the coefficient ratio is at most $\min\{s^{b-1}, \binom{m-1}{b-1}\}$. Its supremum over dimensions is s^{b-1}, attained at $m = b - 1 + s^{b-1}$.*

The proof uses an s-ary address and the same simulation (Appendix C.3.2); ordinal features therefore need not be dichotomized for the theory.

Consider the four-bit Boolean contrast

$$f(x) = \tfrac{1}{2}(x_1 x_2 + x_1 x_3 + x_4 x_2 - x_4 x_3). \tag{C.3}$$

If $x_2 = x_3$ then $f = x_1 x_2$, and otherwise $f = x_4 x_2$. This is the four-bit sortedness function up to sign and permutation (O'Donnell, 2014, Chapter 1). Indeed, in the order (x_3, x_1, x_2, x_4) it is positive exactly when the sequence changes sign at most once. On k disjoint blocks let $d_k = \prod_{r=1}^{k} f(x^{(r)})$. The difference is realized by $p_1 = (1 + d_k)/2$ and $p_0 = (1 - d_k)/2$. Mixing a common rule into both predictors scales the contrast and its risks. The 4^k Fourier terms of d_k all have degree $2k$ and coefficient magnitude 2^{-k}.

Theorem C.3 (Complete query-budget curve)**.** *For d_k, both risks equal one for $b < 2k$. For $2k \le b \le 4k$ define*

$$C_{\text{fixed}} = 2^{4k-b}, \qquad C_{\text{branch}} = 4^{(3k-b)_+}.$$

In either class the minimax risk is $\Theta(\min\{1, C/n\})$, with absolute constants, and the exact first-order coefficient V is C. At $b \ge 3k$, branching attains the full-profile risk $1/(1+\sqrt{n})^2$ at every n.

At $b = 2k$ both interview types identify the gain and both pay a 4^k sample factor for fixed small error. At $b = 3k$ branching needs only the full-data number of people, while fixed forms still pay 2^k. These comparisons allow a different randomly chosen fixed form for every respondent. The penalty comes from committing to this person's questions before seeing their answers.

Figure 4 in Appendix C.8.1 plots this exact synthetic budget curve.

Where the exponents come from. In each block, two, three and four fixed coordinates capture at most $1/4$, $1/2$ and all of its uniform Fourier mass. The masses multiply across blocks. Randomized signed-term scores attain the reciprocal second moments. A branching path with nonzero conditional mean must query at least two coordinates per block; at budget $2k + r$, at least $k - r$ blocks receive only two and each contributes at most $1/2$ in absolute conditional mean. These bounds yield the matching witnesses and scores (Appendix C.8.1).

Two rounds suffice for these extremal examples. The address construction asks all address bits in one batch and the selected data bit in a second. The block family also attains its entire branching curve in two rounds. Consequently the sharp supremum of V_{fixed}/V_R is already 2^{b-1} for every allowed round count $R \ge 2$ (Appendix D.2). This concerns fixed versus two-round interviews; a general separation between two rounds and unrestricted branching is a different question.

An instance-specific comparison. If a depth-b tree τ computes d, then $V_{\text{branch}} = M^2$ for $M = \|d\|_\infty$. Its number L_τ of padded path sets improves the universal upper bound. Write $E_b = \max_{|T|\le b} \sum_{S\subseteq T} \widehat{d}(S)^2$ using uniform Fourier coefficients. Then

$$\max\left\{1, \frac{\|d\|_{2,\mathsf{U}}^4}{M^2 E_b}\right\} \le \frac{V_{\text{fixed}}}{M^2} \le \min\{L_\tau, K_{m,b}\}.$$

Appendix C.6 proves the interval.

C.3. Finite-Sample and Nonbinary Extensions

C.3.1. When the Address Gap Becomes Visible

Consider the address contrast with D data coordinates, either the binary construction of Theorem 2 or the s-ary construction below. The data coordinates take values in $\{-1, 1\}$. The fixed score $H = DYZ_J\mathbf{1}\{J = A\}$ has $\mathbb{E}H = \Delta$ and $\mathbb{E}H^2 = D$ under every population. With $\gamma = \sqrt{n}/(1+\sqrt{n})$, its shrunken sample mean has risk

$$\frac{\gamma^2 D}{n} + \left((1-\gamma)^2 - \frac{\gamma^2}{n}\right)\Delta^2 = \frac{D}{(1+\sqrt{n})^2}.$$

The zero estimator also has risk at most one. Branching recovers the sign Yd and has exact minimax risk $(1+\sqrt{n})^{-2}$ by Appendix C.8.

For a converse take independent uniform address coordinates and independent fair data bits, with profile law μ. On the subfamily $P_\theta(x, y) = \mu(x)(1 + \theta y d(x))/2$, the gain is θ. For a fixed mask T, put $u_T = \mathbb{E}_\mu[d \mid X_T]$. A transcript (x_T, y) has density proportional to $1 + \theta y u_T(x_T)$, so its Fisher information is

$$I_T(\theta) = \mathbb{E}_\mu\left[\frac{u_T^2}{1-\theta^2 u_T^2}\right] \le \frac{1}{D(1-\eta^2)}, \qquad |\theta| \le \eta < 1.$$

Here $|u_T| \le 1$ and the address projection calculation gives $\mathbb{E}u_T^2 \le 1/D$. The bound is uniform in the chosen mask. Conditional scores are martingale differences, so choosing masks from earlier respondents cannot increase total information beyond $n/[D(1-\eta^2)]$. The same bound holds if the current label selects the mask: for either selected mask the law of u_T is symmetric under flipping all data bits, and $\mathbb{E}[u_T^2/(1 \pm \theta u_T)] = \mathbb{E}[u_T^2/(1-\theta^2 u_T^2)]$.

The prior $\eta^{-1}\cos^2(\pi\theta/(2\eta))$ on $[-\eta,\eta]$ has Fisher information π^2/η^2. The van Trees argument in Appendix B.3 therefore gives

$$R_{\text{fixed}}(b,n) \geq \frac{1}{n/[D(1-\eta^2)] + \pi^2/\eta^2}.$$

Minimizing the denominator over $\eta^2 \in (0,1)$ gives $(\sqrt{n/D}+\pi)^2$. Consequently

$$\frac{D}{(\sqrt{n}+\pi\sqrt{D})^2} \leq R_{\text{fixed}}(b,n) \leq \min\left\{1, \frac{D}{(1+\sqrt{n})^2}\right\}.$$

This proves Proposition C.1. The ratio is at least $D/2$ whenever $n \geq 60D$, since $\sqrt{60}/(\sqrt{60}+\pi) > 1/\sqrt{2}$. This threshold is sufficient, not necessary; the result does not identify the exact finite-sample fixed-form risk.

C.3.2. Features with More Than Two Answers

Let each of the m coordinates have at most $s \geq 2$ possible answers. Pad a depth-b tree to b distinct coordinates as in Appendix C.1. Its first $b-1$ answers determine the entire query set, giving at most s^{b-1} sets. Every set contains the root coordinate, so there are also at most $\binom{m-1}{b-1}$. The same simulation and the same conditional second-moment argument give Corollary C.2 and its finite-program counterpart.

For sharpness put $\kappa = b-1$ and $D = s^\kappa$. Use κ independent uniform s-valued address coordinates and D independent fair data bits; features with fewer than s possible answers are permitted. The contrast is $d = Z_A$. A mask with u address coordinates and r data coordinates has

$$\left\|\mathbb{E}[d \mid X_T]\right\|_2^2 = rs^{u-2\kappa} \leq s^{-\kappa}, \qquad u + r \leq \kappa + 1,$$

because $rs^{1-r} \leq 1$ for integers $r \geq 1$. Thus the variance dual gives $V_{\text{fixed}} \geq D$. Querying the complete address and one independent uniform data coordinate yields a pointwise unbiased score with second moment D, while branching evaluates d with second moment one. The coordinate count is $\kappa + D$. If a common alphabet of exactly s symbols is required, map each data symbol to a sign and use a lower-bound population supported on two symbols of opposite signs. The bounds allow arbitrary population laws.

Both complete menus span sums of functions of at most b coordinates: each adaptive leaf indicator depends on at most b coordinates, and fixed forms span every such function. Thus the identification spaces still coincide. The binary Fourier-degree language is replaced by interaction order in a product basis.

C.3.3. Approaching an Identification Floor

For any d, define $\delta = \inf_{v\in\mathcal{H}_\mathcal{A}} \|d-v\|_\infty$. The indistinguishable pair in Appendix B.2 proves $R_\mathcal{A}(d,n) \geq \delta^2$ for every n. For each $v \in \mathcal{H}_\mathcal{A}$, use a design with conditional score mean $Yv(X)$ and maximal second moment approaching $V(v,\mathcal{A})$. Relative to the original target $Yd(X)$, its mean has absolute bias at most $\|d-v\|_\infty$. Hence

$$\delta^2 \leq R_\mathcal{A}(d,n) \leq \inf_{v\in\mathcal{H}_\mathcal{A}} \left\{\|d-v\|_\infty^2 + \frac{V(v,\mathcal{A})}{n}\right\}. \tag{C.4}$$

A best uniform approximant exists in this finite-dimensional subspace and has finite V. For any such $v_\star$, the excess risk is at most $V(v_\star,\mathcal{A})/n$. On a common binary cap, the two menus have the same δ; applying Theorem 2 to each approximant compares their variance terms by $K_{m,b}$. This is an upper bound on approach to the floor, not a matching excess-risk rate.

For example, on twenty binary features let $d(x) = 0.8x_1 + 0.2\prod_{j=1}^{20} x_j$ and set $b = 5$. Its degree exceeds the cap, but $v = 0.8x_1$ has uniform approximation error 0.2. The uniform signed witness proportional to $\prod_{j=1}^{20} x_j$ annihilates every degree-five function, proving that this error is optimal. Both menus satisfy

$$0.04 \leq R_\mathcal{A}(d,n) \leq 0.04 + 0.64/n.$$

For a general dense logistic update, determining an accurate approximant can itself be difficult. More respondents alone cannot remove a positive identification floor. Additional structural assumptions or a larger per-person cap are required for consistency.

C.4. Products of Addresses and Equality Certificates

C.4.1. Nearly Maximal Gaps in Polynomial Dimension

For $\kappa \geq 0$, an address block consists of κ address bits and 2^κ data bits, and returns the addressed data bit. At $\kappa = 0$ it is just one data bit. Let f_j be address blocks on disjoint coordinates, with address lengths $\kappa_1, \dots, \kappa_k$, and put

$$d = \prod_{j=1}^{k} f_j, \qquad b = \sum_{j=1}^{k} (\kappa_j + 1), \qquad m = \sum_{j=1}^{k} (\kappa_j + 2^{\kappa_j}), \qquad D = 2^{\sum_j \kappa_j}.$$

Proposition C.4 (Product address coefficient)**.** *For this contrast, $V_{\mathrm{branch}}(d, b) = 1$ and $V_{\mathrm{fixed}}(d, b) = D = 2^{b-k}$. Two rounds attain the branching coefficient.*

Proof. Query all address bits first, and the k selected data bits second. This evaluates d within budget b, giving coefficient one by Jensen's inequality. For a fixed design, independently guess one data index in each block, query those k bits and all address bits, and return Dd if every guess matches, zero otherwise. Completion has probability $1/D$ at every profile, so the conditional mean is d and second moment is D.

For the converse let μ be uniform on the full cube and use $h = d$ in the variance dual. An arbitrary fixed form can allocate its queries among blocks in any way. If it observes s_j address bits and r_j data bits in block j, independence gives

$$\|\mathbb{E}_\mu[d \mid X_T]\|_2^2 = \prod_{j=1}^{k} r_j 2^{s_j - 2\kappa_j}.$$

If some $r_j = 0$, this is zero. Otherwise $r_j \leq 2^{r_j - 1}$ and $\sum_j (s_j + r_j) \leq b$, whence

$$\prod_j r_j 2^{s_j - 2\kappa_j} \leq 2^{\sum_j (s_j + r_j) - k - 2\sum_j \kappa_j} \leq 2^{-\sum_j \kappa_j} = D^{-1}.$$

Since $\mathbb{E}_\mu[dh] = 1$, the variance dual gives $V_{\mathrm{fixed}} \geq D$. This treats all fixed forms, including those that redistribute queries across blocks. The general minimax theorem then covers arbitrary biased estimators and history-dependent form selection. □

For any fixed $\kappa \geq 1$, equal blocks give

$$m = \frac{\kappa + 2^\kappa}{\kappa + 1} b, \qquad \frac{V_{\mathrm{fixed}}}{V_{\mathrm{branch}}} = 2^{\kappa b/(\kappa+1)} \quad \text{when } \kappa + 1 \text{ divides } b.$$

In particular $m = 3b/2$ permits $2^{b/2}$. More generally, for any fixed $\zeta > 0$ one can choose a constant κ so that the exponent is at least $(1 - \zeta)b$ while $m = O_\zeta(b)$; the hidden dimension multiplier grows with $1/\zeta$. This does not solve the sharp frontier for a prescribed ratio m/b, such as $4/3$.

For every $b \geq 2$, choose $\kappa = \lfloor \log_2 b \rfloor$, $k_0 = \lfloor b/(\kappa + 1) \rfloor$, and $\rho_0 = b - k_0(\kappa + 1)$. Use k_0 blocks of address length κ and ρ_0 blocks of address length zero. Then $m = k_0(\kappa + 2^\kappa) + \rho_0 \leq b^2$, and

$$\log_2(V_{\mathrm{fixed}}/V_{\mathrm{branch}}) = k_0\kappa = b - k_0 - \rho_0 \geq b - \frac{b}{\kappa + 1} - \kappa.$$

Thus polynomial dimension already permits $2^{b - O(b/\log b)}$. The construction uses the standard product method; the content is the exact coefficient and the lower bound uniform over all query allocations.

C.4.2. When the Full-Information Coefficient Is Attained

Write $M = \|d\|_\infty > 0$ and let $\mathcal{H}_\iota$ be the space of functions observable under interview ι. Define the bounded-decomposition quantity

$$\mathcal{C}_{\mathcal{A}}(d) = \inf \left\{ \sum_\iota \|g_\iota\|_\infty : g_\iota \in \mathcal{H}_\iota,\ \sum_\iota g_\iota = d \right\}.$$

It is computed by a linear program: minimize $\sum_\iota u_\iota$ subject to $\sum_\iota g_\iota = d$ and $-u_\iota \leq g_\iota(c) \leq u_\iota$ on every cell. Its size is that of the enumerated profiles and menu, not polynomial in a general compact representation.

Proposition C.5 (Equality and a bounded-score certificate)**.** *For any identifiable d, $M \le \mathcal{C}_{\mathcal{A}}(d)$ and*

$$M^2 \le V(d, \mathcal{A}) \le \mathcal{C}_{\mathcal{A}}(d)^2.$$

Consequently $\mathcal{C}_{\mathcal{A}}(d) = M$ suffices for $V(d, \mathcal{A}) = M^2$. Equivalently, it suffices that d/M be a convex combination of observable functions with values in $[-1, 1]$. If $|d(x)| = M$ at every profile, then $V(d, \mathcal{A}) = M^2$ if and only if one interview in $\mathcal{A}$ computes d exactly. In particular, a Boolean contrast on the full cube has $V_{\mathrm{fixed}}(d, b) = 1$ if and only if it depends on at most b coordinates.

Proof. The triangle inequality gives $\mathcal{C}_{\mathcal{A}}(d) \ge M$. For a decomposition of total norm C, choose $q_\iota = \|g_\iota\|_\infty / C$, dropping zero terms. Each returned score g_ι/q_ι is bounded by C, giving $V \le C^2$. Jensen gives $V \ge M^2$.

For the last assertion the finite program attains its optimum: a bounded objective bounds every g_ι, the simplex is compact, and the quadratic perspectives are closed at $q_\iota = 0$. At an optimum with value M^2, for each profile,

$$d(x)^2 = \left(\sum_\iota q_\iota H_\iota(x) \right)^2 \le \sum_\iota q_\iota H_\iota(x)^2 \le M^2.$$

If $|d(x)| = M$ everywhere, equality in Jensen forces $H_\iota(x) = d(x)$ for every positive-weight interview and every profile. Any such interview computes d. The converse follows by using that interview alone. A fixed coordinate set computes d precisely when it contains all of its relevant coordinates. □

For general real-valued d, equality in Jensen is forced only on the extremal set $E = \{x : |d(x)| = M\}$. Every used form must distinguish opposite-sign extremal profiles. That condition is necessary but does not by itself control second moments away from E. The bounded-decomposition test is a sufficient certificate; it is not asserted to be necessary for general d. Deterministic classifier differences take values in $\{-1, 0, 1\}$, so they are not covered by the constant-magnitude equivalence when zero contrasts occur.

Few extremal profiles alone do not imply a small gap. Take an even number $k \ge 2$ of equal address blocks of length $\kappa \ge 1$, let f be their product, and put $\ell(x) = m^{-1} \sum_{i=1}^m x_i$. For $0 < \epsilon < 1$ define $d_\epsilon = (1 - \epsilon) f + \epsilon \ell$. The only profile with $|d_\epsilon| = 1$ is the all-positive profile: $\ell = \pm 1$ only at the two constant profiles, and $f = 1$ at both when k is even. A mixture of the two-round exact score for f and a uniformly sampled one-coordinate score for ℓ has magnitude one and conditional mean d_ϵ, so $V_{\mathrm{branch}} = 1$. Under the uniform law, $\mathbb{E}[f\ell] = 0$ because any coordinate lies in only one block and another independent block has zero mean. The witness $h = f$ from Proposition C.4 therefore gives

$$V_{\mathrm{fixed}}(d_\epsilon, b) \ge (1 - \epsilon)^2 2^{k\kappa}.$$

An exponentially large gap is therefore compatible with a non-Boolean contrast having a unique absolute maximum. Empirical near-equality should be established by certificates rather than inferred from extremal-profile counts.

C.4.3. Optimizing a Root Augmentation

With $U = (X_{\mathrm{root}}, Y)$ and $w = q_S^{-1} - 1$, the variance identity in Appendix E.6 implies that, for fixed q, the population-optimal root function is

$$c^*(U) = \frac{\mathbb{E}[wt \mid U]}{\mathbb{E}[w \mid U]},$$

with any value when the denominator is zero. This follows by minimizing the conditional quadratic $\mathbb{E}[w(t - c)^2 \mid U]$. The weighted calibration counterpart uses the same form law as the original comparator and a single pseudo-observation at the calibration grand mean, weighted by the mean calibration weight. It remains globally unbiased. A reduction in its calibration criterion does not guarantee a reduction under the held-out law.

C.5. Expected Budgets and a Local Efficiency Comparison

C.5.1. The Expected-Budget and Hybrid Address Coefficients

A mean feature budget b bounds the conditional expected number of queries on each new respondent, for every past history and every input profile. A fixed form has no input-dependent length, so its distribution satisfies $\sum_T q_T |T| \le b$. In the

hybrid model it also obeys $|T| \le B$ almost surely; branching procedures obey the corresponding expected-length and depth constraints. The label is observed in both models and is charged separately. These conventions impose a uniform design constraint, not a budget that holds only under an unknown favorable population. Write $R^{b,B}_{\mathrm{fixed}}(n)$ and $R^{b,B}_{\mathrm{branch}}(n)$ for the minimax risks with these constraints.

Proposition C.6 (Exact address interpolation)**.** *Let $d(A, Z) = Z_A$ have $\kappa \ge 1$ address bits and $D = 2^\kappa$ data bits, so $m = \kappa + D$. For $\kappa + 1 \le b \le B \le m$, with integer B,*

$$\lim_{n\to\infty} nR^{b,B}_{\mathrm{fixed}}(n) = \frac{DB}{b(B-\kappa)}, \qquad \lim_{n\to\infty} nR^{b,B}_{\mathrm{branch}}(n) = 1.$$

In particular the mean-budget comparison ($B = m$) has exact ratio m/b. For an arbitrary nonzero contrast on m binary features and $0 < b \le m$, the mean-budget fixed-to-branching first-order ratio is at most m/b. If $m \ge 3$ and $\lfloor \log_2 m \rfloor + 1 \le b \le m$, the supremum over contrasts is between $m/(2b)$ and m/b.

Proof. For the upper construction, with probability b/B ask all κ address bits and a uniformly sampled set of $B - \kappa$ data bits; otherwise ask no features. Return $DB/[b(B-\kappa)]$ times Yd when the selected set contains the addressed bit, and zero otherwise. The nonzero-score probability is $b(B-\kappa)/(BD)$ at every profile. Hence the score is pointwise unbiased, its second moment is $DB/[b(B-\kappa)]$, its maximum query count is B, and its expected count is b. Branching evaluates the address in $\kappa + 1$ queries. The usual full-information Bernoulli submodel gives its matching coefficient one.

For the fixed-form lower bound use uniform independent feature bits and the label submodel $P_\theta(x, y) = 2^{-m}(1+\theta y d(x))/2$. A form observing s address bits and r data bits has information at zero $J_T = r2^{s-2\kappa}$. Put $c = s + r$. We claim

$$J_T \le \frac{c(B-\kappa)}{BD} \quad (c \le B). \tag{C.5}$$

The empty form has zero information. If $1 \le c \le \kappa + 1$, the inequality $r2^{1-r} \le 1$ gives $J_T \le 2^{c-1-2\kappa}$; the sequence $2^{c-1}/c$ is nondecreasing, so $J_T/c \le 1/[D(\kappa+1)] \le (B-\kappa)/(BD)$. If $c \ge \kappa+1$, write $j = \kappa - s \ge 0$ and $r_0 = c - \kappa \ge 1$. Then $(r_0 + j)2^{-j} \le r_0$, giving $J_T \le (c-\kappa)/D$. The ratio $(c-\kappa)/c$ is increasing in c, proving (C.5) in this case as well.

Conditionally on any previous records, the randomized form therefore has information at zero at most $b(B-\kappa)/(BD)$. For $|\theta| \le \eta < 1$, its information is at most this quantity divided by $1 - \eta$, since the conditional score is bounded by one. The sequential likelihood score is a martingale sum. Applying the scalar van Trees argument of Appendix B.3, then taking $n \to \infty$ and $\eta \downarrow 0$, proves the converse for arbitrary estimators. The same argument allows a label-dependent form distribution when its conditional expected cost is bounded for each label. Thus the bound does not rely on unbiasedness or an iid allocation of forms.

For general d, query the entire profile with probability b/m and return $(m/b)Yd$ on that event, zero otherwise. Its second moment is at most $(m/b)\|d\|^2_\infty$, whereas even full information has first-order minimax coefficient $\|d\|^2_\infty$. For the stated factor-two lower bound, choose the largest $\kappa \ge 1$ such that $m' = \kappa + 2^\kappa \le m$, and ignore the remaining coordinates. Then $m' > m/2$ and $\kappa + 1 \le b$. If $b \le m'$, the address ratio is m'/b; if $b > m'$, the universal lower bound one is already at least $m/(2b)$. This proves both assertions. Existence of the general first-order coefficients also follows by the same finite-menu program with its compact, linear expected-cost constraints. □

The interpolation does not claim a sharp hybrid supremum over all contrasts. It identifies an exact family and the order of the unrestricted mean-budget supremum in the indicated budget range. For example, $\kappa = 3$ gives $m = 11$: at mean budget $b = 4$, increasing the permitted maximum from $B = 4$ to 11 reduces the exact address coefficient from 8 to $11/4$. Paying for labels does not invalidate either experiment. If one label and each feature cost one and the hard total-answer cap is b_{total}, replace the feature cap by $b_{\mathrm{total}} - 1$. With a label cost c_Y, a padded b-feature protocol costs $c_Y + b$ per person; that common cost is positive and need not be small.

C.5.2. Why Population-Specific Efficiency Also Constrains Biased Estimators

The result below is a finite-state application of local asymptotic efficiency and optimal design, rather than a new general efficiency principle (Robins et al., 1994; Tsiatis, 2006; Hahn et al., 2011; Barnes et al., 2020). It supplies the local interpretation missing from a comparison based only on globally unbiased estimators.

Write $Z = (X, Y)$, $t(Z) = Yd(X)$, and let P assign positive probability to every state in a specified finite space $\mathcal{Z}$. A fixed interview ι reveals $O_\iota = (\ell_\iota(X), Y)$. The menu is finite and the interview for the current respondent is chosen before their answers, although all previous respondents may be used. Put

$$W_{\mathcal{A}}(P) = \inf_{q,H:\,\sum_\iota q_\iota H_\iota(O_\iota(z))=t(z)\ \forall z} \mathrm{Var}_{P,q}(H_A(O_A)).$$

For $c > 0$ define the local risk, with the center P available to the procedure,

$$L_{n,c}(P) = \inf_{\pi,\widehat{\Delta}} \sup_{\substack{Q \text{ a probability law} \\ \|Q-P\|_2 \le c/\sqrt{n}}} n\mathbb{E}_Q[(\widehat{\Delta} - \mathbb{E}_Q t)^2].$$

Proposition C.7 (Local minimax counterpart)**.** *If t is identifiable from the menu, then*

$$\lim_{c\to\infty} \liminf_{n\to\infty} L_{n,c}(P) = \lim_{c\to\infty} \limsup_{n\to\infty} L_{n,c}(P) = W_{\mathcal{A}}(P).$$

The infimum permits biased and nonlinear estimators and history-dependent form choices. If one allowed branching interview computes t, its local constant is $\mathrm{Var}_P(t)$; the local fixed-to-branching ratio is therefore $W_{\mathrm{fixed}}(P)/\mathrm{Var}_P(t)$ when the denominator is nonzero.

Proof. Choose an orthonormal matrix U spanning the zero-sum tangent space of the probability simplex and parameterize $P_\theta = P + U\theta$. The derivative of the target is the vector $v = U^\mathsf{T} t$. If A_ι is the incidence matrix of the observed cells, the information matrix of one observation is

$$I_\iota(P) = U^\mathsf{T} A_\iota^\mathsf{T} \,\mathrm{diag}(A_\iota P)^{-1} A_\iota U.$$

Every nonempty observed cell has positive mass. For a design q, elementary least squares for its score functions gives

$$\inf_{H:\,\sum_\iota q_\iota H_\iota = t} \mathrm{Var}_{P,q}(H) = v^\mathsf{T} \left(\sum_\iota q_\iota I_\iota(P)\right)^\dagger v,$$

with the value $+\infty$ when v is outside the matrix range. Indeed, a centered observation function ϕ must satisfy $\mathbb{E}[\phi\, s] = v$, where s is the observation score vector. Its minimum squared norm is the displayed quadratic form. Because P has full support, these moment constraints and $\mathbb{E}\phi = 0$ are equivalent to $\sum_\iota q_\iota \phi_\iota(O_\iota(z)) = t(z) - \mathbb{E}_P t$ for every z.

For the converse place a smooth product prior on a cube of radius $c/(\sqrt{\dim(\theta)}\sqrt{n})$ in tangent coordinates, vanishing at its boundary. Its information matrix is nC_0/c^2 for a fixed positive-definite C_0. The prior is inside the required Euclidean ball and, for sufficiently large n, inside the probability simplex. Uniformly on it, $I_\iota(P_\theta) = I_\iota(P) + O(c/\sqrt{n})$ in operator norm. Under any sequential protocol the likelihood scores form martingale differences. After averaging over both the prior and histories, their total information divided by n is $\sum_\iota \bar{q}_\iota I_\iota(P) + O(c/\sqrt{n})$ for some $\bar{q}$ in the simplex. Integration by parts and matrix Cauchy–Schwarz (the multivariate van Trees inequality) bound the normalized Bayes risk below by

$$v^\mathsf{T} \left(\sum_\iota \bar{q}_\iota I_\iota(P) + C_0/c^2 + \epsilon_n I\right)^{-1} v, \qquad \epsilon_n \to 0$$

for fixed c, with a nonnegative uniform error bound ϵ_n. This uses no unbiasedness or regularity of the estimator. Minimize over $\bar{q}$, take $n \to \infty$, then $c \to \infty$. Compactness of the design simplex and the variational identity $v^\mathsf{T} M^\dagger v = \sup_u (2v^\mathsf{T} u - u^\mathsf{T} M u)$, with the same infinite-value convention, show that this limit is $\min_q v^\mathsf{T} (\sum_\iota q_\iota I_\iota(P))^\dagger v = W_{\mathcal{A}}(P)$.

For the upper bound choose a minimizing fixed design and its globally unbiased score at the center P. Its iid mean has normalized risk $\mathrm{Var}_{Q,q}(H)$, which converges uniformly to $W_{\mathcal{A}}(P)$ on each shrinking ball. A procedure approaching the infimum gives the same argument if desired. Full observation has constant $\mathrm{Var}_P(t)$ and cannot be improved by coarsening; a branching interview computing t attains it. □

The center-dependent local design is an oracle, not a calibration algorithm. The full-support qualification is essential to this regular argument; this is why every population in Section 6 mixes the empirical law with a uniform mass, whose size is varied in Appendix F.1. The finite-sample globally unbiased bound applies at boundary empirical laws. Neither result says that a biased estimator cannot improve at a single distribution or at a small finite sample size.

C.6. Instance Certificates

C.6.1. Instance-Specific Coefficients

Suppose a depth-b tree τ computes d exactly, let $M = \|d\|_\infty > 0$, and let L_τ be its number of distinct padded path sets. Under the uniform law on the binary cube, put

$$E_b(d) = \max_{|T|\le b} \sum_{S\subseteq T} \widehat{d}(S)^2.$$

Then

$$\max\left\{1, \frac{\|d\|_{2,\mathsf{U}}^4}{M^2 E_b(d)}\right\} \le \frac{V_{\mathrm{fixed}}(d,b)}{V_{\mathrm{branch}}(d,b)} \le \min\{L_\tau, K_{m,b}\}, \qquad V_{\mathrm{branch}}(d,b) = M^2. \tag{C.6}$$

Indeed the tree score d has maximum second moment M^2, while Jensen gives that lower bound for every unbiased representation. The variance dual with $\mu = \mathsf{U}, h = d$ gives the first inequality; Parseval identifies its denominator as $E_b(d)$. Simulating τ gives the L_τ bound, and the universal theorem supplies $K_{m,b}$. This certificate is instance-specific but still concerns worst-case populations. It does not bound the ratio $W(P)/\operatorname{Var}_P(Yd)$ at an individual population by the same instance coefficient ratio: the denominator of that latter ratio is a population variance, not M^2.

The path-set bound need not be attainable for a given tree. On four coordinates with cap three, take the chain querying X_4, stopping on its negative branch, then querying X_3, stopping on its negative branch, and otherwise querying X_2. Padding early leaves with the smallest unused coordinates yields the three sets $\{1,2,4\}$, $\{1,3,4\}$ and $\{2,3,4\}$, so $L_\tau = 3$. Nevertheless every assignment of values to the original leaves depends only on X_2, X_3, X_4. One fixed triple computes it, and the coefficient ratio is one for every nonzero assignment. Thus L_τ is a simulation upper bound and can depend on padding; it is not a sharp invariant of every underlying tree.

For four features, our coefficient solver minimizes v subject to $\sum_\iota M_\iota g_\iota = d$, $\sum_\iota q_\iota = 1$, and $\sum_\iota g_\iota(x)^2/q_\iota \le v$ on all sixteen profiles. Here M_ι lifts a triple-score table. An implementable score supplies an upper bound. A nonnegative probability vector μ and arbitrary vector h supply the lower bound

$$\frac{(h^T d)^2}{\max_\iota \sum_C (\sum_{x\in C} h_x)^2/\mu(C)},$$

where C ranges over form ι's cells. This follows directly from Cauchy–Schwarz applied to every unbiased decomposition. We normalize the nonnegative state dual, add 10^{-12} mass per state, and evaluate this expression; no solver-reported objective is treated as a certificate without reconstruction. The returned primal tables are projected onto the exact observation-span constraint. Both bounds and the numerical equality residual are recorded with the code.

C.7. An Exact Four-Feature Example Beyond the Block Gap

The ratio two in the four-bit block is not extremal at $(m, b) = (4, 3)$.

Proposition C.8. *There is a Boolean contrast on four features, computable by a depth-three tree, with*

$$V_{\mathrm{branch}}(d,3) = 1, \qquad V_{\mathrm{fixed}}(d,3) = \frac{7+2\sqrt{2}}{4} = 2.457106\ldots.$$

Proof. Write $\zeta = \sqrt{2}$ and $\alpha = (7+2\zeta)/4$. Define

$$f_-(x) = \frac{(1-x_2)x_1 - (1+x_2)x_3}{2}, \qquad f_+(x) = -\frac{(1-x_3)x_2 + (1+x_3)x_1}{2},$$

$$d(x) = \frac{(1-x_4)f_-(x) + (1+x_4)f_+(x)}{2}.$$

Query x_4 first. If its value is -1, query x_2 and then x_1 or x_3; if its value is $+1$, query x_3 and then x_2 or x_1. This computes $d \in \{-1, 1\}$ with three queries. Jensen's inequality and the exact tree score give $V_{\mathrm{branch}} = 1$.

For the fixed upper bound use only the forms 124, 134 and 234, with respective probabilities $(2-\zeta)/2$, $(2-\zeta)/2$ and $\zeta - 1$. In a form $T = (i, j, k)$ index its answers by $c = (1+x_i)/2 + 2(1+x_j)/2 + 4(1+x_k)/2$. The following table specifies twice the unbiased decomposition table g_T; the returned score is Yg_T/q_T.

c	$2g_{124}$	$2g_{134}$	$2g_{234}$
0	$\zeta-3$	$4-3\zeta$	$2\zeta-3$
1	$3-\zeta$	$2-\zeta$	$2\zeta-1$
2	$\zeta-1$	$\zeta-2$	$3-2\zeta$
3	$1-\zeta$	$3\zeta-4$	$1-2\zeta$
4	$2-\zeta$	$1-\zeta$	$2\zeta-1$
5	$4-3\zeta$	$\zeta-1$	$1-2\zeta$
6	$3\zeta-4$	$3-\zeta$	$2\zeta-3$
7	$\zeta-2$	$\zeta-3$	$3-2\zeta$

Number full profiles lexicographically by $i = 8(1+x_1)/2 + 4(1+x_2)/2 + 2(1+x_3)/2 + (1+x_4)/2$, and put $E = \{1,2,3,6,8,12,13,15\}$. Substitution gives $\sum_T g_T = d$ on all sixteen profiles. The conditional second moment equals α on E, and $(23-10\zeta)/4$ outside E. The latter is smaller because their difference is $3\zeta - 4 > 0$. Thus $V_{\mathrm{fixed}} \le \alpha$.

For the matching lower bound, let μ be uniform on E. Set $\xi = 3 + 2\zeta$ and define a signed vector η, in the same full-profile order, by

$$32\eta = (-\xi, 3, -5, 5, \xi, -\xi, -3, \xi, 5, \xi, \xi, -\xi, 3, -3, -\xi, -5).$$

For a form T let $I_T = \sum_C \eta(C)^2/\mu(C)$ over its cells, with $0/0 = 0$. Every zero-mass cell in this example also has zero signed mass. Direct summation yields

$$\eta^T d = \alpha, \qquad I_{123} = \alpha - \frac{1}{2}, \qquad I_{124} = I_{134} = I_{234} = \alpha.$$

The Cauchy–Schwarz certificate above therefore gives $V_{\mathrm{fixed}} \ge (\eta^T d)^2 / \max_T I_T = \alpha$. Forms with fewer coordinates are coarsenings of triples and cannot increase I_T. All displayed identities are exact in $\mathbb{Q}(\sqrt{2})$; the supplied symbolic script verifies them without a numerical optimizer. □

Consequently the worst ratio at $(4,3)$ lies in $[(7+2\sqrt{2})/4, 3]$. This excludes two as its value, but neither proves that three is attainable nor determines the exact supremum over all identifiable real contrasts.

Corollary C.9. *For the same contrast and budget three, $V_2(d,3) \ge 4/3 > V_{\mathrm{branch}}(d,3) = 1$.*

Proof. Use the uniform law and witness $h = d$. A deterministic two-round protocol can be represented by a first set S of size one or two, followed by a set of $3 - |S|$ remaining coordinates depending on the first answers. Padding covers protocols using fewer queries; a single fixed triple is included by splitting it into a first pair and a final fixed coordinate. For a given S, its largest possible projection energy is

$$J(S) = 2^{-|S|} \sum_{u \in \{-1,1\}^S} \max_{\substack{T \subseteq [4]\setminus S \\ |T| = 3-|S|}} \mathbb{E}\big[\mathbb{E}[d \mid X_S = u, X_T]^2 \mid X_S = u\big].$$

Substitution of the displayed contrast gives the complete table:

First set S	$J(S)$
$\{1\},\{4\},\{1,2\},\{1,3\},\{1,4\},\{2,3\}$	$1/2$
$\{2\},\{3\},\{2,4\},\{3,4\}$	$3/4$

Thus every allowed two-round observation has projection energy at most $3/4$. The variance dual with $\mathbb{E}[d^2] = 1$ gives $V_2 \ge 4/3$, including mixtures of these protocols. The symbolic script also verifies all ten entries using rational arithmetic. This is a strict constant separation; an exponential two-round-versus-full separation is not established here. □

C.8. The Binomial Value and a Synthetic Budget Curve

The binomial minimax value. For any nonzero d set $M = \|d\|_\infty$. With full profiles a randomized sign with conditional mean $Yd(X)/M$ has population mean Δ/M. For n iid signs W_i with unknown mean μ_0, the estimate $\sqrt{n}\,\overline{W}/(1+\sqrt{n})$ has variance $(1-\mu_0^2)/(1+\sqrt{n})^2$ and squared bias $\mu_0^2/(1+\sqrt{n})^2$. For the lower bound fix a profile at which $|d| = M$ and vary the sign label. A $\mathrm{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ prior on the success probability of the sign makes this constant-risk rule its posterior mean on the mean-sign scale. The Bayes risk equals the displayed frequentist risk, proving minimaxity. The full-profile value is therefore $M^2/(1+\sqrt{n})^2$. The one-feature construction in Proposition B.3 simulates these same signs, and the constant-profile lower bound applies. For $M = 0$ return zero.

C.8.1. Proof of Theorem C.3

If $b < 2k$, under uniform X the laws $Y = \pm d_k(X)$ give the same observations on every allowed leaf. The risk is at least one and zero attains it. For $b \geq 2k$ we show that $J_{\mathsf{U}}(d_k) = 1/C$ and construct a per-person score H with $\mathbb{E}[H \mid X, Y] = Y d_k(X)$ and $\mathbb{E}[H^2 \mid X, Y] = C$ for each design type.

Fixed forms. Under the uniform law, projecting a multilinear expansion onto a mask retains exactly its contained Fourier terms. In a block of size four a mask with $j = 2, 3, 4$ coordinates captures at most $1/4, 1/2, 1$ of the total squared coefficient mass, respectively. The first maximum is achieved by a supported pair and the other two by any mask of the indicated size. A block with fewer than two coordinates captures zero mass. Independence of blocks gives total mass at most 2^{b-4k}, and choosing a supported pair in every two-coordinate block attains it for any budget allocation assigning between two and four coordinates to every block. For two coordinates sample one of four supported pairs uniformly and return twice its signed product. For three, choose $\{1, 2, 3\}$ or $\{2, 3, 4\}$ with equal probability. Return $2f$ when the first form is chosen and $x_2 = x_3$, or when the second is chosen and $x_2 \neq x_3$; return zero otherwise. For four the score is f itself. The scores have conditional means f and constant second moments $4, 2, 1$. Multiply independently randomized block scores and Y to obtain the desired H.

Branching forms. When $b = 2k + r$, $0 \leq r \leq k$, evaluate r blocks via branching, using three queries each; use a random signed pair score of magnitude two in every other block. Then H has constant absolute value 2^{k-r} and square $C = 4^{k-r}$. When $b \geq 3k$, evaluate every block. For the lower witness, at any deterministic leaf under uniform X, an unqueried coordinate remains uniform independent of the others. Nonzero conditional mean for the product requires that every block has at least two exposed coordinates. At most r blocks have more than two, so $|\mathbb{E}_{\mathsf{U}}[d_k \mid \ell]| \leq 2^{-(k-r)}$. Averaging yields the bound on J. It is attained by evaluating r blocks exactly and querying one predetermined supported pair in each other block. The absolute conditional mean is then $2^{-(k-r)}$ at every leaf.

Exact program and leading coefficients. The witness $\mu = \mathsf{U}, h = d_k$ in (B.2) gives $\Phi_n \geq 1/(1 + n/C)$. Randomized query procedures can be represented as mixtures over deterministic interviews by drawing all their random coins in advance. Taking their unbiased transcript scores with shrinkage $\gamma = n/(n + C)$ gives $s_g = \gamma d_k$ and a second-moment contribution at most $\gamma^2 C$ at every profile. Thus $\Phi_n \leq (1-\gamma)^2 + \gamma^2 C/n = C/(n + C)$. The same witness in (B.3) and the unshrunk scores give $V = C$. Theorem B.1 yields the nonasymptotic rates. At $b \geq 3k$, evaluating d_k yields a sign of mean Δ and the binomial minimax value above.

Exact binary comparison for branching. For $2k \leq b \leq 3k$ put $\lambda = 2^{b-3k}$. The branching score λH is a sign with mean $\lambda\Delta$ for every P. Conversely consider the subfamily of uniform X with conditional label law $(1 + \theta y d_k(x))/2$. Then $\Delta = \theta \in [-1, 1]$. At each leaf, conditionally on the history and independent random coins, $p_\theta(z) = p_0(z)(1 + \theta s_z)$ with $|s_z| \leq \lambda$ and $\sum_z p_0(z) s_z = 0$. A sign w of mean $\lambda\theta$ can generate this leaf via the probability kernel $K(z \mid w) = p_0(z)(1 + w s_z/\lambda)$. Simulating the leaves successively reproduces the complete protocol. This is an explicit experiment comparison in the sense of Blackwell (1953): on the indicated subfamily an arbitrary interview can be simulated using the scalar sign. Together with the construction for every P, it gives exact equality of minimax risks with the binary-mean experiment, even for biased estimators.

Why the population supremum matters. A uniform-law calculation can be uninformative even when an update is identifiable. For example, take an extra switch $z \in \{-1, 1\}$ and set $\widetilde{d} = 1 + \mathbf{1}\{z = 1\}(d_k - 1)$. This remains Boolean, has degree $2k + 1$, and has uniform mean $1/2$, so even an empty form captures at least $1/4$ of the squared mean signal in the uniform witness $h = \widetilde{d}$. At the identifiable width $2k + 1$, however, a population supported on $z = 1$ contains the original

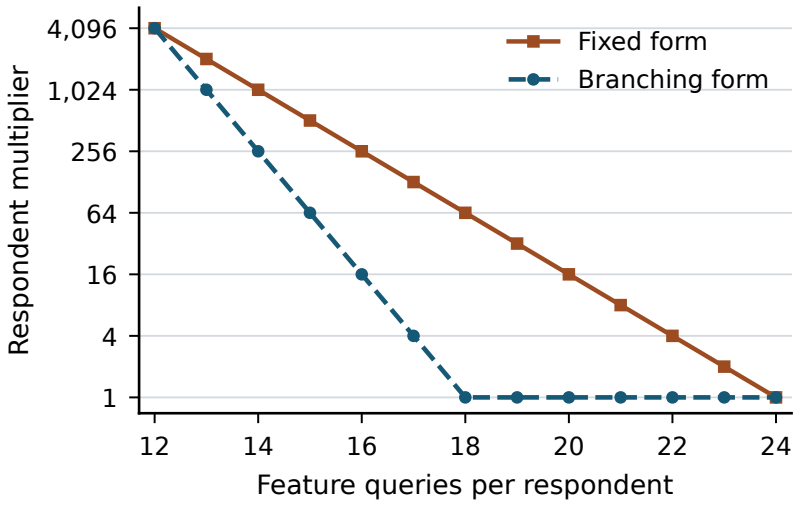


Figure 4. Exact synthetic budget curve for the same update on 24 features ($k = 6$). The fixed coefficient is 64 and the branching coefficient is one at 18 queries.

hard experiment at that width. An auditor given the switch for free still has branching risk of order $\min\{1, 4^{k-1}/n\}$ for $k \geq 1$. Optimizing over profile laws and perturbations in (B.2) accounts for this possibility.

D. Rounds of Adaptivity

D.1. Rounds of Adaptivity

Interviews with r rounds. An r-round interview chooses coordinate batches $B_1,\ B_2(x_{B_1}),\ \ldots,\ B_r(x_{B_1\cup\cdots\cup B_{r-1}})$. Each batch is a function of the answers to earlier batches only, and $\sum_s |B_s| \leq b$ on every input. Let $\mathcal{A}_r(b)$ be the finite menu of deterministic r-round interviews and $V_r(d,b) = V(d, \mathcal{A}_r(b))$; randomized interviews are mixtures and are already represented by the design probabilities in (2). Fixed forms are $\mathcal{A}_1(b)$. Sequentializing each batch shows that every r-round interview is a depth-b tree, and a depth-b tree is a b-round interview with one query per round. Hence $\mathcal{A}_1(b) \subseteq \mathcal{A}_2(b) \subseteq \cdots \subseteq \mathcal{A}_b(b) = \mathcal{A}_{\text{branch}}(b)$ and $V_1 \geq V_2 \geq \cdots \geq V_b = V_{\text{branch}}$. All menus identify the same contrasts, $\deg(d) \leq b$, because the two extreme menus do (Appendix B.2). Theorems B.1 and B.2 apply to each finite menu $\mathcal{A}_r(b)$, so the lower bounds below hold against biased, sequential and label-dependent procedures.

Theorem D.1 (Rounds of adaptivity, full statement of Theorem 3)**.** (a) *For $1 \leq r \leq b \leq m$ and every nonzero contrast of degree at most b on $\{-1,1\}^m$,*

$$V_{\text{branch}}(d,b) \leq V_r(d,b) \leq \min\left\{2^{b-r}, \binom{m-r}{b-r}\right\} V_{\text{branch}}(d,b).$$

(b) *For integers $r, \kappa \geq 1$ put $D = 2^\kappa$ and $b = r\kappa + 1$. The pointer-chasing contrast $N_{r,\kappa}$ defined below satisfies*

$$V_{r+1}(N_{r,\kappa}, b) = V_{\text{branch}}(N_{r,\kappa}, b) = 1, \qquad \frac{D}{b} \leq V_r(N_{r,\kappa}, b) \leq D.$$

Corollary D.2 (Two rounds can be exponentially worse than branching)**.** *For every $b \geq 3$,*

$$\sup_{m,d} \frac{V_2(d,b)}{V_{\text{branch}}(d,b)} \geq \frac{2^{\lfloor (b-1)/2 \rfloor}}{b}, \qquad \sup_{m,d} \frac{V_r(d,b)}{V_{r+1}(d,b)} \geq \frac{2^{\lfloor (b-1)/r \rfloor}}{b} \quad (1 \leq r < b).$$

Proof. Take $\kappa = \lfloor (b-1)/r \rfloor \geq 1$ and $N_{r,\kappa}$, which uses $r\kappa + 1 \leq b$ queries along every path. Branching and $(r+1)$-round interviews still compute it exactly under the larger cap b, while the hit bound in the proof of part (b) uses only the total number b of queries. For the first display use $r = 2$ and $V_{\text{branch}} \leq V_3$. □

The constant separation $V_2 \geq 4/3$ of Corollary C.9 is therefore not the end of the hierarchy. Part (b) and the corollary place the exponent for $2 \leq r < b$ between $\lfloor (b-1)/r \rfloor - \log_2 b$ and $b - r$; the binary pointer tree (Theorem D.3) raises the lower end to $b - 1 - r \log_2(3b)$, and Appendix D.4 computes the constants of that tree exactly at small caps.

Proof of part (a). The left inequality is menu inclusion. For the right inequality pad a depth-b tree τ as in Appendix C.1, so that every path has b distinct coordinates. In rounds $1, \dots, r-1$ ask one question per round and follow τ exactly. This reaches a node v of depth $r-1$, whose padded subtree has depth $b - r + 1$. Its path sets all contain the subtree root and are determined by the first $b - r$ answers below v; hence their number L_v is at most 2^{b-r}, and also at most $\binom{m-r}{b-r}$ because the remaining $b - r$ coordinates are chosen from the $m - r$ coordinates not on the followed prefix or at v. In round r choose one of these L_v sets uniformly, independently of the person, and attempt to complete the path. On completion return L_v times the tree score and otherwise zero. Conditionally on x the mean equals the tree score and the second moment is L_v times its square. The mixture and merging argument of Appendix C.1 applies verbatim to randomized branching designs, so every feasible branching design is simulated with second moment inflated by at most $\max_v L_v$.

The pointer-chasing contrast. Levels $\ell = 1, \dots, r$ consist of registers. Level ℓ has $D^{\ell-1}$ registers $R_\ell(i_1, \dots, i_{\ell-1})$, indexed by $(i_1, \dots, i_{\ell-1}) \in [D]^{\ell-1}$, and each register stores κ binary coordinates encoding an element of $[D]$. There are D^r data coordinates $Z_{i_1 \cdots i_r}$. The pointer path is $j_1 = R_1$ and $j_\ell = R_\ell(j_1, \dots, j_{\ell-1})$, and

$$N_{r,\kappa}(x) = Z_{j_1 \cdots j_r} = \sum_{i \in [D]^r} Z_i \prod_{\ell=1}^{r} \mathbf{1}\{R_\ell(i_{<\ell}) = i_\ell\}.$$

Each register indicator has degree κ, so $\deg N_{r,\kappa} = r\kappa + 1 = b$. Reading the κ coordinates of the level-ℓ register on the pointer path in round ℓ, and the selected data coordinate in round $r+1$, computes the contrast with b queries in $r+1$ rounds. With Jensen's inequality this gives $V_{r+1} = V_{\mathrm{branch}} = 1$. For the upper bound $V_r \leq D$, read the pointer registers of levels $1, \dots, r-1$ in the first $r-1$ rounds. In round r, read the level-r register together with one data coordinate $Z_{j_1 \cdots j_{r-1} J}$, where J is uniform on $[D]$ and drawn in advance. The score $D Z_{j_1 \cdots j_{r-1} J} \mathbf{1}\{J = j_r\}$ is unbiased with second moment D at every profile.

The lower bound $V_r(N_{r,\kappa}, b) \geq D/b$. Let U be the uniform law and write $c^\star = (j_1, \dots, j_r)$ for the selected data cell. Since $\mathbb{E}_{\mathsf{U}}[N^2] = 1$, the dual formula (B.3) with witness $h = N$ gives $V_r \geq 1/\max_{\iota \in \mathcal{A}_r(b)} E_\iota$, where $E_\iota = \mathbb{E}_{\mathsf{U}}[\mathbb{E}_{\mathsf{U}}[N \mid \ell_\iota(X)]^2]$. Fix a deterministic r-round interview ι.

Step 1: a generous transcript. Let ι^+ ask the same questions as ι, but at the end of every round additionally reveal the full content of every register touched in that round. The transcript of ι is a function of that of ι^+, and conditional expectation onto a larger σ-field cannot decrease the second moment, so $E_\iota \leq E_{\iota^+}$.

Step 2: only the selected data coordinate carries signal. Let Obs be the set of data coordinates queried by ι^+. For a data cell c, on the transcript event $\{c \notin \mathrm{Obs}\}$ the transcript and the pointer path are functions of the coordinates other than Z_c, because the interview never reads Z_c. Since Z_c is an independent uniform sign, $\mathbb{E}[Z_c \mathbf{1}\{c^\star = c\} \mid \ell_{\iota^+}]$ vanishes there. Hence

$$\mathbb{E}[N \mid \ell_{\iota^+}] = \sum_{c \in \mathrm{Obs}} Z_c \, \mathbb{P}(c^\star = c \mid \ell_{\iota^+}),$$

so $|\mathbb{E}[N \mid \ell_{\iota^+}]| \leq \mathbb{P}(c^\star \in \mathrm{Obs} \mid \ell_{\iota^+}) \leq 1$ and

$$E_{\iota^+} \leq \mathbb{E}\big[\mathbb{P}(c^\star \in \mathrm{Obs} \mid \ell_{\iota^+})\big] = \mathbb{P}(c^\star \in \mathrm{Obs}).$$

Step 3: observing $c^\star$ in r rounds requires a lucky guess. Let $\mathcal{G}_t$ be the σ-field of the ι^+ transcript before round t, and let K_t be the largest k such that the pointer registers $R_1, R_2(j_1), \dots, R_k(j_{<k})$ are all revealed before round t; thus $K_1 = 0$ and K_t is nondecreasing. A query of round t is *deep* if it touches a register of level at least $K_t + 2$ or a data coordinate while $K_t \leq r - 1$; it is a *hit* if, in addition, the touched register or data coordinate lies on the pointer path. Every deep location is indexed by a tuple whose $(K_t + 1)$st entry is $\mathcal{G}_t$-measurable, and lies on the pointer path only if this entry equals j_{K_t+1}. The latter is the content of the register $R_{K_t+1}(j_{\leq K_t})$, which is unrevealed before round t. For an adaptive sequence of queries to independent uniform coordinates, the unrevealed coordinates remain independent and uniform given the transcript. Hence j_{K_t+1} is uniform on $[D]$ given $\mathcal{G}_t$, and every deep query is a hit with conditional probability at most $1/D$.

If no hit occurs in rounds $1, \dots, t$, then $K_{t+1} \le K_t + 1$. Indeed, $K_{t+1} \ge K_t + 2$ requires the pointer register of level $K_t + 2$ to be revealed by the end of round t, say in round $t' \le t$. Since $K_{t'} \le K_t$, that query was deep in round t' and lay on the pointer path, so it was a hit. By induction, without hits $K_t \le t - 1 \le r - 1$ for every $t \le r$. Then every query of a data coordinate in rounds $1, \dots, r$ is deep, and $c^\star$ can be queried only by a hit. Since the total number of queries is at most b,

$$\mathbb{P}(c^\star \in \mathrm{Obs}) \le \mathbb{P}(\text{some hit}) \le \mathbb{E}\Big[\sum_{t=1}^{r} \sum_{\text{deep queries of round } t} \mathbb{P}(\text{hit} \mid \mathcal{G}_t)\Big] \le \frac{b}{D}.$$

Combining the three steps gives $E_\iota \le b/D$ for every $\iota \in \mathcal{A}_r(b)$, and therefore $V_r \ge D/b$. □

A binary pointer tree: few rounds lose almost the whole exponent. The pointer-chasing family separates consecutive numbers of rounds, but its lower bound $2^{\lfloor (b-1)/r \rfloor}/b$ leaves the exponent for a fixed small number of rounds far below the upper bound $b - r$ of part (a). The following contrast closes most of this gap. Let $L \ge 1$ and $b = L + 1$. Arrange $2^L - 1$ *pointer coordinates* as the internal nodes u of a complete binary tree with levels $0, \dots, L-1$ and attach 2^L *data coordinates* Z_v to its leaves v, so that $m = 2^b - 1$. Encode the pointer coordinate of node u as a bit $x_u \in \{0, 1\}$ that selects one of the two children of u. The *pointer path* starts at the root and descends through the selected children; let $c^\star(x)$ be the leaf it reaches, and

$$B_L(x) = Z_{c^\star(x)} = \sum_v Z_v \prod_{u \in \mathrm{path}(v)} \mathbf{1}\{x_u \to v\},$$

where $\mathrm{path}(v)$ is the set of the L internal nodes above v and $\{x_u \to v\}$ is the event that the bit of u selects the child on the way to v. Each indicator has degree one, so $\deg B_L = L + 1 = b$, and following the path with one question per round computes B_L with b queries. The function is the tree evaluation problem of Cook et al. (2012); a deterministic adversary argument on the same instance shows that exact evaluation with at most B queries per round needs $L/\lceil \log_2(B+1) \rceil$ rounds (Li et al., 2026). Here the total number of queries is capped at b, exact evaluation in fewer than b rounds is impossible for every strategy, and we quantify the variance that the missing rounds cost.

Theorem D.3 (Binary pointer tree)**.** *Let $L \ge 1$, $b = L + 1$ and let B_L be defined on $\{0,1\}^{2^b - 1}$ as above.* (a) *$V_{\mathrm{branch}}(B_L, b) = V_b(B_L, b) = 1$ and $V_r(B_L, b) \le 2^{b-r}$ for $1 \le r \le b$.* (b) *$V_1(B_L, b) \ge 2^{b-1}/b$.* (c) *For $2 \le r \le b$,*

$$V_r(B_L, b) \;\ge\; \frac{2^{b-1}}{r\, b \left(1 + \frac{2b}{r-1}\right)^{r-1}} \;\ge\; \frac{2^{b-1}}{(3b)^r}.$$

Corollary D.4 (Few rounds lose almost the whole exponent)**.** *For every $b \ge 2$,*

$$\frac{2^{b-2}}{b(2b+1)} \;\le\; \sup_{m,d} \frac{V_2(d,b)}{V_{\mathrm{branch}}(d,b)} \;\le\; 2^{b-2}, \qquad \frac{2^{b-1}}{(3b)^r} \;\le\; \sup_{m,d} \frac{V_r(d,b)}{V_{\mathrm{branch}}(d,b)} \;\le\; 2^{b-r} \quad (2 \le r \le b).$$

Hence the exponent of the r-round gap lies between $b - 1 - r\log_2(3b)$ and $b - r$: for every fixed r it equals $b - O(r \log b)$, and an r-round protocol that is within a factor b^c of full branching on every contrast of degree at most b needs $r \ge (b - 1 - c\log_2 b)/\log_2(3b)$ rounds.

Proof. The upper bounds are Theorem D.1(a). The lower bounds are Theorem D.3(c) with $V_{\mathrm{branch}}(B_L, b) = 1$; for $r = 2$ the first expression in (c) equals $2^{b-1}/(2b(1+2b))$. □

The lower bounds are informative only once $2^{b-2} > b(2b+1)$, that is for $b \ge 10$; for smaller b the constant separation $V_2 \ge \frac{4}{3} V_{\mathrm{branch}}$ of Corollary C.9 at $b = 3$ remains the best statement we have. Like the address contrast, B_L uses a number of coordinates exponential in b. Whether the polynomial factors in (c) can be removed, so that the r-round gap is 2^{b-r} up to constants, is open.

Proof of Theorem D.3. (a) Following the pointer path returns B_L itself, so $V_b \le 1$; every unbiased design has $\sum_\iota g_\iota(x) = B_L(x) = \pm 1$, and Cauchy–Schwarz gives $\sum_\iota g_\iota(x)^2/q_\iota \ge 1$ at every x, so $V_{\mathrm{branch}} \ge 1$. The remaining inequality is Theorem D.1(a).

(b) and (c). Let U be the uniform law on all coordinates, under which $\mathbb{E}_{\mathsf{U}}[B_L^2] = 1$ and the pointer path is a uniformly distributed root-to-leaf path. By the dual formula (B.3) with witness $h = B_L$, $V_r(B_L, b) \ge 1/\max_{\iota \in \mathcal{A}_r(b)} E_\iota$ with

$E_\iota = \mathbb{E}_{\mathsf{U}}[\mathbb{E}_{\mathsf{U}}[B_L \mid \ell_\iota(X)]^2]$. Fix a deterministic r-round interview ι. Let S_t be the set of coordinates it reads in round t; S_t is a function of the transcript T_{t-1} of rounds $1, \dots, t-1$, and we may assume that no coordinate is read twice. Write $a_t = |S_t \cap U|$ and $\lambda_t = |S_t \cap V|$ for the numbers of pointer and data coordinates read in round t, so that $\sum_t (a_t + \lambda_t) \le b$, and let I_t be the set of pointer coordinates read in rounds $1, \dots, t$ and Obs the set of leaves whose data coordinate is read at all.

Step 1: only the selected leaf carries signal. On the event $\{c^\star \notin \mathrm{Obs}\}$, the transcript and the pointer path are functions of coordinates other than $Z_{c^\star}$, so $\mathbb{E}_{\mathsf{U}}[Z_v \mathbf{1}\{c^\star = v\} \mid \ell_\iota] = 0$ for every unread leaf v and

$$\mathbb{E}_{\mathsf{U}}[B_L \mid \ell_\iota] = \sum_{v \in \mathrm{Obs}} Z_v \, \mathbb{P}(c^\star = v \mid \ell_\iota), \qquad E_\iota \le \mathbb{E}\big[\mathbb{P}(c^\star \in \mathrm{Obs} \mid \ell_\iota)^2\big] \le \mathbb{P}(c^\star \in \mathrm{Obs}).$$

Step 2: the conditional law of the selected leaf. For a leaf v let $R_t(v) = |\mathrm{path}(v) \cap I_t|$ be the number of pointer coordinates on its path read by the end of round t, and let $C_t(v)$ be the event that every one of them selects the child towards v. The event $\{T_t = \tau\}$ fixes the values of the coordinates read under τ and nothing else, so conditionally on T_t the unread coordinates remain independent and uniform. Since $c^\star = v$ exactly when all L pointer coordinates on $\mathrm{path}(v)$ select the child towards v,

$$\mathbb{P}(c^\star = v \mid T_t) = \mathbf{1}\{C_t(v)\}\, 2^{R_t(v) - L}. \tag{D.1}$$

For $r = 1$ this gives $\mathbb{P}(c^\star \in \mathrm{Obs}) \le \lambda_1 2^{-L} \le b 2^{-L}$, which with Step 1 proves (b).

Step 3: a potential for the most likely leaf. Put $M_t = \max_v \mathbf{1}\{C_t(v)\} 2^{R_t(v)}$, so that $M_0 = 1$ and, by (D.1), $\max_v \mathbb{P}(c^\star = v \mid T_t) = M_t 2^{-L}$. We claim

$$\mathbb{E}[M_t \mid T_{t-1}] \le (1 + 2a_t)\, M_{t-1}. \tag{D.2}$$

Condition on T_{t-1}, which fixes S_t. Let $v^\star$ attain M_t. If no coordinate of S_t lies on $\mathrm{path}(v^\star)$, then $R_t(v^\star) = R_{t-1}(v^\star)$ and $C_t(v^\star) = C_{t-1}(v^\star)$, so $M_t \le M_{t-1}$. Otherwise let u be the deepest node of $\mathrm{path}(v^\star) \cap S_t$, write $\mathrm{path}(u)$ for the nodes above u together with u itself, and define

$$Q(u) = \mathbf{1}\{\text{every node of } (\mathrm{path}(u) \setminus \{u\}) \cap I_t \text{ selects towards } u\}\, 2^{|\mathrm{path}(u) \cap I_t|},$$

$$B(u) = \max_{w \text{ below } u} \mathbf{1}\{\text{every node of } (\mathrm{path}(w) \setminus \mathrm{path}(u)) \cap I_{t-1} \text{ selects towards } w\}\, 2^{|(\mathrm{path}(w) \setminus \mathrm{path}(u)) \cap I_{t-1}|}.$$

No node of $\mathrm{path}(v^\star)$ strictly below u is read in round t, so every read node there belongs to I_{t-1} and $\mathbf{1}\{C_t(v^\star)\} 2^{R_t(v^\star)} \le Q(u)B(u)$: the factor $Q(u)$ accounts for the nodes of $\mathrm{path}(u)$ other than u and for the count including u, the bound $\mathbf{1}\{x_u \to v^\star\} \le 1$ drops u's own selection, and $B(u)$ dominates the contribution of the nodes below u because $w = v^\star$ is admissible in its maximum. Consequently $M_t \le M_{t-1} + \sum_{u \in S_t \cap U} Q(u)B(u)$. The quantity $B(u)$ is T_{t-1}-measurable. Given T_{t-1}, the $|\mathrm{path}(u) \cap S_t| - 1$ newly read nodes strictly above u select towards u independently with probability $1/2$ each, so

$$\mathbb{E}[Q(u) \mid T_{t-1}] = \mathbf{1}\{\text{every node of } \mathrm{path}(u) \cap I_{t-1} \text{ selects towards } u\}\, 2^{|\mathrm{path}(u) \cap I_{t-1}|} \cdot 2.$$

Multiplying by $B(u)$ reassembles, for the leaf w below u attaining $B(u)$, exactly $2 \cdot \mathbf{1}\{C_{t-1}(w)\} 2^{R_{t-1}(w)} \le 2M_{t-1}$, because $u \notin I_{t-1}$. Summing over the a_t nodes of $S_t \cap U$ proves (D.2).

Step 4: reading the selected leaf. Since S_t is T_{t-1}-measurable, (D.1) gives $\mathbb{P}(c^\star \in S_t \cap V \mid T_{t-1}) \le \lambda_t M_{t-1} 2^{-L}$.

Step 5: combination. Let $\Psi_t = M_t \prod_{s \le t} (1 + 2a_s)^{-1}$. As a_t is T_{t-1}-measurable, (D.2) says $\mathbb{E}[\Psi_t \mid T_{t-1}] \le \Psi_{t-1}$, hence $\mathbb{E}[\Psi_t] \le \Psi_0 = 1$. For $2 \le t \le r$ the pathwise constraint $\sum_{s<t} a_s \le b$ and the inequality between arithmetic and geometric means give $\lambda_t \prod_{s<t}(1 + 2a_s) \le b\,(1 + 2b/(t-1))^{t-1} \le b\,(1 + 2b/(r-1))^{r-1}$, because $k \mapsto (1 + 2b/k)^k$ is increasing; the empty product at $t = 1$ is 1. Therefore

$$\mathbb{P}(c^\star \in \mathrm{Obs}) = \sum_{t=1}^{r} \mathbb{E}\big[\mathbb{P}(c^\star \in S_t \cap V \mid T_{t-1})\big] \le 2^{-L} \sum_{t=1}^{r} \mathbb{E}\Big[\lambda_t \prod_{s<t} (1 + 2a_s)\, \Psi_{t-1}\Big] \le 2^{-L}\, r\, b \left(1 + \frac{2b}{r-1}\right)^{r-1}.$$

With Step 1 and the dual formula this proves the first inequality in (c). For the second, $1 + 2b/(r-1) \le 3b/(r-1)$ because $r - 1 \le b$, so the denominator is at most $rb(3b)^{r-1}/(r-1)^{r-1} \le (3b)^r$, since $r \le 3(r-1)^{r-1}$ for $r \ge 2$. □

The argument is a potential-function version of the lucky-hit bound: a round can raise the conditional probability of the most likely leaf only by a factor linear in the number of pointer coordinates it reads, whereas following the path raises it by a

factor two per question. As a numerical check, the released code simulates the recursion: on trees of depth 3 and 4 with random previously read sets and random new batches, the largest observed ratio of $\mathbb{E}[M_t \mid T_{t-1}]$ to $(1+2a_t)M_{t-1}$ was 0.73; for greedy two- and three-round strategies with total budget b on trees of depth 6, 8 and 10, the sample mean of Ψ_r was between 0.32 and 0.57, and the hit probabilities (0.25, 0.09 and 0.04 for two rounds) lay far below the bound of Step 5, as they must.

Interpretation. Each round can advance the known part of the pointer path by at most one level unless a question lands on an unknown pointer target, which happens with probability at most $1/D$ per question. Additional questions within a round therefore do not substitute for a round. Practical counterparts of rounds are questionnaire pages, clinic visits and batches of laboratory orders. The address contrast of Theorem 2 is th

D.2. Two Rounds Already Attain the Universal Extremum

An R-round interview uses at most R batches, choosing each batch of coordinates using only the preceding batches' answers, with at most b coordinates in total. Let V_R use all such interviews. One round is the fixed-form menu and b rounds suffice for full branching. For any $R \geq 2$,

$$1 \leq V_{\text{fixed}}/V_R \leq K_{m,b}, \qquad \sup_{m,d\neq 0} \frac{V_{\text{fixed}}(d,b)}{V_R(d,b)} = 2^{b-1},$$

where the supremum is over identifiable binary contrasts. The upper bound follows from menu inclusion. The address construction attains equality using two batches: all address bits, then the addressed data bit. Thus the universal extremum measures the value of permitting a second batch; it does not require b sequential rounds.

The block construction also needs at most two rounds throughout its identifiable budget curve. At $b = 2k + R$, $0 \leq R \leq k$, use the paper's three-query strategy on R blocks and two-query strategy on the others. Ask the first two queries in every block together; then ask the selected third coordinate in the R adaptive blocks. At $b \geq 3k$ this already evaluates the contrast exactly. The address and block examples do not separate two rounds from unrestricted branching. Corollary C.9 supplies such a constant separation with a different four-feature contrast; a general exponential hierarchy remains unsettled here.

D.3. Labels and Across-Person Adaptation

Let R_{pre} require selecting the interview independently of the current label, and let R_{label} allow observing that label first, while both permit adaptation across people. For an identifiable contrast, Theorem B.2 and the iid upper construction imply

$$\lim_{n\to\infty} nR_{\text{pre}}(d,n) = \lim_{n\to\infty} nR_{\text{label}}(d,n) = V(d,\mathcal{A}).$$

Thus neither access to earlier people nor early access to the current label reduces the first-order worst-case coefficient relative to an optimal iid design.

There is also an exact statement for the *finite program*. If one extends Φ_n to separate form laws and score tables for $y = -1, +1$, its objective is the maximum over these labels of their individual worst-profile quadratic objectives. Multiplying the negative-label tables by -1 turns both subproblems into the definition of $\Phi_n(d,\mathcal{A})$. Each has value at least Φ_n, and using the same form law and sign-adjusted tables attains it. Early labels therefore do not improve this program. This is not an equality claim for the exact finite-n minimax risks, which the program bounds only within a constant.

D.4. Exact Round Constants of the Binary Pointer Tree at Small Caps

Theorem 3(c) is informative only for $b \geq 10$. For small caps we compute the round constants of B_{b-1} directly. Under the uniform law U with a label independent of the profile, $M_r(\mathsf{U})$, the optimal second moment over r-round designs, is a lower bound on V_r, because $V_r = \sup_P M_r(P)$ by Sion's theorem applied to (2). For $b \leq 4$ the cube is small enough to evaluate the uniform-optimal design at every profile; its worst profile is an upper bound on V_r. The solver of Appendix E.3 computes $M_r(\mathsf{U})$ with certified bounds that agree to 10^{-7} or better. at $b \in \{3,4\}$ fixed forms lose exactly 4 and 8, two rounds lose 1.80 and 3.32–3.33 against the upper bounds 2 and 4 of part (a), and $b-1$ rounds lose 1.80 and 1.80 (Appendix D.4).APP

Table 4. Round constants of the binary pointer tree B_{b-1} ($m = 2^b - 1$). Entries are $M_r(\mathsf{U})$, a lower bound on V_r; for $b \le 4$ the brackets give the worst profile of the uniform-optimal design, an upper bound on V_r. The last column is the upper bound 2^{b-2} of Theorem 3(a) for $r = 2$.

b	m	fixed	$r = 2$	$r = 3$	$r = 4$	2^{b-2}
3	7	4.000	1.800			2
4	15	8.000	3.324 [3.334]	1.800		4

E. Population Constants and Their Computation

e case $r = 1$, where two rounds already attain the branching coefficient. The pointer-chasing family shows that this is not a general property of two-stage designs.

E.1. The Value of Adaptivity at a Given Population

For a menu $\mathcal{A}$ and a law P on $\mathcal{X} \times \{-1, 1\}$ write $M_{\mathcal{A}}(P) = W_{\mathcal{A}}(P) + \Delta(P)^2$, the smallest second moment of a score that is unbiased for $t(x, y) = yd(x)$ at every profile and label. At a full-support law, $W_{\mathcal{A}}(P)$ is also the local asymptotic minimax risk over arbitrary estimators (Proposition C.7). For a deterministic depth-b tree τ that computes d, let $S_\tau(x)$ be its padded path set and $\mathcal{S}_\tau$ the collection of these sets (Appendix C.1).

Theorem E.1 (Population-specific bracket, full statement of Theorem 4)**.** (a) *For every menu $\mathcal{A}$, law P and function h on $\mathcal{X} \times \{-1, 1\}$ with $\mathbb{E}_P[th] \neq 0$,*

$$M_{\mathcal{A}}(P) \;\ge\; \frac{\mathbb{E}_P[th]^2}{\max_{\iota \in \mathcal{A}} \mathbb{E}_P\big[\mathbb{E}_P[h \mid \ell_\iota(X), Y]^2\big]}.$$

In particular $M_{\mathcal{A}}(P) \ge \mathbb{E}_P[t^2]/\Gamma_{\mathcal{A}}(P)$, where $\Gamma_{\mathcal{A}}(P) = \max_\iota \mathbb{E}_P[\mathbb{E}_P[t \mid \ell_\iota(X), Y]^2]/\mathbb{E}_P[t^2]$ is the largest fraction of the target's energy explained by a single interview.

(b) *If a depth-b tree τ computes d, then $M_{\mathrm{branch}}(P) = \mathbb{E}_P[t^2]$ and*

$$\frac{1}{\Gamma_{\mathrm{fixed}}(P)} \;\le\; \frac{M_{\mathrm{fixed}}(P)}{M_{\mathrm{branch}}(P)} \;\le\; N_P(\tau) := \frac{\Big(\sum_{S \in \mathcal{S}_\tau} \mathbb{E}_P\big[t^2 \mathbf{1}\{S_\tau(X) = S\}\big]^{1/2}\Big)^2}{\mathbb{E}_P[t^2]} \;\le\; |\mathcal{S}_\tau| \le K_{m,b}.$$

(c) *Both inequalities in (b) are equalities, with value D, for the address contrast with $\kappa = b - 1$ under the uniform law with independent uniform labels.*

The quantity $N_P(\tau) = \exp H_{1/2}(w)$ is the exponential of the Rényi entropy of order $1/2$ of the weights $w_S = \mathbb{E}_P[t^2 1\{S_\tau = S\}]/\mathbb{E}_P[t^2]$: it is an effective number of question sets that the tree actually uses at P. Adaptivity can be valuable only when the tree's questions diverge across many well-populated paths. It is valuable when, in addition, no fixed set of b questions explains most of the target, so that $\Gamma_{\mathrm{fixed}}(P)$ is small; correlated features and shared coordinates raise Γ_{fixed}. The variance ratio is recovered as $\rho_P = (M_{\mathrm{fixed}} - \Delta^2)/(\mathbb{E}_P t^2 - \Delta^2)$.

Proof. (a) Let H be any feasible score with interview probabilities q_ι and transcript tables g_ι, so $\sum_\iota g_\iota(\ell_\iota(x), y) = t(x, y)$ everywhere and $\mathbb{E}[H^2] = \sum_\iota \|g_\iota\|_P^2/q_\iota \ge (\sum_\iota \|g_\iota\|_P)^2$ by Cauchy–Schwarz in q. For every h,

$$\mathbb{E}_P[th] = \sum_\iota \mathbb{E}_P\big[g_\iota\, \mathbb{E}_P[h \mid \ell_\iota(X), Y]\big] \le \Big(\sum_\iota \|g_\iota\|_P\Big) \max_\iota \big\|\mathbb{E}_P[h \mid \ell_\iota(X), Y]\big\|_P,$$

because g_ι is a function of $(\ell_\iota(X), Y)$. Squaring and taking the infimum over feasible scores proves the display; $h = t$ gives the second statement.

(b) Following τ returns t itself, and Jensen's inequality gives $\mathbb{E}[H^2] \ge \mathbb{E}[\mathbb{E}[H \mid X, Y]^2] = \mathbb{E}_P[t^2]$ for every unbiased score. The lower bound is (a) with $\mathcal{A}$ the fixed forms. For the upper bound choose $q_S > 0$ on $\mathcal{S}_\tau$, draw S before the person answers, and return $t(X, Y)/q_S$ if the tree's path completes inside S and zero otherwise. Completion occurs exactly when $S = S_\tau(X)$ (Appendix C.1), so the score is unbiased at every profile and label, and its second moment is $\sum_S q_S^{-1} \mathbb{E}_P[t^2 \mathbf{1}\{S_\tau(X) = S\}]$. Minimizing over the simplex gives $q_S \propto \mathbb{E}_P[t^2 1\{S_\tau = S\}]^{1/2}$ and the value $N_P(\tau)\mathbb{E}_P[t^2]$. Sets of zero weight are handled by letting their probability tend to zero. The last inequalities are the power-mean inequality and $|\mathcal{S}_\tau| \le K_{m,b}$.

(c) Under the uniform law the weights are equal over the D path sets, so $N_P = D$. With independent labels, $\mathbb{E}[t \mid X_T, Y] = Y\mathbb{E}[d \mid X_T]$, whose energy is at most $2^{-\kappa} = 1/D$ by (6). Hence $1/\Gamma_{\text{fixed}} \geq D$, and the universal bound $M_{\text{fixed}} \leq D$ forces equality throughout. □

The bracket on the variance scale. Section 6 reports the number of records a design needs relative to branching, that is the ratio of variances $W_{\mathcal{A}}(P) = M_{\mathcal{A}}(P) - \Delta(P)^2$ of pointwise-unbiased designs rather than the ratio of second moments. The two scales differ when $\Delta(P) \neq 0$, and the bracket transfers as follows.

Corollary E.2 (Bracket on the variance scale)**.** *Under the assumptions of Theorem E.1(b), with* $\sigma_P^2 = \mathbb{E}_P[t^2] - \Delta(P)^2 = W_{\text{branch}}(P) > 0$,

$$\frac{\mathbb{E}_P[t^2]/\Gamma_{\text{fixed}}(P) - \Delta(P)^2}{\sigma_P^2} \;\leq\; \frac{W_{\text{fixed}}(P)}{W_{\text{branch}}(P)} \;\leq\; \frac{N_P(\tau)\,\mathbb{E}_P[t^2] - \Delta(P)^2}{\sigma_P^2}.$$

When the second-moment endpoints exceed one, the variance-scale endpoints are larger than them, so the variance ratio is bracketed by a wider interval than the second-moment ratio.

Proof. Every design in (4) is unbiased for $\Delta(P)$, so its variance is its second moment minus $\Delta(P)^2$; in particular $W_{\text{fixed}} = M_{\text{fixed}} - \Delta^2$ and $W_{\text{branch}} = \mathbb{E}_P[t^2] - \Delta^2$. Subtract Δ^2 from the three terms of Theorem E.1(b) multiplied by $\mathbb{E}_P[t^2]$ and divide by σ_P^2. The last statement is the inequality $(a-c)/(b-c) \geq a/b$ for $a \geq b > c \geq 0$. □

The tables of Section 6 and Appendix F.3 report this variance-scale bracket, and "inside the bracket" refers to it.

Contrasts that branching cannot compute. For an ensemble whose contrast is not computed by one depth-b tree, part (a) applies to both menus. Every depth-b branching interview is a coarsening of a complete depth-b tree, and the energy of a tree is a sum over its leaves. Hence $\max_\iota \mathbb{E}_P[\mathbb{E}_P[h \mid \ell_\iota, Y]^2]$ over all depth-b trees is computed exactly by dynamic programming over partial assignments, in time $O(m^b 2^b)$. This yields certified lower bounds on $M_{\text{branch}}(P)$ for every dual witness h. Appendix F.2 uses these bounds inside column generation to compute the branching coefficient over all depth-three trees.

E.2. A Computable Upper End: the Optimized Path-Set Design

The upper end $N_P(\tau)$ of Theorem 4 is the second moment of completion-only path guessing, which discards every answer on a form except a completed path. Optimizing the full score program on the same small menu tightens it without enumerating the $\binom{m}{b}$ forms.

Proposition E.3 (Optimized path-set design)**.** *Let a depth-b tree τ compute d, let $S_1, \ldots, S_L$ be its distinct padded path sets ($L \leq K_{m,b}$), and let $M_{\text{PS}}(P)$ be the optimum of the population program (4) over the fixed-form menu $\{S_1, \ldots, S_L\}$, with all pointwise-unbiased tables. Then*

$$M_{\text{fixed}}(P) \;\leq\; M_{\text{PS}}(P) \;\leq\; N_P(\tau)\, M_{\text{branch}}(P),$$

and the same chain holds on the variance scale after subtracting $\Delta(P)^2$ and dividing by σ_P^2.

Proof. The menu $\{S_1, \ldots, S_L\}$ is a set of fixed forms, so its optimum is at least the optimum over all forms. The menu also identifies d: every leaf indicator of τ depends only on that leaf's path coordinates, which lie in its padded set, so d is a sum of tables on the S_j. Completion-only path guessing with $q_j \propto w_j^{1/2}$ is a feasible design in this program, and its second moment is $(\sum_j w_j^{1/2})^2 \mathbb{E}_P[t^2] = N_P(\tau)\mathbb{E}_P[t^2]$ by (E.2); Theorem 4 gives $M_{\text{branch}}(P) = \mathbb{E}_P[t^2]$. Every design in the program is pointwise unbiased, so all three quantities exceed their variances by the same $\Delta(P)^2$. □

The program has at most $2L2^b$ score variables and is solved from reference marginals; Appendix F.3 compares M_{PS} with the all-forms optimum on every screening run. The two coincide whenever an optimal fixed design uses only path sets of τ; in general the gap measures the value of forms that no path of the tree reads.

E.3. Computing Optimal Designs for Every Protocol Class

All population constants in Section 6 are computed by one column-generation solver whose cost depends on $\binom{m}{b}2^b$, not on the 2^m profiles.

Cells and masses. Pad every transcript of an interview to exactly b coordinates. An interview of $\mathcal{A}_r$ is then a partition of the cube into cells $C(T,u) = \{x : x_T = u\}$ with $|T| = b$; for a fixed form all cells share T, for an r-round interview T varies across cells. Padding refines a transcript and therefore never increases the optimal second moment. For a law P that mixes records with a uniform mass α over profiles and labels, the masses $P(T,u,z)$ of all $\binom{m}{b}2^{b+1}$ cells are computed once from the records.

Master problem. For a list of interviews, minimize $\sum_{\iota,c,z} P(c,z)g_{\iota,z}(c)^2/q_\iota$ subject to $\sum_\iota q_\iota = 1$ and the Fourier identities $\sum_{\iota,c} \widehat{\mathbf{1}}_c(S)g_{\iota,z}(c) = (2z-1)\widehat{d}(S)$ for all $|S| \le b$, where $\widehat{\mathbf{1}}_{C(T,u)}(S) = 2^{-b}\chi_S(u)$ for $S \subseteq T$ and zero otherwise. Because cell indicators and d have degree at most b, the identities are equivalent to pointwise unbiasedness at every profile. The problem is a second-order cone program; we solve it with Clarabel through CVXPY at tolerance 10^{-10}. Inside column generation the solver's solution steers the iteration; the final design is repaired to exact feasibility by the least-norm correction weighted by the interview probabilities, which adds the least second moment, so every reported upper bound is the second moment of an exactly pointwise-unbiased design. Looser solver tolerances produced visibly suboptimal masters in our tests, which the consistency checks of Appendix F (the fixed optimum may not exceed the path-set or fitted designs) detect.

Pricing. Let $\eta_z(S)$ be the multipliers of the identities. The information of a cell is

$$e(T,u) = \sum_z \frac{\psi_z(T,u)^2}{P(T,u,z)}, \qquad \psi_z(T,u) = 2^{-b}\sum_{S\subseteq T}\eta_z(S)\chi_S(u),$$

a Walsh transform of η restricted to T; all cells cost $O(\binom{m}{b}b2^b)$ operations. The information of an interview is the sum over its cells. The most informative interview in $\mathcal{A}_r$ is found by dynamic programming over partial assignments (A,a), $|A| < b$:

$$V_1(A,a) = \max_{T\supseteq A,|T|=b}\ \sum_{u:\,u_A=a} e(T,u), \qquad V_\rho(A,a) = \max_{B\supsetneq A,\,|B|\le b}\ \sum_{a':\,a'_A=a} V_{\rho-1}(B,a'),$$

with $V_\rho(T,u) = e(T,u)$ for $|T| = b$; the best r-round interview has information $V_r(\emptyset,\emptyset)$, the best tree is the special case of singleton batches, and the best form is $\max_T \sum_u e(T,u)$. The program visits $O(\sum_{k\le b}\binom{m}{k}4^k)$ entries, and an optimal interview is reconstructed by following the maximizers.

Certificates and stabilization. Normalize η so that $\sum_{z,S}\eta_z(S)(2z-1)\widehat{d}(S) = 1$. By the last display of Theorem 4, $1/\max_{\iota\in\mathcal{A}_r} e(\iota)$ is a lower bound on $M_{\mathcal{A}_r}(P)$ for every such η, and the master value is an upper bound. Every pointwise-unbiased design also satisfies $M_{\mathcal{A}}(P) \ge \mathbb{E}_P[t^2]$ by Jensen's inequality, which certifies designs that attain full information. Each iteration adds up to six (twelve for the forests and the 21-item study) interviews whose information exceeds the master's maximum, and the solver stops when no interview of the class improves the master or when the bounds agree to a relative error of 10^{-7} (10^{-4} for the forest tasks at $b \ge 4$). Master duals of column generation are degenerate, and unstabilized pricing can stall for many iterations; we price at a convex combination, with equal weights, of the current dual and the dual with the best dual bound so far (Wentges, 1997), and fall back to the current dual when the smoothed point finds no improving interview. Removing unused columns from the master made the iteration cycle in our tests, so the master keeps every generated interview. The master starts from the support of the optimal fixed design (and, for the forests, the padded component trees), so it stays small.

Validation. On stored tasks the solver reproduces the fixed constants, Γ_P and the certified branching constants of an independent dense implementation (released with the code) to six significant digits, and every generated interview is checked to partition the cube. Branching and r-round designs at $m \le 12$ converge in tens to a few hundred iterations; pricing is no longer the bottleneck, and the master problem dominates the running time (Appendix F.2).

E.4. The Form Program and the Path-Set Menu

A form can reveal useful coordinates even when it fails to complete a tree path. For a frozen depth-b tree, take the coordinate set on each root-to-leaf path, extend it to size b with fixed unused coordinates, and retain the L distinct sets as a tractable menu. Let $g_{T,y}(u)$ be a table on *all* answers u to form T, and let $\widehat{P}_T(u,y)$ be calibrated marginal frequencies. For binary

features solve

$$\begin{aligned} \min_{q,g} \quad & \sum_{T,u,y} \widehat{P}_T(u,y)\frac{g_{T,y}(u)^2}{q_T}, \\ \text{subject to} \quad & \sum_{T\supseteq S} \widehat{g}_{T,y}(S) = y\widehat{d}(S) \quad \text{for every } S, y, \\ & \sum_T q_T = 1, \qquad q_T \geq \beta/L. \end{aligned} \tag{E.1}$$

Here $\widehat{g}_{T,y}(S)$ is the Fourier coefficient of the table on its own form; it is a linear transform of g. Fitted designs in Section 6 use $\beta = 0.05$. These constraints are exactly equivalent to $\sum_T g_{T,y}(x_T) = yd(x)$. Returning $g_{T,Y}(X_T)/q_T$ is therefore globally unbiased even when the reference frequencies are inaccurate; the reference affects only the variance.

There are at most $2L2^b$ score entries and coefficient constraints, and the program avoids enumerating the 2^m full profiles. Pairwise shared-coordinate corrections span the complete unbiased score space on the supplied menu over the full binary cube (Proposition E.4). A correction adds $f(X_{T\cap T'}, Y)/q_T$ on form T and subtracts it divided by $q_{T'}$ on form T'. The optimization therefore includes every such control variate. In the language of coarsening at random, (E.1) chooses the optimal augmentation of the inverse-probability-weighted score under the reference law, jointly with the form probabilities. For ordinal data the same program applies to tensor contrasts on the categories defined by the frozen tree's thresholds.

For an existing randomized archive, keep its known q fixed and optimize only the tables. Form-specific marginal frequencies can be estimated from incomplete training records; they need not be assembled into a joint reference distribution. Identification then requires coverage by the actual form menu, not only the degree bound for the menu of all size-b forms.

A *completion-only* score returns an inverse-probability weighted target on a matching padded path and zero otherwise; the experiments call its uniform version path guessing.

Proper scoring rules. For Brier improvement, with $z = (Y+1)/2$,

$$(p_0 - z)^2 - (p_1 - z)^2 = d(1 - p_0 - p_1) + Yd.$$

In the form program, replace Yd by this known target and impose separate coefficient identities for each label. The same unbiasedness guarantee holds. For ensembles of degree r, Brier loss can require degree $2r$; log loss can introduce still higher interactions. Appendix G.3 gives the identification conditions and explains why AUC requires a separate pairwise design.

E.5. Shared-Coordinate Corrections and the Complete Score Space

Fix a menu of coordinate sets $\mathcal{T}$. For binary features write $g_{T,y}(x_T) = \sum_{S\subseteq T} \gamma_{T,y,S}\chi_S(x)$. Uniqueness of the Fourier expansion gives the following elementary representation.

Proposition E.4 (All shared-coordinate corrections)**.** *The score tables satisfy $\sum_T g_{T,y}(x_T) = yd(x)$ on the full binary cube if and only if*

$$\sum_{T\supseteq S} \gamma_{T,y,S} = y\widehat{d}(S) \quad \textit{for every } S.$$

For a fixed positive design q, every change to a globally unbiased score is a sum of pairwise transfers of functions on shared coordinates: add $f(X_{T\cap T'}, Y)/q_T$ to form T and subtract the same function divided by $q_{T'}$ from form T'.

Proof. Equate the coefficients of each χ_S. For the second claim, any score difference has, for each S and each label, coefficients summing to zero over forms containing S. Choose one of those forms as an anchor and transfer every other coefficient to or from the anchor. Each transfer is observable under both forms. Terms appearing in only one form must have zero difference. Summing the transfers proves the statement. □

Thus including all such terms uses the entire unbiased score space on the supplied menu. This is a product-basis implementation of standard control variates and known-coarsening influence-function optimization, not a new general missing-data efficiency bound.

The implementation works in cell values. A block of the constraint matrix is the normalized Walsh transform of a form's table, placed in the rows of its contained coordinate subsets. If F is the resulting matrix, the constraints are $Fg_y = y\widehat{d}$. Its rows are orthogonal: FF^{T} is diagonal, with entry $\sum_{T \supseteq S} 2^{-|T|}$ for subset S. We use this identity to repair numerical equality residuals explicitly. Rotated second-order cones represent $g_{T,y}(u)^2 \le q_T v_{T,y}(u)$.

For a given population, marginal form–label frequencies are sufficient for the objective. No full-profile enumeration is needed. With L forms of size at most b, there are at most $2L2^b$ score entries and $2L2^b$ coefficient constraints. A tree provides the target coefficients through its leaves, or by Fourier transforming its known completion-score decomposition. Choosing an unrestricted menu can still require $\binom{m}{b}$ forms; this implementation does not solve that combinatorial design problem.

For ordinal features one splits each coordinate at all thresholds used for it in the frozen tree, with a fixed middle cut for otherwise unused padding coordinates. We use orthonormal Helmert contrasts on those categories and their tensor products. The same argument is exact on the resulting product space. This retains every observed tree-threshold category, but not within-category distinctions between raw ordinal answers. Its cell count is $\sum_T \prod_{j \in T} s_j$, where s_j is the category count.

E.6. Efficiency and Augmented Fixed Forms

E.6.1. Relation to the Missing-Data Efficiency Bound

Fix a known form law q and a full-support law P on the finite complete-record space. The observed record is $O = (A, o_A(Z))$. Its score along a centered complete-data perturbation h is $\mathbb{E}_P[h(Z) \mid O]$. A mean-zero observed influence function $\phi(O)$ for $\Delta = \mathbb{E}_P t$ must therefore satisfy

$$\mathbb{E}[\phi(O) \mid Z] = t(Z) - \Delta.$$

Conversely this equation gives the required pairing with every complete-data score, and hence an influence function in this finite regular model. Minimizing $\mathbb{E}\phi^2$ is the efficiency bound. Substituting $\phi = H - \Delta$ identifies it with the fixed-design population variance $W_q(P)$. This is the usual known-coarsening influence-function geometry (Robins et al., 1994; Tsiatis, 2006), not a new asymptotic efficiency formula. The proposition's additional statement is finite-n pointwise optimality over globally unbiased nonlinear estimators and sequential form assignments, together with its extension to boundary laws. It does not assert a finite-sample bound for all locally regular biased estimators.

Fixed random forms implement planned missingness. A tree's terminal cell is generated using observed answers, so its known acquisition mechanism is a designed coarsening-at-random mechanism (Heitjan & Rubin, 1991; Gill et al., 1997). This description does not cover arbitrary unrecorded selection or nonresponse.

E.7. Learning a Design from a Compact Tree

Fix a tree that recovers $t(z) = yd(x)$ within b distinct queries. Pad each path deterministically to b coordinates. Enumerate its distinct path sets $\mathcal{S} = \{S_1, \dots, S_L\}$ and let $S(z)$ denote the reached set. If an assigned fixed form S_j permits following the padded tree to completion, then $S(z) = S_j$, as in Appendix C.1. For any strictly positive form law q, the score

$$H_q(z, J) = \frac{t(z)}{q_J} \mathbf{1}\{S(z) = S_J\}, \qquad J \sim q,$$

is computable from the selected form and label. It is unbiased conditional on every full record. With $w_j = \mathbb{E}_P[t(Z)^2 \mathbf{1}\{S(Z) = S_j\}]$,

$$\operatorname{Var}_P H_q = \sum_{j=1}^{L} \frac{w_j}{q_j} - \Delta(P)^2. \tag{E.2}$$

Within this importance-score family the unconstrained optimum is $q_j \propto \sqrt{w_j}$, with infimal second moment $(\sum_j \sqrt{w_j})^2$. Zero-mass cells are handled by limits. We enforce $q_j \ge \beta/L$, ensuring global unbiasedness even on path sets absent from calibration. If some $w_j > 0$, the constrained optimum satisfies $q_j = \max\{\beta/L, c\sqrt{w_j}\}$, with c chosen to normalize the probabilities. If all weights vanish, every feasible q is optimal. Iteratively fixing cells below the floor computes the positive-weight solution without profile enumeration. This oracle optimizes a specified score family, not all fixed-form estimators.

E.7.1. A Worked Two-Question Example

Let $d(x) = [(1 + x_1)x_2 + (1 - x_1)x_3]/2$. Branching queries x_1 and then x_2 when $x_1 = 1$, or x_3 when $x_1 = -1$. The two fixed forms are

Form	Probability	Observed score
$\{1, 2\}$	$1/2$	$2Yx_2\mathbf{1}\{x_1 = 1\}$
$\{1, 3\}$	$1/2$	$2Yx_3\mathbf{1}\{x_1 = -1\}$

At every full record the mean score is Yd and its second moment is two. The branching score has second moment one. For an independent fair label both score means vanish, so their sample-mean variances are exactly $2/n$ and $1/n$. This describes these concrete estimators; the minimax results additionally justify their worst-case comparison against all protocols.

E.7.2. A Finite Calibration Guarantee

Suppose the tree is fitted independently of r calibration records and $|t| \le M$. Full calibration records reveal $W_j = t^2\mathbf{1}\{S = S_j\}$ directly. Alternatively, assign each calibration person a uniformly random fixed path set. On successful completion return $W_j = Lt^2\mathbf{1}\{J = j, S = S_j\}$ and return zero on failure. This alternative obeys the same fixed per-person cap and uses only observable quantities. In both cases $\mathbb{E}W = w$. Let $\widehat{w}$ be the mean and let $\widehat{q}$ minimize $\sum_j \widehat{w}_j/q_j$ subject to $q_j \ge \beta/L$.

Proposition E.5 (Learning path-set probabilities)**.** *Let $q^\star_\beta$ be the population optimum with the same probability floor. Set $B = M^2$ for full calibration or $B = LM^2$ for uniformly assigned fixed calibration. With probability at least $1 - \alpha$ over the r independent calibration records,*

$$\operatorname{Var}_P H_{\widehat{q}} - \operatorname{Var}_P H_{q^\star_\beta} \le \frac{2LB}{\beta}\sqrt{\frac{2(L\log 2 + \log(1/\alpha))}{r}}.$$

The resulting audit mean on fresh respondents remains globally unbiased for every realized calibration sample.

Proof. Each calibration vector has at most one nonzero entry and $\|W\|_1 \le B$. For any sign vector $\sigma \in \{-1, 1\}^L$, $\sigma^T W \in [-B, B]$. Hoeffding's inequality and a union bound over 2^L vectors give

$$\|\widehat{w} - w\|_1 = \max_\sigma \sigma^T(\widehat{w} - w) \le B\sqrt{2(L\log 2 + \log(1/\alpha))/r}.$$

The difference between the empirical and population second-moment objective is uniformly at most $(L/\beta)\|\widehat{w} - w\|_1$. Empirical minimization costs at most twice this uniform deviation. Subtracting the common Δ^2 proves the bound. Pointwise conditional unbiasedness holds for each positive q, hence also conditional on the entire calibration sample. □

The bound is conservative and becomes informative only with sufficiently large calibration samples. It proves consistency relative to a constrained family oracle, not a calibration-free optimality claim. Form enumeration takes time polynomial in the given tree's size and m; that tree may itself have exponentially many leaves in b. A deployment needs only its form probabilities and the tree, not a table over all s^m profiles.

E.7.3. Confidence Intervals and Costs

Conditional on calibration, each path score has magnitude at most $M/q_{\min}$. Thus the ordinary score mean has a distribution-free $(1 - \alpha)$ confidence interval with half-width

$$\frac{M}{q_{\min}}\sqrt{\frac{2\log(2/\alpha)}{n}},$$

intersected with $[-M, M]$. The branching score has the same bound with $q_{\min} = 1$. These Hoeffding intervals are valid but conservative; they are not claims of optimal interval width or power. The paper's risk comparisons concern MSE and do not order intervals for biased estimators.

For a menu with known per-person costs $c_\iota > 0$, let $c(q) = \sum_\iota q_\iota c_\iota$. An iid unbiased score mean with n respondents uses expected cost $nc(q)$, so its worst-case MSE-times-cost upper criterion is $c(q)\max_x \sum_\iota g_\iota(x)^2/q_\iota$. Set $r_\iota = q_\iota/c(q)$. Joint

cost-weighted design becomes the convex program

$$U_c(d,\mathcal{A}) = \inf_{\substack{r,g:\,\sum_\iota c_\iota r_\iota = 1 \\ \sum_\iota g_\iota = d}} \max_x \sum_\iota \frac{g_\iota(x)^2}{r_\iota}.$$

Conversely $q_\iota = r_\iota / \sum_j r_j$ recovers the design, proving equivalence of the two criteria. For fixed forms, c_ι may include recruitment, the label and item-specific costs. For a branching tree, taking c_ι as its maximum path cost gives a conservative criterion. This uses expected total expenditure and continuous sample-size scaling; it does not impose an almost-sure stopping budget or establish a sharp adaptivity ratio for arbitrary costs.

E.8. Learning the Design from the Audit Itself

The population-specific constants $W_\mathcal{A}(P)$ are attained by designs tuned to P. Section 6 tunes them on separate reference records. No reference is needed asymptotically: because every cell mass of every interview is estimable from the records assigned that interview, a pilot phase of the audit itself can supply the design, and pointwise unbiasedness keeps every intermediate design exactly unbiased.

For a finite menu $\mathcal{A}$, a law P on $\mathcal{X} \times \{-1,1\}$ and a design (q,g) feasible for (4), write $J_P(q,g) = \sum_\iota \sum_{c,y} P_\iota(c,y) g_\iota(c,y)^2/q_\iota$ for its second moment, where $P_\iota(c,y) = P(\ell_\iota(X) = c, Y = y)$ are the cell masses, so that $M_\mathcal{A}(P) = \inf_{q,g} J_P(q,g)$. For $\beta \in (0,1]$ let $M_\mathcal{A}^\beta(P)$ denote the infimum over designs with $q_\iota \geq \beta/|\mathcal{A}|$ for all ι.

Proposition E.6 (Two-phase self-calibration)**.** *Let $n_1 = n_1(n) \to \infty$ with $n_1/n \to 0$, and let $\alpha_n, \beta_n \in (0,1]$ tend to zero with $\alpha_n n_1^{1/2} \to \infty$. Phase one interviews n_1 records with the uniform design on $\mathcal{A}$ and forms, for every interview ι, the Horvitz–Thompson estimates $\widehat{P}_\iota(c,y) = (|\mathcal{A}|/n_1) \sum_{i \leq n_1} \mathbf{1}\{\iota_i = \iota,\ \ell_\iota(X_i) = c,\ Y_i = y\}$ and the smoothed masses $\widetilde{P}_\iota = (1-\alpha_n)\widehat{P}_\iota + \alpha_n u_\iota$, where u_ι is uniform on the cells and labels of ι. Let $(\hat{q},\hat{g})$ minimize $J_{\widetilde{P}}$ over feasible designs with $q \geq \beta_n/|\mathcal{A}|$, and interview the remaining $n_2 = n - n_1$ records with $(\hat{q},\hat{g})$. Then the mean $\widehat{\Delta}$ of their scores is unbiased for $\Delta(P)$ conditionally on phase one, for every P, and*

$$n_2 \operatorname{Var}(\widehat{\Delta} \mid \textit{phase one}) = J_P(\hat{q},\hat{g}) - \Delta(P)^2, \qquad J_P(\hat{q},\hat{g}) \xrightarrow{\mathbb{P}} M_\mathcal{A}(P), \qquad n \operatorname{Var}(\widehat{\Delta} \mid \textit{phase one}) \xrightarrow{\mathbb{P}} W_\mathcal{A}(P).$$

Proof. Unbiasedness holds because $(\hat{q},\hat{g})$ satisfies the pointwise constraints $\sum_\iota \hat{g}_\iota(\ell_\iota(x),y) = t(x,y)$ for every (x,y), whatever the phase-one data. Conditionally on phase one the phase-two scores are iid with mean $\Delta(P)$ and second moment $J_P(\hat{q},\hat{g})$, which gives the variance identity. For the limit put $P_\iota^\alpha = (1-\alpha_n)P_\iota + \alpha_n u_\iota$. Each $\widehat{P}_\iota(c,y)$ is an average of n_1 iid bounded terms with mean $P_\iota(c,y)$, so $\max_{\iota,c,y} |\widehat{P}_\iota(c,y) - P_\iota(c,y)| = O_\mathbb{P}(n_1^{-1/2})$ over the finitely many cells. Since $\widetilde{P}_\iota - P_\iota^\alpha = (1-\alpha_n)(\widehat{P}_\iota - P_\iota)$ and $P_\iota^\alpha \geq \alpha_n u_\iota$,

$$\rho_n := \max_{\iota,c,y} \left| \frac{\widetilde{P}_\iota(c,y)}{P_\iota^\alpha(c,y)} - 1 \right| \leq \frac{\max_{\iota,c,y} |\widehat{P}_\iota(c,y) - P_\iota(c,y)|}{\alpha_n \min_\iota u_\iota} = O_\mathbb{P}\big((\alpha_n n_1^{1/2})^{-1}\big) \to 0.$$

Let (q^α, g^α) be optimal for J_{P^α} under the floor β_n. Because J is linear in the masses,

$$(1-\alpha_n) J_P(\hat{q},\hat{g}) \leq J_{P^\alpha}(\hat{q},\hat{g}) \leq \frac{J_{\widetilde{P}}(\hat{q},\hat{g})}{1-\rho_n} \leq \frac{J_{\widetilde{P}}(q^\alpha,g^\alpha)}{1-\rho_n} \leq \frac{1+\rho_n}{1-\rho_n} M_\mathcal{A}^{\beta_n}(P^\alpha).$$

Finally $M_\mathcal{A}^\beta(P^\alpha) \to M_\mathcal{A}(P)$ as $\alpha, \beta \to 0$: for any feasible (q,g), mixing q with the uniform design at weight β keeps the tables feasible and multiplies J by at most $(1-\beta)^{-1}$, so $M_\mathcal{A}^\beta(P^\alpha) \leq (1-\beta)^{-1} J_{P^\alpha}(q,g) \leq (1-\beta)^{-1}[(1-\alpha)J_P(q,g) + \alpha J_u(q,g)]$, whose infimum over (q,g) tends to $M_\mathcal{A}(P)$; conversely $M_\mathcal{A}^\beta(P^\alpha) \geq (1-\alpha) M_\mathcal{A}(P)$. Combining the displays gives $\limsup J_P(\hat{q},\hat{g}) \leq M_\mathcal{A}(P)$ in probability, and $J_P(\hat{q},\hat{g}) \geq M_\mathcal{A}(P)$ holds for every feasible design. Since $n_2/n \to 1$, the last claim follows from $W_\mathcal{A}(P) = M_\mathcal{A}(P) - \Delta(P)^2$. □

The proposition is an asymptotic statement about a finite menu: phase one must estimate the masses of all $\sum_\iota |\mathrm{cells}(\iota)|$ cells, which requires n_1 large compared with $|\mathcal{A}|2^b$, and in the experiments of Section 6 reference records play the role of phase one. A sequential version that refits the design after every batch of records inherits exact unbiasedness by the same argument; its rate is not analyzed here. Learning the design across records is standard adaptive importance sampling; the point here is that the pointwise constraints make every fitted design exactly unbiased, so that the price of learning is paid in variance only.

F. Experimental Details

Every experiment of Section 6 is described here; no other experiment is used anywhere in the paper.

F.1. Data, Populations and Software

Data. WDBC (569 records, 30 continuous features) comes from scikit-learn (Wolberg et al., 1993); Spambase (4,601 records) and Ionosphere (351 records) from the PMLB mirror of the UCI files (Hopkins et al., 1999; Sigillito et al., 1989; Romano et al., 2022); the CDC Diabetes Health Indicators (253,680 respondents of the 2015 Behavioral Risk Factor Surveillance System) from a public mirror of the Kaggle release referenced by the UCI entry (Centers for Disease Control and Prevention, 2017); and the thyroid files (7,200 patients) from public mirrors of the UCI release (Quinlan, 1986). Each mirror was checked by reproducing values computed from the original downloads to nine decimals, which also confirms the record order.

Populations. Every population P is an empirical law of held-out or target records mixed with a uniform mass α over all profiles and both labels, $\alpha = 1\%$ unless stated. Full support makes $W_{\mathcal{A}}(P)/n$ a local minimax risk (Proposition C.7) and keeps every cell mass positive. Oracle designs are optimized at P; fitted designs are optimized on a reference law (training or source records, smoothed with $\alpha = 10\%$) with a floor of $5\%/L$ on each of the L interview probabilities of the final master problem, and evaluated at P. Every design is pointwise unbiased, so oracle and fitted designs differ only in variance.

Software. Python 3.13.5, NumPy 2.5.4, SciPy 1.18.1, scikit-learn 1.9.1, pandas 3.1.1, CVXPY 1.9.3 with Clarabel 0.11.1, single-threaded. All constants are computed by the solver of Appendix E.3; it reproduces the values of an independent dense implementation on every overlapping task.

F.2. Forest Updates

Protocol. For each dataset and seed, a random permutation assigns n_{old} records to train the old forest, the next n_{new} to train the new forest, and the remaining n_{eval} records to the evaluation law: $(n_{\mathrm{old}}, n_{\mathrm{new}}, n_{\mathrm{eval}}) = (100, 200, 269)$ for WDBC, $(500, 1000, 3101)$ for Spambase and $(60, 120, 171)$ for Ionosphere. Continuous features are binarized at their training medians; m features are chosen at equally spaced positions of the eligible feature list, without labels. Each forest has twenty trees of depth b, $\sqrt{m}$ features per split and minimum leaf size $\max\{3, n_{\mathrm{old}}/30\}$. The contrast d is the difference of the two forests' predicted probabilities on the cube; it has degree at most b. The cap is binding for $m > b$. At $b = 3$ twenty splits per $(m, \text{dataset})$ cell are used; at $b \in \{2, 4, 5\}$ five splits. At $m = 12$ the held-out records occupy a median of 171, 388 and 121 of the 4,096 profiles for WDBC, Spambase and Ionosphere (ranges 164–180, 373–405 and 113–130 over twenty splits). Spambase, whose 3,101 held-out records cover few profiles densely, gives a median exact ratio of 1.58 at $b = 3$, $m = 12$, close to WDBC (1.66) and Ionosphere (1.73), so the gains are not an artifact of sparse held-out sets; Appendix F.1 also varies the smoothing mass and resamples the held-out records.

Designs and certificates. The fixed constant optimizes over all $\binom{m}{b}$ forms; the branching constant over all depth-b trees by column generation. The restricted menu adds only the forty component trees to the forms; at $b \geq 4$ their short paths are padded to depth b, a refinement within the cap, so the restricted-menu ratios at $b \geq 4$ are slightly more favourable to branching than at $b \leq 3$. At $b \in \{4, 5\}$ column generation stops when the certified bounds agree to relative error 10^{-4} or no tree improves the master; over the 15 tasks computed with the new solver the largest relative gap of the reported ratio is 2e-08, the median number of iterations 28 and the median time 40 seconds per task on one core. At $b \in \{2, 3\}$ the dense implementation certified every task to 10^{-6}.

Smoothing mass and resampling. For the fifteen $b = 3$ tasks with $m = 12$ (five splits per dataset) the exact constants are recomputed with $\alpha \in \{0.1\%, 1\%, 5\%, 10\%\}$ and, at $\alpha = 1\%$, on bootstrap resamples of the held-out records with the forests fixed.

Variation	tasks	median ratio	median rel. change	max rel. change	median $\lvert \log \rvert$ change
$\alpha = 0.1\%$	15	1.654	+0.0%	2.0%	0.003
$\alpha = 5\%$	15	1.660	+0.8%	2.7%	0.008
$\alpha = 10\%$	15	1.690	+1.8%	5.3%	0.018
bootstrap of held-out records ($\alpha = 1\%$)	27	1.602	-1.1%	7.5%	0.012

Table 5. Independent forest updates, cap $b = 3$, 20 splits per cell. Exact fixed/branching variance ratio over all forms and all depth-3 trees (median and 10th–90th percentiles). Restr.: branching restricted to the 40 component trees and all triples. Fix/full: fixed-form variance relative to the full-information variance. Fitted: designs optimized on training records, evaluated held-out ($m = 12$, five splits).

Data	m	Exact ratio	Restr.	Fix/full	Fitted
WDBC	6	1.36 [1.27, 1.48]	1.08	2.11	
	9	1.57 [1.43, 1.69]	1.09	3.91	
	12	1.66 [1.53, 1.74]	1.08	5.55	1.67
Spambase	6	1.41 [1.27, 1.52]	1.03	2.31	
	9	1.45 [1.36, 1.59]	1.03	3.09	
	12	1.58 [1.49, 1.66]	1.04	4.75	1.60
Ionosphere	6	1.36 [1.20, 1.46]	1.07	2.06	
	9	1.61 [1.45, 1.77]	1.10	4.19	
	12	1.73 [1.58, 1.78]	1.10	5.94	1.79

Table 6. Exact records ratio by cap, dimension and dataset (median over five splits; $b = 3$ uses twenty splits and is detailed in Table 5). Width: largest relative gap between the certified bounds of the branching constant.

b	m	WDBC	Spambase	Ionosphere	fitted	width
2	6	1.31	1.35	1.31	1.31	1e-07
2	9	1.23	1.25	1.44	1.26	6e-07
2	12	1.27	1.35	1.42	1.35	2e-07
4	6	1.23	1.28	1.20	1.27	2e-08

Predicting the gain from reference data. For every forest task at $b \in \{2, 3, 4\}$ (the grid available when this analysis was run) and every screening run, the lower end of the variance-scale bracket of Theorem 4 is recomputed on the reference law alone and compared with the realized exact ratio.

Tasks	n	Spearman($\widehat{L}$, realized)	median realized/$\widehat{L}$	realized$< \widehat{L}$	rule: $\widehat{L} \le 1.05 \Rightarrow$ realized ≤ 1.25
forests	240	0.52	1.08	75	2/2
screening trees	48	0.78	1.00	0	31/31

The decision rule *lower end at most 1.05 ⇒ do not plan for a gain* has no misses in these data: all 17 screening runs with a realized gain above 1.25 have a reference lower end above 1.05, all 31 runs with a lower end at most 1.05 realize at most 1.25, and none of the 214 forest tasks with a realized gain above 1.25 has a lower end at most 1.05. The lower end is conservative for forests, where no single tree computes the contrast.

F.3. Screening Questionnaires

Items and targets. BRFSS items: high blood pressure, high cholesterol, body-mass index at least 30, smoking, stroke, heart disease, physical activity, heavy drinking, fair or poor general health, difficulty walking, age 60 or older, and income in the five lowest categories; the target is the diabetes-or-prediabetes indicator. Thyroid items: age above its upper quartile, sex, thyroxine treatment, thyroid surgery, query hypothyroid, goitre, TSH above its upper quartile, T3, TT4 and FTI below their lower quartiles, T4U above its upper quartile, and sickness, with quartiles from the training file and no labels; the target is any abnormal class.

Splits and trees. For the random-half target, a random half of the records is the source; for the shifted target, the source is one sex and the target the other. Up to 20,000 source records train a class-balanced CART tree of depth b (minimum leaf size 50 for BRFSS, 10 for thyroid); up to 20,000 further source records are the reference for fitted designs. The estimand is the target accuracy of the tree's decision, $d \in \{-1, 1\}$, so $M_{\mathrm{branch}} = \mathbb{E}[t^2] = 1$ and branching attains the full-information variance. Seeds $0, 1, 2$ vary the split and the tree.

Designs. Best: the optimal fixed design over all $\binom{12}{b}$ forms. Fitted: the same program on the source reference. PS: the optimized path-set design of Proposition 5, at the target law and fitted. Guess: uniform path guessing with completion-only scores, second moment $K\mathbb{E}_P[t^2]$ for K path sets. Two (heuristic): follow the root question, then use the best fixed form of the remaining size in the reached subcube. r-round (exact): the certified optimum over all r-round interviews. Lower: $1/\Gamma_P$

on the variance scale. Upper: N_P on the variance scale.

Table 7. All screening settings (medians over three seeds). Ratios are records needed relative to branching. Best/Fitted: optimal and fitted fixed designs; PS: optimized path-set design (oracle/fitted); Guess: uniform path guessing; H2: heuristic two-round design; r: exact optimal r-round design; Lower, N_P: ends of Theorem 4.

Data	b	Target	Acc.	Paths	Best	Fitted	PS	PS fit	Guess	H2	r=2	r=3	Lower	N_P
BRFSS	3	random half	0.629	2	1.00	1.04	1.00	1.00	2.1	1.00	1.00	–	1.00	1.5
BRFSS	3	other sex	0.741	1	1.00	1.04	1.00	1.00	1.0	1.00	1.00	–	1.00	1.0
BRFSS	4	random half	0.664	4	1.46	1.52	1.50	1.52	4.3	1.14	1.00	1.00	1.17	3.3
BRFSS	4	other sex	0.681	4	1.52	1.61	1.57	1.62	4.5	1.13	1.00	1.00	1.18	3.9
BRFSS	5	random half	0.675	7	1.28	1.32	1.28	1.29	7.8	1.15	1.02	1.00	1.04	5.1
BRFSS	5	other sex	0.734	6	1.43	1.52	1.44	1.50	7.6	1.27	1.00	1.00	1.13	6.6
BRFSS	6	random half	0.719	14	1.53	1.58	1.53	1.56	17.4	1.40	–	–	1.13	11.8
BRFSS	6	other sex	0.722	13	1.60	1.70	1.65	1.70	16.0	1.39	1.00	1.00	1.13	10.9
Thyroid	3	random half	0.823	3	1.00	1.04	1.00	1.00	4.4	1.00	1.00	–	1.00	3.3
Thyroid	3	other sex	0.819	3	1.00	1.04	1.00	1.01	4.4	1.00	1.00	–	1.00	3.6
Thyroid	4	random half	0.823	3	1.00	1.04	1.00	1.00	4.5	1.00	1.00	1.00	1.00	3.4
Thyroid	4	other sex	0.848	4	1.00	1.04	1.00	1.01	6.8	1.00	1.00	1.00	1.00	5.0
Thyroid	5	random half	0.823	6	1.00	1.03	1.00	1.00	9.7	1.00	1.00	1.00	1.00	6.3
Thyroid	5	other sex	0.848	5	1.00	1.04	1.00	1.01	8.7	1.00	1.00	1.00	1.00	5.8
Thyroid	6	random half	0.823	7	1.00	1.03	1.00	1.00	11.4	1.00	1.00	1.00	1.00	6.7
Thyroid	6	other sex	0.848	5	1.00	1.04	1.00	1.01	8.7	1.00	1.00	1.00	1.00	6.1

F.4. All Twenty-One BRFSS Variables

All 21 variables of the CDC file are binarized at pre-specified cut points: high blood pressure, high cholesterol, cholesterol check, body-mass index at least 30, smoking, stroke, heart disease, physical activity, fruit, vegetables, heavy drinking, any health coverage, cost barrier to a doctor visit, fair or poor general health, at least 14 days of poor mental health, at least 14 days of poor physical health, difficulty walking, sex, age 60 or older, no college education, and income in the five lowest categories. Splits, trees and references are as for the twelve-item study (random-half target). The form program has $\binom{21}{b}$ forms, 1,330 at $b = 3$ and 5,985 at $b = 4$, and the profile cube has 2^{21} elements; the solver never enumerates the cube, and at $b = 4$ one master problem over all forms certifies the optimum. Exact two-round optima and the cap $b = 5$ were not completed at $m = 21$ within the compute budget of this version.

Table 8. All 21 BRFSS variables (random-half target; medians over three seeds). Forms: $\binom{21}{b}$. Time: median minutes for the certified fixed design.

b	forms	Acc.	items used	Best	Fitted	r=2	PS	Lower	N_P	time
3	1330	0.629	4	1.00	1.00	1.00	1.00	1.00	1.5	0.0
4	5985	0.664	6	1.46	1.54	–	1.50	1.17	3.8	6.1

F.5. Imputation From Complete Source Records

Simulated audits. Each replication draws $n = 1000$ target respondents with replacement; 400 replications per setting. Branching averages t. The fixed-form score draws a form from the fitted design and averages its score. Regression imputation uses the same observed forms and answers and replaces t by its conditional mean given the asked items and the label among the complete source records, with a naive standard error. Multiple imputation draws, for each of ten imputations, Bayesian-bootstrap weights over the source records, imputes the unasked items of each respondent from a donor that agrees on the asked items and the label, and combines by Rubin's rules. Truth is the contrast on the full target sample. Regression imputation and MI use information that the capped audit does not collect: complete records whose conditional law of unasked items is assumed to transport to the target.

Pre-specified shifts. Before any result was computed we fixed four shifts defined by variables outside the twelve items: sex (female source, male target), education (college graduates, target without education beyond high school), cost barrier (no cost barrier to a doctor visit, target with a cost barrier) and physical-health burden (no poor physical-health days in the past month, target with at least 14). All four are reported. Table 9 gives the random-half target and Table 10 the four shifts.

Under the random-half target, MI intervals over-cover (0.975–0.993), so Rubin's variance is conservative there. For a frozen imputation bias $\mathfrak{b}_I$ and per-record variances σ_I^2 and σ_U^2 of the imputation and design-based estimators, imputation has the

Table 9. Audit simulations with $n = 1000$ target respondents and 400 replications. Bias, standard deviation and RMSE are multiplied by 100; coverage of nominal 95% intervals has Monte Carlo standard error about 0.011. Regression imputation: conditional mean of t given the asked items and the label from the complete source records, naive standard error. MI: ten proper hot-deck imputations from the same source records with Rubin's rules.

Data, b, target	Design	Bias	SD	RMSE	Coverage	Width
BRFSS, 4, random half	branching	-0.14	2.96	2.96	0.955	11.70
	fitted fixed form	-0.11	3.60	3.60	0.965	14.42
	fixed form + regression imputation	-0.45	2.59	2.63	0.955	10.27
	fixed form + MI	-0.42	2.58	2.61	0.993	13.21
BRFSS, 4, other sex	branching	0.05	3.06	3.05	0.953	11.69
	fitted fixed form	0.06	3.12	3.11	0.948	11.89
	fixed form + regression imputation	0.32	3.01	3.03	0.940	11.50
	fixed form + MI	0.33	3.02	3.03	0.955	11.87
BRFSS, 5, random half	branching	-0.05	2.92	2.92	0.950	11.60
	fitted fixed form	-0.06	3.19	3.18	0.955	12.85
	fixed form + regression imputation	-0.06	2.67	2.67	0.958	10.84
	fixed form + MI	-0.06	2.68	2.68	0.975	12.41
BRFSS, 5, other sex	branching	0.08	2.89	2.89	0.965	11.42
	fitted fixed form	0.11	3.44	3.44	0.938	13.12
	fixed form + regression imputation	-0.45	2.80	2.84	0.948	10.72
	fixed form + MI	-0.47	2.81	2.85	0.983	12.22

smaller mean squared error exactly when $n\mathfrak{b}_I^2 < \sigma_U^2 - \sigma_I^2$, so a biased imputation can win at small n and lose as the audit grows.

F.6. When Asking Changes the Answer

Let P describe latent complete records and let K_T be the distribution of the recorded transcript given a latent record under form T. A frozen score then estimates $\mathbb{E}_{P,q,K} H_T(O_T)$, which can differ from Δ; a branching policy has its own sequential kernel. We compute the resulting biases exactly on the screening trees of Appendix F.3. The answer at query position $j = 1, \ldots, b$ flips independently with probability $\epsilon(j-1)/(b-1)$, so the first answer is exact and the last flips with probability ϵ; labels are unchanged. Branching routes on the recorded answers; fixed forms ask their coordinates in increasing index order and use the fitted design's scores. Summing over all 2^b flip patterns gives the exact mean and second moment of each design at the target law. This is an illustrative position model, not an estimated response-error process.

F.7. Cost Regimes Under a Relaxable Cap

Suppose each feature costs one unit and each label c units, and the cap may be relaxed. A complete record costs $c + m$ and attains the full-information variance σ_P^2; a capped record costs at most $c + b$. For a screening tree, branching attains σ_P^2 at cost $c + b$ and therefore dominates complete records. Complete records are cheaper than the best fixed design exactly when $(c + m)/(c + b) < W_{\text{fixed}}/\sigma_P^2$, that is, when $c > (m - \rho b)/(\rho - 1)$ with $\rho = W_{\text{fixed}}/\sigma_P^2$. For the BRFSS trees ($m = 12$) the median ρ is 1.49, 1.35, 1.58 at $b = 4, 5, 6$, so complete records undercut the best fixed design once a label costs more than about 12, 15, 4 feature questions, respectively, while branching is never undercut. Under an immutable cap the complete-record alternative is unavailable.

G. Beyond Low-Degree Accuracy Contrasts

G.1. A Feasibility-First Workflow for Nonlinear Updates

Fix a profile space, an acquisition menu, a confidence level $1 - \alpha > 1/2$, and a target half-width e. The workflow separates two certificates: a dual lower bound δ_L on the identification floor and a feasible approximation $v \in \mathcal{H}_{\mathcal{A}}$ with residual at most ε. Only the second certificate permits an accuracy guarantee.

Rejecting an insufficient cap. If $\delta_L > e$, no interval with half-width at most e can have uniform coverage $1 - \alpha$, for any number of records. To see this on a finite space, choose a dual signed measure h with $\|h\|_1 = 1$, $h \perp \mathcal{H}_{\mathcal{A}}$, and

Table 10. Pre-specified shifts on BRFSS: simulated audits with $n = 1000$ target respondents and 400 replications. Bias, SD and RMSE are multiplied by 100; coverage of nominal 95% intervals has Monte Carlo standard error about 0.011. TV: total variation distance between source and target item profiles.

Shift	b	TV	Design	Bias	SD	RMSE	Coverage	Width
sex	4	0.21	branching	0.05	3.06	3.05	0.953	11.69
			fitted fixed form	0.06	3.12	3.11	0.948	11.89
			regression imputation	0.32	3.01	3.03	0.940	11.50
			multiple imputation	0.33	3.02	3.03	0.955	11.87
sex	5	0.21	branching	0.08	2.89	2.89	0.965	11.42
			fitted fixed form	0.11	3.44	3.44	0.938	13.12
			regression imputation	-0.45	2.80	2.84	0.948	10.72
			multiple imputation	-0.47	2.81	2.85	0.983	12.22
education	4	0.47	branching	0.08	3.12	3.12	0.948	12.28
			fitted fixed form	-0.06	3.60	3.60	0.960	14.20
			regression imputation	2.39	2.79	3.67	0.863	11.28
			multiple imputation	2.38	2.83	3.69	0.920	13.27
education	5	0.47	branching	0.04	3.11	3.11	0.950	12.25
			fitted fixed form	-0.03	3.39	3.38	0.953	13.28
			regression imputation	1.67	2.96	3.39	0.902	11.76
			multiple imputation	1.66	2.97	3.39	0.935	12.73
cost barrier	4	0.41	branching	-0.09	3.23	3.22	0.945	12.01
			fitted fixed form	-0.18	3.80	3.80	0.943	14.75
			regression imputation	0.39	2.82	2.85	0.930	10.76
			multiple imputation	0.39	2.84	2.86	0.980	13.31
cost barrier	5	0.41	branching	0.01	3.13	3.12	0.935	11.75
			fitted fixed form	-0.10	3.89	3.89	0.938	14.45
			regression imputation	-0.72	2.84	2.93	0.925	10.58
			multiple imputation	-0.70	2.92	3.00	0.960	13.01
poor health	4	0.68	branching	0.21	3.17	3.17	0.950	12.39
			fitted fixed form	0.23	3.23	3.23	0.950	12.59
			regression imputation	0.65	3.15	3.21	0.960	12.23
			multiple imputation	0.65	3.15	3.22	0.965	12.57
poor health	5	0.68	branching	0.09	3.17	3.16	0.945	12.37
			fitted fixed form	0.14	3.94	3.94	0.940	15.04
			regression imputation	5.49	2.96	6.24	0.515	11.44
			multiple imputation	5.48	2.96	6.23	0.667	13.23

$\sum_x h(x)d(x) > e$. Define

$$P_{\pm}(x, y) = \frac{|h(x)| \pm yh(x)}{2}.$$

Every interview cell has the same joint probability with Y under the two laws because its sum of h is zero. This also gives identical experiments under sequential design. Their target values are opposite and more than $2e$ apart. Under the common observation law an interval of that width cannot cover both, so the two coverage probabilities sum to at most one. This rules out uniform coverage greater than one half. The conclusion assumes arbitrary joint laws on the stated profile space; a substantively justified restriction on those laws changes the problem.

Passing with a certified approximation. Suppose an observable score H has conditional mean $Yv(X)$ and satisfies $|H| \le C$. Conditional on all model fitting and calibration, fresh iid records satisfy

$$\Pr_P\left\{|\overline{H} - \Delta(P)| \le \varepsilon + C\sqrt{2\log(2/\alpha)/n}\right\} \ge 1 - \alpha.$$

This is Hoeffding's inequality plus $|\mathbb{E}_P[Y(v - d)]| \le \varepsilon$. It is valid for every population on the certified space, not merely the reference population used to fit a score. For $\varepsilon < e$, the sufficient count in Section 2 follows by inversion. If the lower and upper certificates straddle e, they have not resolved feasibility: improve the approximation or compare a larger cap.

On a binary cube, write $v = \sum_{|S|\le b} \gamma_S \chi_S$ and $C = \sum_S |\gamma_S|$. Query term S with probability $|\gamma_S|/C$ and return $CY \operatorname{sgn}(\gamma_S)\chi_S(X)$. The empty term uses no feature; if $C = 0$, return zero. This yields the required observable score and bound. The finite-sample radius can therefore be minimized by a linear program:

$$\min_{\gamma, \varepsilon \ge 0} \varepsilon + \sqrt{2\log(2/\alpha)/n}\|\gamma\|_1, \qquad |d - \mathsf{F}\gamma| \le \varepsilon \mathbf{1},$$

where only ε is constrained nonnegative and γ is free. This is a conservative confidence criterion, distinct from the MSE criterion in Appendix G.2.

Table 11. Position effects on the screening trees: median absolute bias in units of the clean branching standard error at $n = 1000$, and the share of runs in which branching keeps the smaller RMSE at $n = 1000$ (the last answer flips with probability ϵ).

Data	ϵ	branching	fitted form	oracle form	branching RMSE smaller
BRFSS	0.01	0.15	0.21	0.22	24/24
BRFSS	0.03	0.45	0.64	0.65	24/24
BRFSS	0.05	0.75	1.08	1.09	24/24
BRFSS	0.1	1.54	2.17	2.20	24/24
Thyroid	0.01	0.02	0.24	0.23	24/24
Thyroid	0.03	0.06	0.71	0.70	24/24
Thyroid	0.05	0.11	1.18	1.17	24/24
Thyroid	0.1	0.21	2.37	2.34	23/24

What scales without full form enumeration. The all-form search certifies an offline benchmark; deployment of an already fitted score requires no such search. If a frozen contrast has a supplied decomposition $d = \sum_j w_j f_j$, with each f_j computable within b queries and $|f_j| \le M_j$, choose j with probability $|w_j|M_j/C$ for $C = \sum_j |w_j|M_j$ and return $CY \operatorname{sgn}(w_j) f_j/M_j$. It is exactly unbiased and bounded by C. This includes differences of probability-averaging forests with depth at most b, even when their combined prediction uses many coordinates and the inputs are continuous. A fixed form can simulate a sampled component by its path-set construction. Neither method asserts optimal efficiency. Nonlinear transformations of ensemble scores, including the logistic link in ordinary GBDT probability outputs, need a separate approximation certificate.

For an arbitrary continuous model, restrict selected coordinates to two values and hold the others fixed. The restriction of any b-coordinate observable function has Boolean degree at most b. Thus a bias-floor lower certificate on that finite slice is also a lower bound on the full product-domain problem. The slice supplies no full-domain upper bound. This is a one-sided rejection tool whose scope must be stated explicitly.

G.1.1. Independent GBDT and Neural-Network Versions

WDBC contains 569 diagnostic records and 30 image-derived continuous features (Wolberg et al., 1993). The target is malignant versus benign. The data therefore supply a non-questionnaire outcome and continuous model inputs, but their image-derived acquisition does not instantiate our survey cap. This experiment asks about mathematical feasibility for ordinary model classes.

Five fixed seeds, 20261108–20261112, permute the data. The first 100 records train the old model; the next 300 train the new model; the last 169 determine its evaluated accuracy gain. GBDT uses 30 versus 100 trees, depth two, learning rate 0.05 and minimum leaf size five. The MLP uses one ReLU hidden layer of width eight versus sixteen, ℓ_2 penalty one, and L-BFGS. Each model's standardization uses its own training records. No evaluation record selects model size, variables, thresholds or seed. The splits reuse individuals and are descriptive sensitivity analyses, not independent populations.

The discrete condition takes twelve coordinates at equally spaced indices of the original feature list and thresholds them at medians of the 400 training records. Both probability outputs and deterministic thresholds at one half are evaluated. The admissible domain is the entire binary cube, with $b = 3$. The continuous condition trains on all thirty original features. Its certificate varies the same twelve coordinates between their training first and third quartiles and fixes the others to training medians. Continuous-condition brackets refer to that slice; only their lower endpoints apply to the full domain. Arbitrary product combinations can violate empirical feature dependencies, so these are distribution-free certificates under the declared domain, not support-aware clinical claims.

For a target vector d on the 4,096-profile grid and the degree-three Walsh matrix F, the floor program minimizes $\|d - \mathsf{F}\gamma\|_\infty$. Its floating-point solution is converted to certificates: round primal coefficients to multiples of 2^{-28}; round a dual vector to integers z on scale 2^{40}; then form

$$h_{\mathrm{int}} = 4096z - \mathsf{F}\mathsf{F}^T z.$$

Because $\mathsf{F}^T\mathsf{F} = 4096I$, the integer vector has exactly zero degree-at-most-three coefficients. Dividing by its exact ℓ_1 norm gives a feasible rational dual. Both its inner product with the stored binary64 predictions and the rounded primal's maximum residual are computed with rational arithmetic. The certificate treats those stored software predictions as the defined finite target, without claiming formal verification of a real-arithmetic neural network.

Table 12. Cap-three nonlinear-model diagnostic. Binary rows bracket the full binary-domain floor. Continuous rows bracket a twelve-coordinate slice; only the lower endpoints transfer to the full continuous domain. The last column compares the certified floor directly with the requested half-width.

Model condition	Splits	Median LB	Median UB	LB>0.025
Binary, probability, GBDT	5	0.0552	0.0552	5/5
Binary, probability, MLP	5	0.2780	0.2780	5/5
Binary, threshold, GBDT	5	0.3409	0.3409	5/5
Binary, threshold, MLP	5	0.6572	0.6572	5/5
Continuous slice, GBDT	5	0.0065	0.0065	0/5
Continuous slice, MLP	5	0.1286	0.1286	5/5

The binary probability condition also minimizes the bias-expanded Hoeffding radius at $n = 1000$, $\alpha = .05$. The exact certificates do not depend on an optimizer's success flag. A bounded-time implementation can fall back to a feasible low-degree projection and its orthogonal residual; solver status and certificate width are released in every row. All thirty floor programs terminated optimally in the released run. The largest certified bracket width is below 10^{-7}, and the fitted models emitted no convergence warnings.

The continuous slice results are weaker statements than the complete binary-cube results. A small slice lower bound cannot certify an acceptable full-domain bias. Similarly, reporting a good empirical MSE does not replace the uniform residual required for a bias-expanded interval.

G.2. A Computable Workflow for Dense Updates

On a finite binary cube let F contain the p parity columns of degrees at most b. Compute the identification floor by the linear program

$$\delta = \min_{\gamma \in \mathbb{R}^p, e \geq 0} \{e : -e\mathbf{1} \leq d - \mathsf{F}\gamma \leq e\mathbf{1}\},$$

where only e is constrained to be nonnegative and γ is unrestricted. A dual witness h with $\mathsf{F}^T h = 0$ and $\|h\|_1 \leq 1$ certifies $\delta \geq h^T d$. For an audit of size n, select a variance-aware approximation by minimizing

$$\|d - \mathsf{F}\gamma\|_\infty^2 + \|\gamma\|_1^2/n.$$

For each respondent draw a parity term S with probability $|\gamma_S|/\|\gamma\|_1$, ask its at most b coordinates, and return $Y \operatorname{sgn}(\gamma_S)\|\gamma\|_1 \chi_S(X)$. The empty term requires no feature. If all coefficients vanish, return zero. The conditional mean is $Y(\mathsf{F}\gamma)(X)$ and the second moment is $\|\gamma\|_1^2$, proving the displayed uniform MSE bound. This converts approximation into an implementable audit while reporting the unavoidable lower floor δ^2 separately.

G.3. Proper Scores and Pairwise Targets

The score-design method applies to a bounded known record-level target $t(x, y)$, without requiring that it be a randomized classification gain. For a label observed separately, write

$$t(x, y) = a(x) + yc(x), \qquad a(x) = \frac{t(x, 1) + t(x, -1)}{2}, \quad c(x) = \frac{t(x, 1) - t(x, -1)}{2}.$$

On a finite domain, identification under arbitrary joint populations is equivalent to $a, c \in \mathcal{H}_\mathcal{A}$. On the full binary cube with cap b, this is $\deg(a), \deg(c) \leq b$. Indeed the observation span on (x, y) is $\mathcal{H}_\mathcal{A} \otimes \operatorname{span}\{1, y\}$: a table may have separate entries for the two labels. Its orthogonal complement gives indistinguishable populations whenever the target lies outside that span. In particular the common identification criterion for fixed forms and branching still applies.

For $z = (y + 1)/2$ and $d = p_1 - p_0$, the improvement in Brier loss is

$$t_{\mathrm{Br}}(x, y) = (p_0 - z)^2 - (p_1 - z)^2 = d(1 - p_0 - p_1) + yd.$$

Thus degree at most r for each probability function implies degree at most $2r$ for this target's feature-only component and at most r for its label component. The bound can be strict because terms cancel. The actual degree, or a certified approximation residual, determines the necessary cap. Doubling is attainable: take $p_0 = 1/2$ and $p_1 = 1/2 + \gamma(x_1x_2 + x_3x_4)$, with

Table 13. PISA secondary analysis on 2,441 held-out students. Brier improvement and 95% intervals are multiplied by 10^3; positive means the newer rule has smaller loss. Survey intervals use the 80 supplied Fay replicates. MI intervals also use Rubin combination under a working imputation model.

Method	Estimate	95% interval
Form score	1.13	$[-2.28, 4.53]$
Block/pair HT	1.24	$[-2.09, 4.57]$
Logistic MI	1.24	$[-2.12, 4.60]$

$0 < \gamma \le 1/4$. The accuracy contrast has degree two, whereas the Brier feature-only component is $-2\gamma^2(1 + x_1x_2x_3x_4)$ and has degree four. For a common depth-b tree whose leaf values determine both probabilities, the complete loss difference is still computable along that same path.

For log loss, clip each probability to a fixed interval $[\eta, 1-\eta]$, $0 < \eta < 1/2$, before defining the target. Then

$$a_{\log} = \tfrac{1}{2} \log \frac{p_1(1-p_1)}{p_0(1-p_0)},$$
$$c_{\log} = \tfrac{1}{2} \log \frac{p_1(1-p_0)}{p_0(1-p_1)}.$$

These formulas estimate improvement of the clipped rules. A logarithm of an ensemble average can introduce interactions of every order, even when each probability has a low-degree representation. A common tree partition again avoids that issue because the transformed values are constant on leaves.

The full-form program in Section E.4 replaces its right-hand side by $\widehat{t}_y(S)$ for each label. Its pointwise identity, population-specific variance objective, and invariance of unbiasedness to calibration error remain unchanged. The symmetric-label minimax formula $V(d, \mathcal{A})$ and its sharp ratio were proved for $Yd(X)$; this extension of the estimator does not assert that the identical formula is the global minimax coefficient for every proper score. The path simulation does preserve any bounded target's conditional mean and multiplies its conditional second moment by at most the number of padded path sets.

AUC needs a pair design. Let $X^+ \sim P(X \mid Y = 1)$ and $X^- \sim P(X \mid Y = -1)$ be independent. For a frozen score s, define $k_s(x^+, x^-) = \mathbf{1}\{s(x^+) > s(x^-)\} + \frac{1}{2}\mathbf{1}\{s(x^+) = s(x^-)\}$. The AUC difference is $\mathbb{E}[k_{s_1} - k_{s_0}]$ under the product of these two conditional laws. Disjoint labeled case–control pairs give iid pair targets; one can design unbiased scores on pair transcripts, retaining a separate cap on each member. This differs from a single-record target: using all case–control pairs introduces U-statistic dependence, and the sharp single-record coefficient ratio cannot be asserted for that design. Class prevalence, label-screening cost and the availability of both classes must also enter an acquisition comparison. We establish the record-level Brier/log-loss extension here and leave pair-design optimality unclaimed.

H. A Real Planned-Missing Archive

H.1. Observed forms, common target and acquisition constraint

We use the national U.S. PISA 2012 student public-use file supplied by NCES (National Center for Education Statistics, 2014). The OECD technical report, Chapter 3, describes random allocation of three student questionnaires, with approximately equal counts. Each has an eight-minute common part and two eleven-minute rotated parts. The U.S. technical report describes the forms' A–B–C spiraling and the required survey-weight analysis (OECD, 2014; Kastberg et al., 2014). Our analysis uses allocation probabilities $q_A = q_B = q_C = 1/3$. The observed form identifier is `QUESTID`; `FORMID` identifies a different, computer-based assessment and must not be used for this purpose.

The nine predictors are the first three response items in each of `ST29` (interest/motivation), `ST42` (anxiety/self-concept), and `ST53` (learning strategies). Call these blocks U, V, W. Forms A, B, C reveal (U, W), (U, V), (V, W), respectively. The released codes confirm that every selected item outside its assigned form has the structural not-applicable code. Within a received form, ST29/ST42 responses 1 or 2 define endorsement; ST53 response 1 defines endorsement. All other received responses, including item nonresponse or invalid codes, define nonendorsement. Thus we estimate a contrast of recorded responses, not the unobserved substantive answers of nonresponders. This convention does not complete any structurally omitted block.

The common target `ST07Q01` asks about grade repetition in grades 1–6. Codes 2/3 define $Z = 1$, and code 1 defines $Z = 0$. We retain 4,752 of 4,978 records with a valid target report. The design-based interpretation is conditional on this respondent population and assumes that common-part response does not break the randomized allocation. A claim about all students, including target nonresponders, would require further assumptions. There are 1,582/1,590/1,580 retained records on A/B/C. No record has all nine selected predictors. These are nine selected indicators within a larger 30-minute questionnaire, not nine independent tests in the actual survey protocol.

H.2. Frozen ensembles trained from incomplete records

We permute the 162 school IDs with seed 20261108. The first forty schools train the old rule, the next forty train the new rule, and the remaining 82 form the audit domain. The corresponding valid-target record counts are 1,115, 1,196 and 2,441. The audit contains 801/821/819 A/B/C forms. School-level separation avoids training and evaluating on students in the same school. Estimates and uncertainty below condition on the fitted rules and the chosen audit domain; they do not include variation from retraining the models.

For version $j \in \{0, 1\}$, let

$$p_j(x) = \{f_{jU}(x_U) + f_{jV}(x_V) + f_{jW}(x_W)\}/3.$$

Each component is a fifty-tree probability forest, trained only on records receiving that block. Old/new component depths are two/three, respectively, with minimum leaf size twenty, all three block features eligible at each split, fixed independent seeds, and survey weights normalized to mean one in fitting. Old block fits use 745/744/741 records; new fits use 805/786/801. There is no complete-record training or calibration sample. The resulting contrast depends on all nine predictors. This is a specified secondary model-comparison task, not a deployed educational prediction system or a causal analysis of grade repetition.

H.3. Why Brier comparison is identifiable on these actual forms

For $K = 3$ blocks, set

$$a_k = \frac{f_{0k}^2 - f_{1k}^2}{9} + \frac{2Z}{3}(f_{1k} - f_{0k}), \qquad b_{k\ell} = \frac{2}{9}(f_{0k}f_{0\ell} - f_{1k}f_{1\ell}).$$

Expanding the squared losses gives the pointwise identity

$$(p_0 - Z)^2 - (p_1 - Z)^2 = \sum_k a_k + \sum_{k<\ell} b_{k\ell}.$$

Every single block is observed with probability $2/3$ and every pair with probability $1/3$. On the form containing k, ℓ, the score $\frac{3}{2}(a_k+a_\ell)+3b_{k\ell}$ is consequently unbiased. For randomized accuracy the corresponding score is $\frac{Y}{2}\{(f_{1k}-f_{0k})+(f_{1\ell}-f_{0\ell})\}$. Both are evaluated entirely on received responses and the common label.

The same argument applies to an average of arbitrary block functions: accuracy requires coverage of each block; Brier additionally requires coverage of each interacting block pair. This is an application of the observation-span criterion. A nonlinear transformation such as log loss can involve three-block interactions that none of these forms observes. The fact that every pair of blocks co-occurs does not identify arbitrary functions of all three blocks.

For the optimized score, keep $q = 1/3$ and solve the quadratic version of (E.1). Each form has 64 binary cells; 169 distinct Fourier terms occur across the three forms. The constraints are imposed separately for the two labels. Estimate each form's joint cell–label frequencies from the pooled old/new training schools, using survey weights within that form and 0.1 uniform shrinkage. No assumption that these estimated marginals form a compatible joint distribution is needed for unbiasedness. The released tables satisfy the target identity on all 512 profiles and both labels; the maximum numerical residual is 3.1×10^{-16}.

H.4. Survey uncertainty and imputation comparator

Let w_i and $w_i^{(r)}$ be the supplied final and eighty replicate weights. For frozen scores H_i, report the weighted mean $\widehat{\Delta} = \sum_i w_i H_i / \sum_i w_i$ in the audit domain and

$$\widehat{\mathrm{se}}^2 = \frac{1}{80(1-0.5)^2} \sum_{r=1}^{80} (\widehat{\Delta}^{(r)} - \widehat{\Delta})^2.$$

This is the documented Fay BRR calculation, not an iid Wald calculation. The resulting normal intervals are survey-asymptotic intervals, not the finite-sample minimax confidence guarantee. Conditional on sampled records and their fixed weights, the weighted form-score mean is unbiased over the stipulated form assignment; survey-population interpretation also uses the survey's weighting and response assumptions.

We implement twenty independent chains of logistic multiple imputation (van Buuren & Groothuis-Oudshoorn, 2011), with fifteen sweeps per chain and fixed seeds. Only the omitted block is imputed. Each conditional regression uses the other eight predictors, observed target, reported sex, grade and log survey weight. Coefficients use Laplace draws around a weakly regularized, weighted logistic fit. This is a transparent working-model implementation, not a claim to run the R `mice` package or an exact posterior sampler. For each completed dataset we evaluate the frozen contrast and its Fay variance. Rubin combination uses within-imputation mean variance $\overline{U}$, between-imputation variance B, total $\overline{U} + (1 + 1/20)B$, and the usual degrees-of-freedom adjustment. These intervals depend on the imputation specification, its compatibility with the target, and survey approximations; nominal coverage is not verified without complete-data truth.

For randomized accuracy, optimized scores give 0.04142 with interval $[0.03785, 0.04500]$, block HT gives 0.04174 with $[0.03814, 0.04534]$, and logistic MI gives 0.04180 with $[0.03825, 0.04535]$. For Brier improvement, all intervals in Table 13 contain zero. Both probabilities stay below 0.5 on the full cube (old range 0.112–0.231, new range 0.040–0.137), so thresholded accuracy has identically zero contrast. The probability-loss comparison is substantive only as a probability-quality target, not as an improvement in these deterministic decisions. The agreement is useful evidence of stability for this specified analysis, not evidence that the imputation model is correct or that optimized scores dominate simpler scores.

Each predictor occurs on two forms, allowing nine marginal response comparisons. On audit records, two unadjusted BRR comparisons have $p < 0.05$ (0.0479 and 0.0449); none meets $0.05/9$. The maximum estimated difference is 0.0482. Random differences, form order, context and nonresponse can all contribute. Failure to reject after correction does not certify measurement invariance, and these marginal checks cannot identify all unobserved joint response differences. No branching benefit, MSE or coverage estimate is claimed from this archive.

I. Reproducibility and Research Methods

The anonymous supplement contains the code of the solver and of every experiment, per-task results with certificates, logs, and the generators of all tables and figures. Source data are public and obtained separately; no individual survey records are redistributed.

Checks. The solver reproduces, on every task that overlaps with an independent dense implementation, the fixed constants, Γ_P and the certified branching constants; every interview it generates is checked to partition the profile cube (for $m \leq 15$). Exact rational checks cover the hybrid-budget inequalities of Proposition C.6, small conic programs are solved independently, and the product counterexample of Appendix C.4 has a rational certificate checked on all 256 profiles. Nonlinear-model floor certificates (Appendix G.1) use exact integer nullspace checks and rational endpoint evaluation. Monte Carlo checks of the potential inequality (D.2) and of greedy r-round strategies on pointer trees of depth 3 to 10 accompany the proof of Theorem D.3. These finite checks support the implementations and derivations; they do not replace the proofs.

Research methods. A generative language model participated in formulating research questions, deriving and checking proof arguments, implementing analyses, and drafting the manuscript; the authors remain responsible for all content.